

## **FlowPilot: A Multi-Agent AI System for Continuous Flow Chemistry Process Design**

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## 1 FlowPilot workflow, data contracts, and run provenance

FlowPilot separates chemical interpretation, engineering calculation, equipment realization, and publication of executable instructions into explicit computational stages. This separation is intended to prevent an unconstrained language-model response from being presented directly as a laboratory procedure. The complete workflow begins with a standardized chemist-facing intake and ends with either a validated, inventory-assigned design or a diagnostic blocked state. Interpretation of design evidence follows an explicit information-authority order: measured experimental evidence, hard safety and inventory constraints, confirmed batch-protocol facts, chemist hypotheses, model inference. Consequently, historical measurements and declared laboratory limits constrain candidate generation and revision, whereas a hypothesis is retained as an idea to test rather than promoted to an observed fact. This evidence hierarchy does not authorize overriding a hard safety or equipment constraint: earlier experimental success outside a declared limit does not make a new proposal compliant.

### 1.1 System architecture and decision authority

A FlowPilot run passes through standardized intake, batch parsing, upstream chemistry analysis, plan-aware retrieval, deterministic calculation and design-space exploration, downstream proposal generation, multi-agent council review, inventory reconciliation, final validation, and artifact publication (**Figure S1a**). The upstream chemistry layer identifies the transformation, mechanistic features, stage structure, stream incompatibilities, phase behavior, and candidate process bottlenecks. The retrieval layer supplies literature analogies and engineering rules relevant to that plan. These model-derived interpretations inform design, but they cannot supersede confirmed protocol facts, measurements, or hard operating limits. Deterministic components calculate and reconcile dependent quantities such as flow rate, reactor volume, residence-time basis, gas delivery, pressure settings, and inventory assignments. The council can criticize a candidate and propose bounded revisions, but a council response is not an executable output. Revised fields are recalculated and checked before publication. The final disposition therefore depends on deterministic closure rather than model confidence. Three user-facing outcomes are distinguished. An executable screening design contains canonical parameters and a fully assigned process topology. An inventory-confirmation state identifies the missing capability required before numerical execution. A blocked state records the failed requirements and diagnostic topology but withholds executable run instructions.

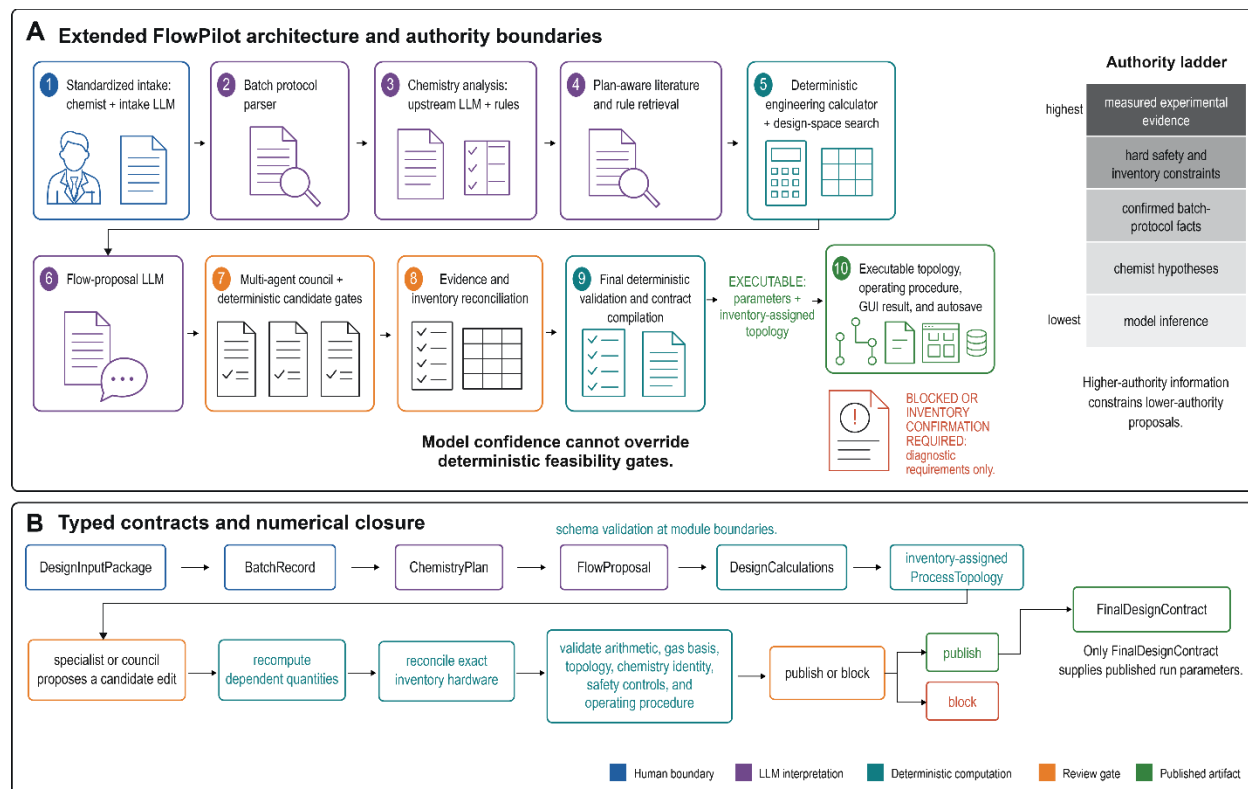
### 1.2 Typed contracts and numerical closure

Major pipeline boundaries are represented by validated data objects rather than prose alone (**Figure S1b**). The principal chain is DesignInputPackage > BatchRecord > ChemistryPlan > FlowProposal > DesignCalculations > inventory-assigned ProcessTopology > FinalDesignContract. Natural-language rationales may accompany these objects, but downstream code consumes typed fields and performs boundary validation.

The FinalDesignContract is rebuilt after engineering realization and inventory reconciliation. A publishable screening result contains canonical parameters, stage definitions, streams, instrument assignments, a process graph, chemistry identity, and the available operating and validation instructions. The implemented gates check consistency among these records, including arithmetic, declared time basis, equipment allocation, and topology. An unresolved blocking check prevents publication of executable parameters and retains intermediate values as diagnostics. Passing these checks establishes compliance with the



implemented contract; it does not establish chemical completeness, physical safety under all transients, or experimental yield.



**Figure S1. FlowPilot architecture, authority boundaries, and typed data contracts.** (a) The workflow separates standardized intake, chemistry interpretation, retrieval, deterministic engineering, council review, inventory reconciliation, and artifact publication. Measured evidence and hard safety or inventory constraints outrank confirmed protocol facts, chemist hypotheses, and model inference. (b) Typed contracts and closure checks ensure that only a validated FinalDesignContract can publish executable parameters.

**Table S1.** FlowPilot components, data contracts, and decision responsibilities.

| Component             | Computational class                        | Principal input                        | Principal output   | Decision responsibility                                                          |
|-----------------------|--------------------------------------------|----------------------------------------|--------------------|----------------------------------------------------------------------------------|
| Standardized intake   | LLM extraction + deterministic state rules | Free-text protocol and chemist answers | DesignInputPackage | Structures user information; fixed IDs and readiness rules remain deterministic. |
| Batch protocol parser | Structured parser + schema validation      | Frozen protocol facts                  | BatchRecord        | Normalizes batch conditions; cannot outrank the original confirmed protocol.     |

| Component                                      | Computational class                          | Principal input                                             | Principal output                         | Decision responsibility                                                                                  |
|------------------------------------------------|----------------------------------------------|-------------------------------------------------------------|------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Chemistry analysis                             | Upstream LLM + retrieved fundamentals        | BatchRecord, intake context                                 | ChemistryPlan                            | Interprets mechanism, stages, sensitivities, and stream logic; does not publish hardware settings.       |
| Plan-aware retrieval                           | Deterministic filtering and ranking          | ChemistryPlan and retrieval query                           | Ranked analogies and rules               | Supplies supporting precedents; retrieved analogies do not override measured evidence or constraints.    |
| Engineering calculator and design-space search | Deterministic                                | BatchRecord, ChemistryPlan, inventory limits                | DesignCalculations and feasible seeds    | Owens arithmetic closure, candidate feasibility calculations, and bounded search outputs.                |
| Flow-proposal agent                            | Downstream LLM                               | Chemistry, calculations, retrieval, inventory context       | FlowProposal                             | Proposes a complete candidate and rationale; proposal remains non-executable until validation.           |
| Multi-agent council                            | Specialist LLMs + deterministic gates        | Candidate proposals and evidence                            | Selected or revised candidate            | Critiques chemistry, engineering, safety, and practicality; revisions are bounded and recalculated.      |
| Inventory reconciliation and topology compiler | Deterministic                                | Selected candidate and LabInventory                         | Inventory allocation and ProcessTopology | Binds required operations to declared equipment and rejects unresolved assignments.                      |
| Final validation and artifact compiler         | Deterministic                                | Reconciled proposal, calculations, topology, safety context | FinalDesignContract                      | Owens executable/blocked disposition and canonical operating parameters.                                 |
| GUI renderer and autosave                      | Deterministic presentation and serialization | FinalDesignContract plus diagnostic traces                  | Result views and stored artifacts        | Publishes canonical outputs; diagnostic information is labelled and kept separate from run instructions. |

### 1.3 Reproducible standardized intake

The intake layer converts an initial free-text protocol and follow-up answers into a frozen DesignInputPackage. The LLM may extract batch fields and conversationally present missing questions, but it selects from a fixed bank of stable identifiers. It cannot invent a new question identifier or mark an incomplete package as ready. The current question bank is listed in **Table S2**. The batch protocol, design objective, and chemistry identity must be answered. Historical data, inventory, operating limits, and hypotheses must either be answered or explicitly marked unavailable. Output preference is optional.

Readiness and the ordered list of pending identifiers are calculated from package state. The frozen package records the raw protocol, extracted batch fields, objective, historical evidence, inventory constraints, operating limits, hypotheses, output preferences, chemistry confirmation, question log, answers, readiness state, and authority order. This design makes the intake boundary reproducible without claiming that every generative phrase is identical across model calls. Reproducibility applies to the allowed question identifiers, resolution rules, serialized chemist responses, readiness decision, and frozen context delivered to downstream modules. Where exact computational replay is required, the frozen package must be combined with model, decoding, software-revision, and environment records.

**Table S2.** Fixed standardized-intake question bank and readiness rules.

| Information category   | Readiness role      | Permitted resolution    | Downstream use                                                                                    |
|------------------------|---------------------|-------------------------|---------------------------------------------------------------------------------------------------|
| Batch protocol         | Required            | Answered only           | Establishes confirmed protocol facts and batch conditions.                                        |
| Design objective       | Required            | Answered only           | Defines how candidates are ranked and what the first design should achieve.                       |
| Chemistry identity     | Required            | Answered only           | Confirms transformation family, substrate/product identity, and bond changes.                     |
| Historical experiments | Required resolution | Answered or unavailable | Provides measured evidence for closed-loop calibration and trend constraints.                     |
| Laboratory inventory   | Required resolution | Answered or unavailable | Supplies hard equipment capabilities and discrete hardware options.                               |
| Operating limits       | Required resolution | Answered or unavailable | Defines pressure, temperature, flow, material, and safety boundaries.                             |
| Chemist hypotheses     | Required resolution | Answered or unavailable | Guides candidate screening but is explicitly treated below measured evidence and confirmed facts. |
| Output preference      | Optional            | Answered or omitted     | Controls presentation and requested screening format without changing physical feasibility rules. |

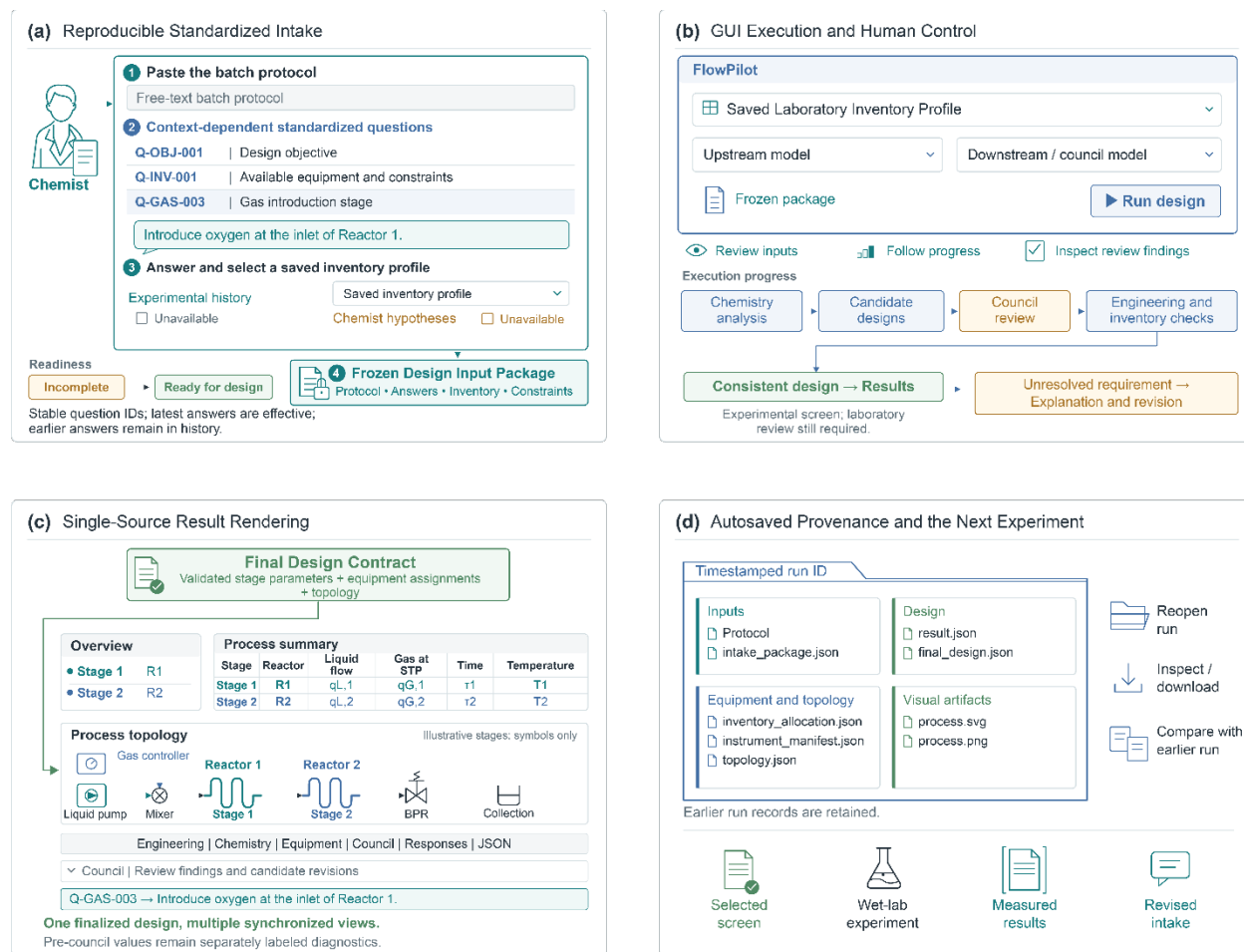
## 1.4 GUI execution and single-source result rendering

In the standardized-intake GUI, the user submits a protocol, resolves the fixed intake questions, and imports or selects a laboratory-inventory profile before initiating design. The run action is enabled only when the package is ready. The finalized package is passed to the same translation pipeline used by programmatic execution, and the resulting package is attached to the result. After final validation, executable numerical views are constructed from the post-realization FinalDesignContract. These include the summary,

engineering design, stream assignments, equipment allocation, process graph, safety controls, and canonical operating procedure. Chemistry-plan, retrieval, council, and experiment-loop records remain available as supporting context or audit traces. They are not independently promoted as authoritative run parameters. For a blocked run, the GUI displays the failure reasons and a requirements or diagnostic topology while withholding an executable procedure.

## 1.5 Run-level provenance and reconstruction

Completed GUI runs are automatically written to a uniquely timestamped directory. The artifact set preserves the standardized input, complete result, canonical final contract, concise summary, compiled topology, equipment allocation, instrument manifest, and available rendered diagrams (**Table S3**). Diagnostic topology and rendering artifacts are stored separately when the run is blocked or requires inventory confirmation. The render manifest can associate diagram files with topology-hash metadata, allowing a stored diagram to be checked against the corresponding graph where that metadata is available. The autosave package supports inspection of what was entered, calculated, validated, displayed, and assigned to hardware. It should not, by itself, be described as a complete computational-replay bundle because separate prompt logs, exact model snapshots, decoding parameters, dependency versions, and software revision identifiers are not guaranteed by the GUI autosave function.



**Figure S2. Standardized intake, GUI execution, synchronized results, and run provenance.** (a) The chemist supplies a protocol, resolves context-dependent questions with stable identifiers, and selects

inventory. The completed input package is frozen for the current run, while earlier answers remain in history. (b) The schematic GUI supports model selection, progress monitoring, and review, routing consistent designs to results and unresolved requirements to revision. (c) Overview, process tables, and equipment topology derive from one Final Design Contract; supporting diagnostics remain separately inspectable. Parameter symbols and the oxygen inlet arrangement are illustrative. (d) Timestamped folders preserve inputs, designs, equipment assignments, topology, and visual artifacts. Measured experimental feedback initiates a new design cycle while earlier records are retained. Computational consistency checks do not constitute experimental validation, and traceable input handling does not guarantee identical model responses.

**Table S3.** GUI autosave artifacts and their provenance function.

| Artifact                                                                              | Creation condition                          | Provenance function                                                                          |
|---------------------------------------------------------------------------------------|---------------------------------------------|----------------------------------------------------------------------------------------------|
| input.txt                                                                             | When raw user input is supplied to autosave | Original GUI text submitted for the run.                                                     |
| intake_package.json                                                                   | When an intake package is used              | Frozen standardized context, answers, readiness state, and authority order.                  |
| result.json                                                                           | Every completed autosave                    | Complete JSON-safe pipeline result, including available intermediate and diagnostic records. |
| final_design.json                                                                     | When the final contract is available        | Canonical executable or blocked FinalDesignContract used by numerical result views.          |
| summary.json                                                                          | Every completed autosave                    | Compact status and principal design fields for indexing and review.                          |
| topology.json                                                                         | When a process topology is compiled         | Stored unit-operation graph and stream connections.                                          |
| inventory_allocation.json                                                             | When inventory allocation is attempted      | Equipment assignments, unresolved requirements, and allocation status.                       |
| instrument_manifest.json                                                              | When a manifest is produced                 | Declared instruments assigned to the final design.                                           |
| process.svg; process.png                                                              | For a rendered executable topology          | Portable visual representation of the canonical process graph.                               |
| render_manifest.json                                                                  | When executable rendering is attempted      | Rendering status, artifact paths, and topology-hash metadata where available.                |
| process_requirements_topology.json                                                    | Conditional: unresolved inventory preflight | Required process structure before complete equipment assignment.                             |
| diagnostic_topology.json; diagnostic_process.svg/png; diagnostic_render_manifest.json | Conditional: blocked or diagnostic run      | Non-executable requirements view and rendering diagnostics; not operating instructions.      |

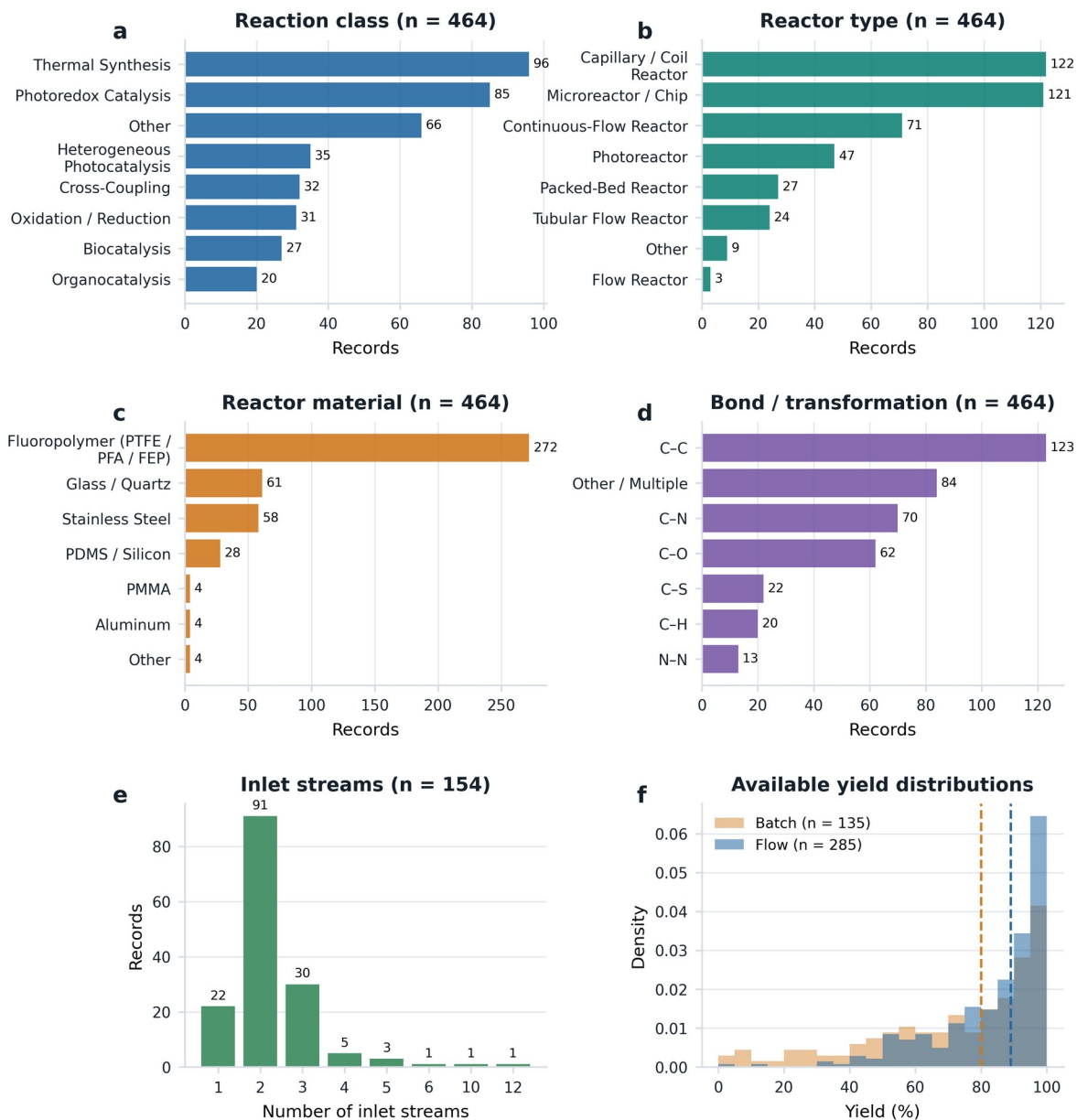
## 2 Literature corpus, rule base, and retrieval

FlowPilot uses two complementary knowledge resources. A structured literature corpus provides empirical batch-to-flow precedents and process metadata, whereas a handbook-derived rule base provides general engineering principles. Plan-aware retrieval selects a small set of literature analogies for the chemistry under consideration, and the upstream chemistry layer receives a prioritized subset of relevant handbook rules. These resources provide evidence and context; they do not replace deterministic calculations, laboratory inventory constraints, or experimental validation.

### 2.1 Frozen literature-corpus composition

The frozen manuscript export contains 464 classified flow-chemistry literature records. The classification spans thermal synthesis, photoredox and heterogeneous photocatalysis, cross-coupling, oxidation and reduction, electrochemistry, biocatalysis, polymer synthesis, crystallization, and other continuous-flow applications. For each record, available information was normalized into fields describing reaction class, reactor type, reactor material, bond or transformation label, operating conditions, number of inlet streams, and reported batch and flow outcomes. The primary categorical exports each contain 464 classified entries, but field completeness is not uniform. Explicit inlet-stream counts are available for 154 processes, batch yields for 135 records, and optimized-flow yields for 285 records (**Table S4**). Unknown, other, multiple, and long-tail classifications are retained rather than being excluded from denominators. This avoids inflating apparent coverage and makes the limitations of the extracted metadata visible.

The available yield distributions are descriptive. They should not be interpreted as a paired estimate of the causal benefit of flow because batch and flow values are missing for different subsets, publication and optimization selection can affect the distributions, and individual records differ in chemistry and objective. **Figure S3** therefore reports field-specific denominators directly in each panel.



**Figure S3. Composition of the flow-chemistry literature corpus.** (a-d) Reaction class, reactor type, reactor material, and bond or transformation labels (n = 464 records in each panel). (e) Number of inlet streams among records with stream-count annotation (n = 154). (f) Available batch-yield (n = 135) and optimized-flow-yield (n = 285) distributions. Categories are descriptive and non-exclusive where the source record contains more than one label.

**Table S4.** Available literature-corpus fields and available denominators.

| Metadata field                | Available n | Coverage               | Leading categories or summary                                                                                                                                    | Interpretation note                               |
|-------------------------------|-------------|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| Reaction class                | 464         | 100% classified export | Thermal Synthesis 96 (20.7%); Photoredox Catalysis 85 (18.3%); Other 66 (14.2%); Heterogeneous Photocatalysis 35 (7.5%); Cross-Coupling 32 (6.9%)                | Other and long-tail classes retained.             |
| Reactor type                  | 464         | 100% classified export | Capillary / Coil Reactor 122 (26.3%); Microreactor / Chip 121 (26.1%); Continuous-Flow Reactor 71 (15.3%); Photoreactor 47 (10.1%); Packed-Bed Reactor 27 (5.8%) | Long-tail hardware labels retained.               |
| Reactor material              | 464         | 100% classified export | Fluoropolymer (PTFE / PFA / FEP) 272 (58.6%); Glass / Quartz 61 (13.1%); Stainless Steel 58 (12.5%); PDMS / Silicon 28 (6.0%)                                    | Classification granularity varies across records. |
| Bond/transformation label     | 464         | 100% classified export | C–C 123 (26.5%); Other / Multiple 84 (18.1%); C–N 70 (15.1%); C–O 62 (13.4%); C–S 22 (4.7%)                                                                      | Other/multiple is an explicit category.           |
| Explicit inlet-stream count   | 154         | 33.2% of 464           | 1 streams: 22 (14.3%); 2 streams: 91 (59.1%); 3 streams: 30 (19.5%); 4 streams: 5 (3.2%); 5 streams: 3 (1.9%)                                                    | Percentages use n = 154, not the full corpus.     |
| Reported batch yield          | 135         | 29.1% of 464           | Median 80%; IQR 56-93%                                                                                                                                           | Available numeric values only.                    |
| Reported optimized-flow yield | 285         | 61.4% of 464           | Median 89%; IQR 75-96%                                                                                                                                           | Not a paired causal comparison with batch yield.  |

## 2.2 Engineering rule-base construction and use

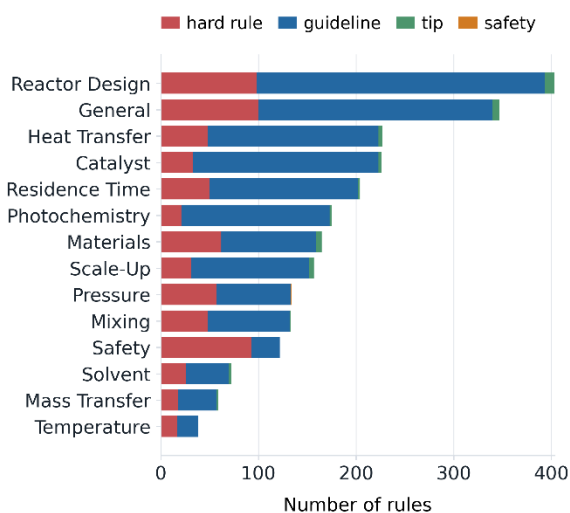
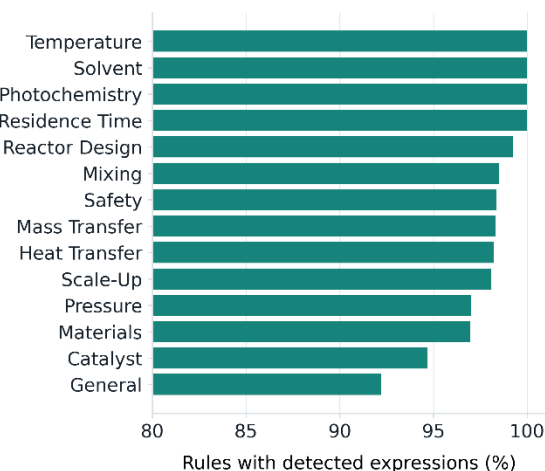
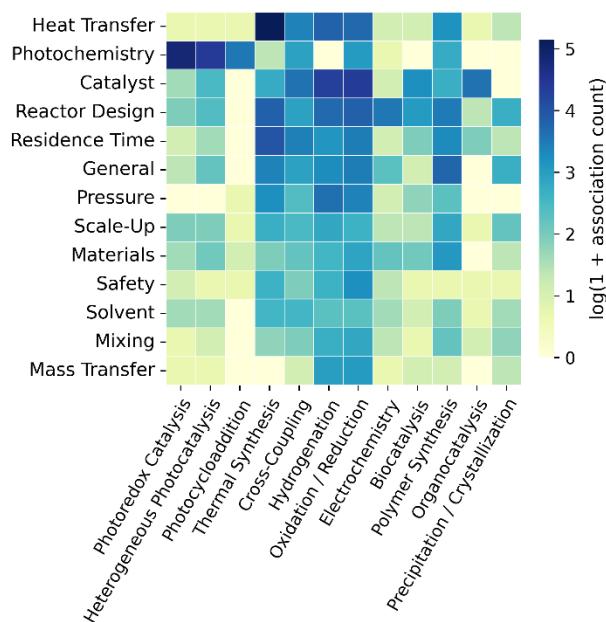
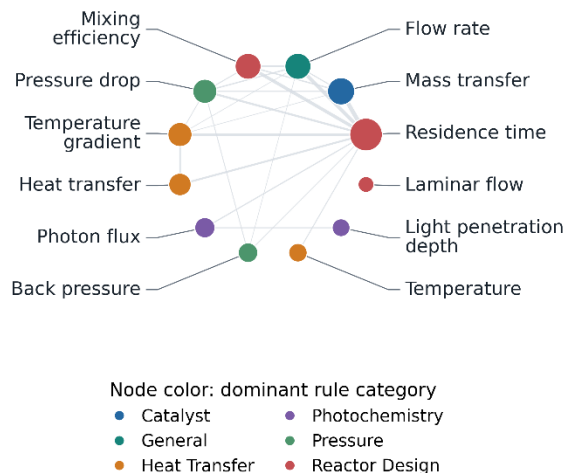
The engineering knowledge store contains 2,537 structured rules extracted from six handbook, guide, and literature sources. Each stored rule includes a rule identifier, category, triggering condition, recommendation, reasoning, optional quantitative expression, exceptions, severity, confidence, source document, source page, and source context when available. The complete export contains 29 rule categories; the 14 largest are summarized in **Table S5**. Across the complete store, 1,772 rules are labelled guideline, 717 hard\_rule, 47 tip, and 1 safety severity. Category and severity are distinct labels. For example, most entries in the Safety category are encoded as hard\_rule rather than with the separate safety



severity label. **Figure S4** therefore displays both dimensions rather than treating the Safety category as a severity class. At run time, the knowledge store selects broad engineering categories from the reaction context and adds keyword matches from the mechanism, catalyst, solvent, phase regime, and operating conditions. Duplicate rule identifiers are removed, hard rules are prioritized, and a compact subset is formatted for the upstream chemistry prompt. The downstream pipeline receives the resulting ChemistryPlan, while executable numerical requirements remain the responsibility of deterministic calculators and validation gates. The rule-based labels should not be overinterpreted. A machine-detected quantitative expression indicates that a rule contains formula-like content; it does not demonstrate independent equation verification. Similarly, a hard\_rule label controls prompt prioritization but is not automatically equivalent to an executable coded constraint. A rule becomes a deterministic gate only when the corresponding calculation or validation check is implemented in code.

**Table S5.** Largest engineering rule categories and severity composition.

| Category       | Total | Hard rule | Guideline | Tip | Safety |
|----------------|-------|-----------|-----------|-----|--------|
| Reactor Design | 403   | 98        | 295       | 10  | 0      |
| General        | 347   | 100       | 240       | 7   | 0      |
| Heat Transfer  | 227   | 48        | 175       | 4   | 0      |
| Catalyst       | 226   | 33        | 190       | 3   | 0      |
| Residence Time | 204   | 50        | 152       | 2   | 0      |
| Photochemistry | 175   | 21        | 152       | 2   | 0      |
| Materials      | 165   | 62        | 97        | 6   | 0      |
| Scale-Up       | 157   | 31        | 121       | 5   | 0      |
| Pressure       | 134   | 57        | 76        | 0   | 1      |
| Mixing         | 133   | 48        | 84        | 1   | 0      |
| Safety         | 122   | 93        | 29        | 0   | 0      |
| Solvent        | 72    | 26        | 44        | 2   | 0      |
| Mass Transfer  | 59    | 18        | 39        | 2   | 0      |
| Temperature    | 38    | 17        | 21        | 0   | 0      |

**a Largest rule categories****b Quantitative-expression coverage****c Associations by chemistry class****d Engineering concept co-occurrence**

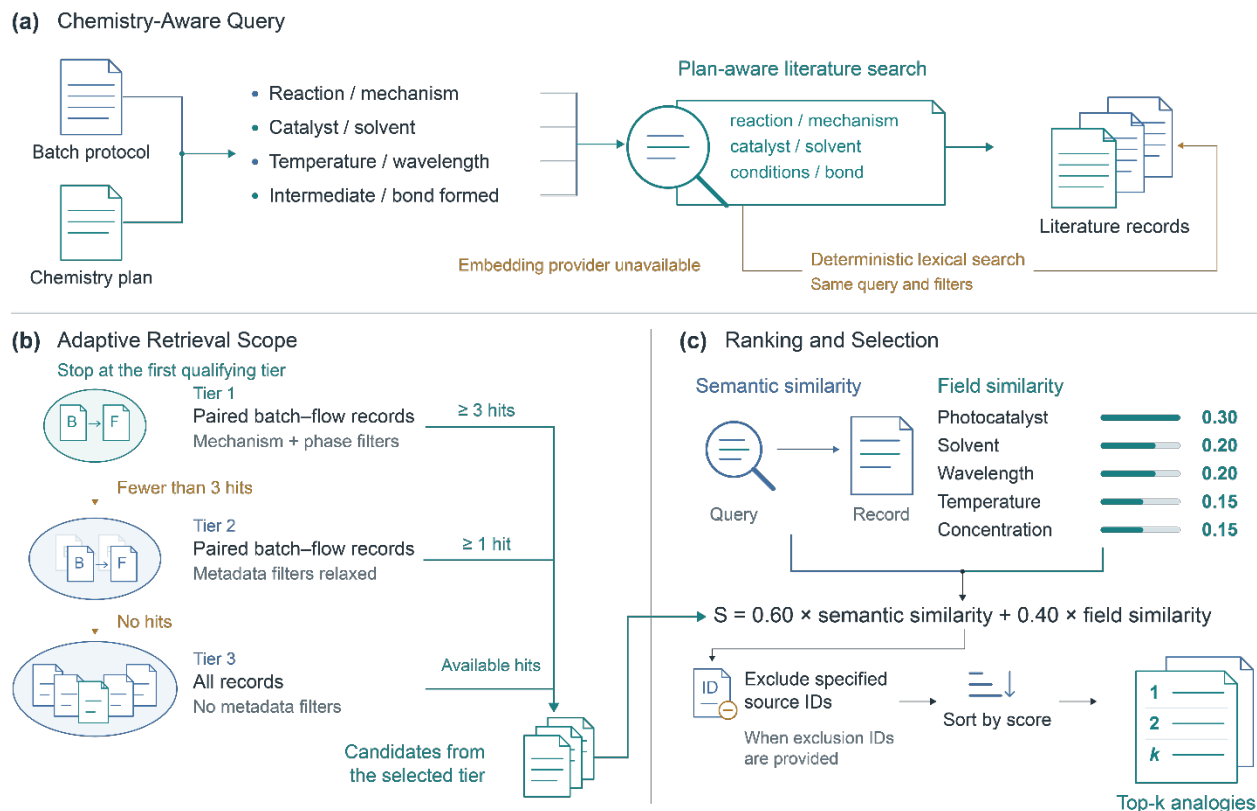
**Figure S4. Engineering rule-base structure.** (a) Counts by category and severity for the 14 largest categories. (b) Fraction of rules containing a machine-detected quantitative expression; this indicates expression coverage, not independent equation verification. (c) Non-exclusive rule-category associations across chemistry classes. (d) Computed concept co-occurrence network using the 12 most frequent concepts and 22 highest-weight links. Node area is  $35 + 420 \sqrt{f_i/f_{\max}}$  points squared, where  $f_i$  is concept frequency; node color is the dominant rule-category label; edge width is  $0.25 + 2.4(w_{ij}/w_{\max})$  points, where  $w_{ij}$  is the rule-level co-occurrence count. The circular layout is used only for legibility.

### 2.3 Plan-aware retrieval and tier transitions

The retrieval query is constructed from the ChemistryPlan rather than from protocol keywords alone. Reaction class, mechanism, catalyst or photocatalyst, solvent, phase regime, wavelength, temperature, and concentration are used when available. Retrieval begins with mechanism and phase filters over paired batch-flow records. If fewer than three candidates are found, the metadata filters are relaxed; if no paired records remain, the search expands to the complete corpus. Candidate records are reranked using 0.60 semantic similarity and 0.40 field similarity. The field score combines photocatalyst, solvent, wavelength, temperature, and concentration terms with weights of 0.30, 0.20, 0.20, 0.15, and 0.15, respectively (**Figure S5**).

**Table S6.** Frozen plan-aware retrieval tiers and reranking weights.

| Component                   | Setting                                   | Scope or transition                     |
|-----------------------------|-------------------------------------------|-----------------------------------------|
| Tier 1                      | Paired records; mechanism + phase filters | Relax when fewer than three hits remain |
| Tier 2                      | Paired records; filters relaxed           | Expand when no hits remain              |
| Tier 3                      | All records; no metadata filter           | Return ranked available records         |
| Semantic similarity         | 0.60                                      | Contribution to final reranking score   |
| Field similarity            | 0.40                                      | Contribution to final reranking score   |
| Photocatalyst               | 0.30                                      | Contribution within field similarity    |
| Solvent / wavelength        | 0.20 / 0.20                               | Contributions within field similarity   |
| Temperature / concentration | 0.15 / 0.15                               | Contributions within field similarity   |



**Figure S5. Plan-aware retrieval workflow.** (a) ChemistryPlan fields enrich the query before semantic retrieval; an embedding-provider failure invokes the deterministic lexical fallback without changing the query or metadata filters. (b) Tiered retrieval starts with paired records and strict mechanism/phase filters, relaxes filters when fewer than three results remain, and expands to all records only when no paired result remains; hidden source identifiers are excluded. (c) Reranking combines 0.60 semantic similarity with 0.40 field similarity before descending top-k selection.

## 2.4 Retrieval benchmark and leakage controls

The retrospective rank analysis used the archived implementation in `visualization/fig3c_score_decomposition.py`. Eighty corpus records were sampled without replacement with seed 42. Each query was compared with its 20 nearest non-self records, giving 1,600 query–result pairs. Embedding similarity was  $1/(1 + d)$ , where  $d$  is Euclidean distance. The offline final score was 0.60 times embedding similarity plus 0.40 times a reduced field score containing photocatalyst, solvent, and wavelength terms weighted 0.30, 0.20, and 0.20. Temperature and concentration terms were not included, and the reduced weights were not renormalized. The source export records 401 changed ranks (25.1%) and no exact query-ID self-hit. This is an offline reranking analysis, not a run of the complete operational query-enrichment and tier-filtering pipeline in **Table S6**.

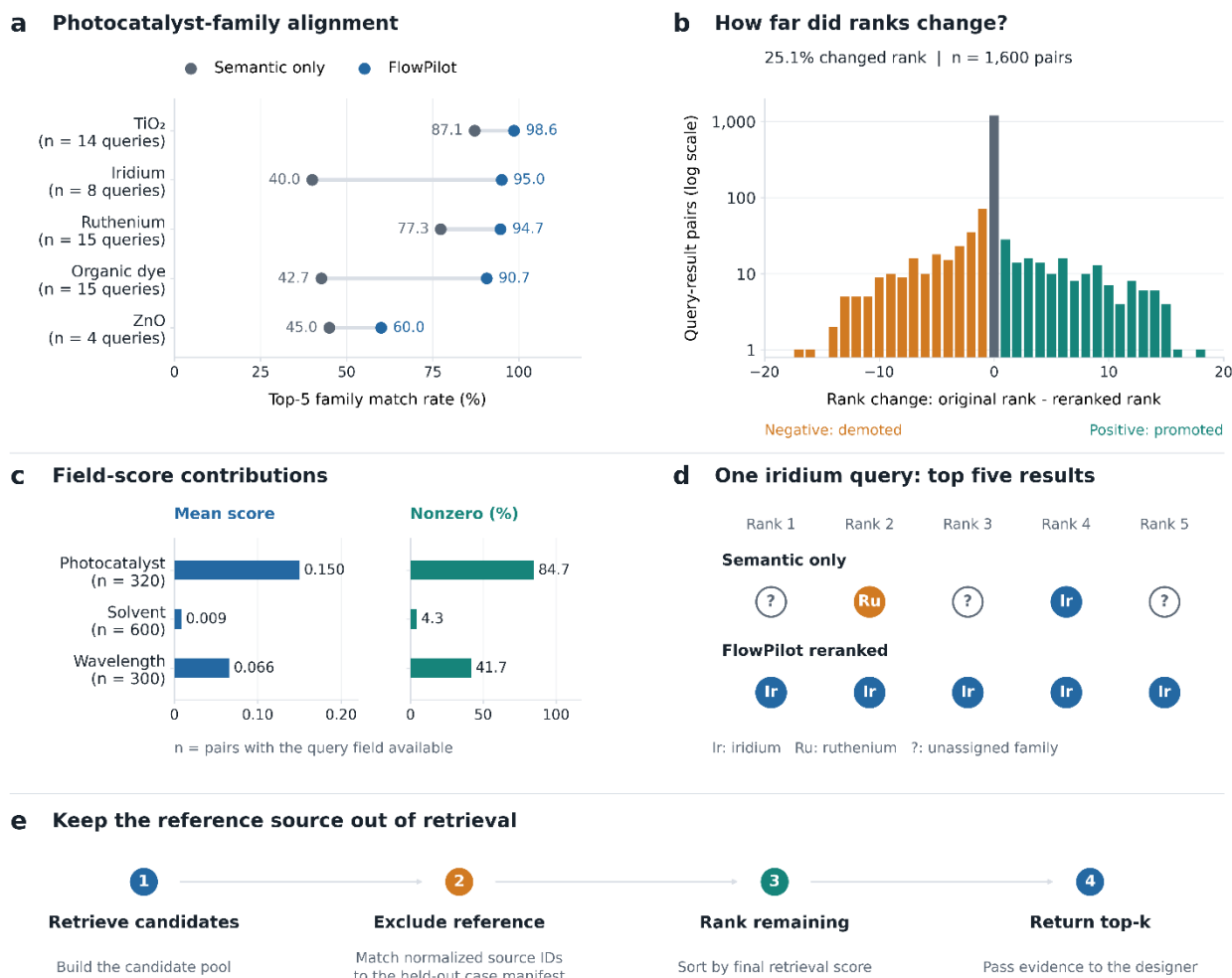
The separate family-alignment analysis used `visualization/fig3d_rag_quality.py`, with seed 42 and up to 15 queries per family: iridium (8), organic dyes (15), ruthenium (15), TiO<sub>2</sub> (14), and ZnO (4). For each query,

the metric was the fraction of five retrieved records with the same name-pattern-derived family label; reported rates are means over queries. The rank and family scripts both used photocatalyst, solvent, and wavelength terms, but their family classifiers and wavelength functions were not identical. In the rank script, wavelength similarity is 1 within 30 nm and decreases linearly to 0 at 100 nm; the family script uses  $\max(0, 1 - |\Delta\lambda|/100)$ . Matching nonempty family labels score 1, nonmatching labels score 0.3, and absent labels score 0. The rank script groups unrecognized nonempty catalyst names as `other_pc`, whereas the family script leaves unrecognized names unassigned. Solvent scoring uses normalized exact-name matches. Because family labels enter both ranking and evaluation, these rates describe metadata alignment rather than independent validation of chemical analogy quality.

Source-exclusion controls in the architecture campaigns remove the designated held-out source identifiers before final retrieval selection. The offline rank-pair CSV independently confirms no exact query self-hit. The family-analysis helper sets the self semantic similarity to zero but does not explicitly mask the self record after adding the metadata score. The archived family aggregates alone therefore do not certify strict self-exclusion for every query. The illustrative ranking contains no exact query self-hit, but is not a substitute for the missing complete family-query rankings. None of these controls establishes that the source publications were absent from model pretraining.

**Table S7.** Retrospective retrieval-analysis summary. Rank changes summarize 80 queries with 20 neighbors each. Family-match rates summarize a separate 56-query sample over five families. The offline scripts use the reduced three-field score described in Section 2.4, not the full operational score in **Table S6**.

| Metric or family              | Semantic retrieval | Plan-aware reranking | Evaluation n |
|-------------------------------|--------------------|----------------------|--------------|
| Query-result pairs reranked   | -                  | 25.1% changed rank   | 1,600 pairs  |
| Iridium family match          | 40.0%              | 95.0%                | 8 queries    |
| Organic-dye family match      | 42.7%              | 90.7%                | 15 queries   |
| Ruthenium family match        | 77.3%              | 94.7%                | 15 queries   |
| TiO <sub>2</sub> family match | 87.1%              | 98.6%                | 14 queries   |
| ZnO family match              | 45.0%              | 60.0%                | 4 queries    |



**Figure S6. Retrospective metadata alignment and source-exclusion controls.** (a) Mean top-five same-family fractions, with query counts. (b) Rank changes among 1,600 query–result pairs: 401 (25.1%) change rank; positive values indicate promotion. (c) Weighted component scores and nonzero fractions among pairs with the query field available. (d) Recorded family labels for an illustrative iridium query; ? denotes unassigned metadata. (e) Source-exclusion procedure used in the architecture campaigns. Panels (a–d) derive from separate offline analyses; Section 2.4 specifies their scoring functions and self-exclusion limitations. The figure does not independently establish reaction relevance or absence from model pretraining.

### 3 Ablation test and architecture comparison

#### Purpose and scope

The fixed-criteria multi-LLM evaluation was designed to compare the quality of final batch-to-flow designs produced by a direct one-shot large language model and by the complete FlowPilot architecture under matched inputs. The comparison was architecture-neutral at the evaluation stage: both methods received the same frozen chemistry, objective, laboratory inventory, and operating constraints, and both were required to return the same normalized final-design contract. The assessed object was the delivered design,

not the internal reasoning trace or the number of intermediate agent calls. Two related experiments are reported in this section. The architecture comparison tested one-shot generation against full FlowPilot using five retained generator models, three chemistry cases, and three independent generation repeats. The internal module-ablation screen fixed the generator to Qwen3.8-27B and selectively removed or altered FlowPilot components to examine their contribution to outcome quality and executability.

The benchmark measures outcome quality and repeatability for the frozen cases and inventory. It does not establish universal superiority across all reaction classes and does not replace prospective wet-lab validation.

### Paired architecture-comparison design

The retained manuscript comparison included the following generator models:

- Qwen3.6-27B
- Qwen3.8-27B
- GPT-4o
- Claude Sonnet 4.6
- Claude Opus 4.6

For every model, each of the three frozen chemistry cases was generated using both architectures in three independent calls. This produced:

$$5 \text{ models} \times 2 \text{ architectures} \times 3 \text{ cases} \times 3 \text{ repeats} = 90 \text{ candidate outcomes}$$

The paired unit was the same generator model, chemistry case, and repeat under the two architectures. The temperature was 0.2, calls were independently recorded, and the stopping rule was fixed before evaluation. The held-out source record for each case was excluded from FlowPilot retrieval. This exclusion prevents direct RAG leakage but does not prove that the underlying publication was absent from a model's pretraining data.

### Compared architectures

**One-shot.** A single model call received the complete frozen design input and a fixed output contract. No FlowPilot specialist agents, council deliberation, deterministic candidate selection, or post-generation correction operated inside this architecture.

**Full FlowPilot.** The same frozen design input entered the complete FlowPilot workflow, including chemistry interpretation, engineering calculations, specialist-agent assessment, council selection and revision, safety analysis, deterministic closure checks, and final schema enforcement. The internal workflow could use several LLM calls, but only the final normalized design was scored.

### Common one-shot wrapper

The one-shot system instruction was:

*You are a general chemistry assistant. Translate a batch chemistry protocol into a practical continuous-flow experiment.*

For each chemistry, the frozen design input reproduced in the three case-study subsections below was inserted after FROZEN DESIGN INPUT:, followed by the common response contract.

### Exact one-shot response contract

*Propose one complete flow design with numeric operating conditions, stream assignments, safety controls, and a brief rationale. Return one JSON object and no markdown with exactly these top-level fields: recommended\_disposition, disposition\_rationale, and proposal. The proposal value must be a JSON object, not a JSON-encoded string. The proposal object uses these fields: residence\_time\_min, flow\_rate\_mL\_min, temperature\_C, concentration\_M, BPR\_bar, reactor\_type, tubing\_material, tubing\_ID\_mm, reactor\_volume\_mL, residence\_time\_basis, residence\_time\_inlet\_min, residence\_time\_in\_channel\_min, light\_setup, wavelength\_nm, streams, mixer\_type, mixing\_order\_reasoning, pre\_reactor\_steps, post\_reactor\_steps, chemistry\_notes, stage\_parameters, reasoning\_per\_field, literature\_analogies, engine\_validated, safety\_flags, confidence. Each stream is an object with stream\_label, pump\_role, contents, solvent, phase, flow\_rate\_mL\_min, concentration\_M, molar\_equiv, gas\_flow\_sccm, and gas\_flow\_actual\_mL\_min where applicable. Distinguish inlet/STP gas flow from pressure-corrected in-channel gas flow. Ensure reactor\_volume\_mL is consistent with flow and residence time. State assumptions rather than inventing missing facts. recommended\_disposition must be exactly EXECUTE, SCREEN, or BLOCK. Use BLOCK when the hard constraints make a safe design physically or operationally impossible. Use SCREEN for a feasible first experiment that still requires experimental validation. Use EXECUTE only for a directly executable design supported by sufficient evidence.*

FlowPilot did not use one monolithic system prompt. It accepted the same frozen design input at its architecture boundary and distributed the information across its intake, chemistry, engineering, council, and validation stages. All internal prompts and stage events were retained in the archived run directories.

### Identity-masked evaluation

Each final outcome was converted to a standardized candidate packet. Generator identity and architecture labels were removed before judging. Complete architectural blinding is not claimed because formatting or output style could reveal provenance. Every judge received the same protocol, objective, inventory, normalized final outcome, held-out reference facts, deterministic verification sheet, and fixed rubric.

Three separate evaluator model families assessed each candidate:



The archived judge configuration for the matched example was Qwen /models/Qwen3.6-27B, OpenAI gpt-5.4-2026-03-05 and Anthropic claude-sonnet-4-6, at temperature 0.0. GPT-5.4 is identified here only as an evaluator; it is not one of the five retained generator models. Exact prompt schemas, seeds, responses and telemetry are retained in the companion source records. A recorded seed does not guarantee that a provider honors deterministic sampling.

- Qwen
- OpenAI
- Claude

The deterministic sheet exposed machine-checkable evidence for schema validity, numerical closure, inventory compliance, reactor and residence-time consistency, gas bookkeeping where applicable, topology coverage, and safety requirements. Deterministic evidence informed the judges but did not itself add an architecture-specific score bonus.

### Benchmark scoring method

The same 14 criteria were applied to every candidate. Each applicable criterion received an integer rating from 0 to 4. All criteria and all judges had equal weight; no architecture bonus or hand-tuned criterion weighting was used.

**Table S8.** Fixed 0-4 scoring anchors used by every independent judge.

| Score | Anchor                                                                                                    |
|-------|-----------------------------------------------------------------------------------------------------------|
| 0     | Fundamentally wrong, unsafe, chemically invalid, or physically impossible.                                |
| 1     | Major deficiencies, multiple material defects, or poor reproducibility; substantial redesign is required. |
| 2     | Partially correct with one material but recoverable defect requiring revision before execution.           |
| 3     | Correct overall with a minor defect that is unlikely to change execution or interpretation.               |
| 4     | Correct, complete, executable, and supported; no material defect.                                         |

**Table S9.** Complete frozen outcome questions, critical-error triggers and applicability rules. Gas use denotes a gas reagent or gas-liquid process; multi-stage denotes more than one reactive stage.

| Criterion                        | Exact evaluation question                                                                                                                                                | Critical-error conditions                                                                     | Applies |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|---------|
| UO-01<br>Transformation fidelity | Does the final design preserve the intended substrate, product, transformation, catalyst strategy, reaction modality, and phase regime without unsupported substitution? | Wrong product or transformation;<br>Chemically incompatible catalyst or modality substitution | Always  |

| Criterion                                    | Exact evaluation question                                                                                                                                                                               | Critical-error conditions                                                                                           | Applies     |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|-------------|
| UO-02<br>Required materials                  | Are all required reagents, catalysts, solvents, additives, gases, and reactive components present and correctly assigned?                                                                               | Required reactive component omitted; Unsafe or transformation-changing material introduced                          | Always      |
| UO-03<br>Stoichiometry and feed chemistry    | Are concentrations, equivalents, feed compositions, dilution, and chemical compatibility explicit and coherent?                                                                                         | Stoichiometry prevents intended reaction; Feed combination creates an unmitigated incompatibility                   | Always      |
| UO-04<br>Condition and stage mapping         | Are temperature, pressure, atmosphere, irradiation, addition order, stage boundaries, and chemistry-critical batch operations translated faithfully?                                                    | Chemistry-critical operation or stage omitted; Condition change makes the intended chemistry implausible or unsafe  | Always      |
| UO-05<br>Executable topology                 | Are feeds, pumps or MFCs, mixers, reactors, controls, separators, quench, collection, and waste connected in the correct executable order?                                                              | Required unit operation absent; Topology cannot physically execute the proposed chemistry                           | Always      |
| UO-06<br>Liquid material balance             | Do liquid concentrations and flow rates reproduce component molar flows, equivalents, total flow, dilution, and throughput?                                                                             | Material balance is dimensionally invalid; Reported feeds cannot produce the stated stoichiometry                   | Always      |
| UO-07<br>Residence-time and geometry closure | Do reactor volume, cumulative flow, geometry, holdup basis, per-stage residence time, and total residence time close numerically?                                                                       | Residence time is physically impossible; Reactor geometry or volume contradicts the executable design               | Always      |
| UO-08<br>Gas bookkeeping                     | For gas-using designs, do gas identity, STP definition, inlet flow, molar flow, equivalents, absolute pressure, channel flow, holdup, and gas-sensitive residence-time bases close and remain distinct? | Gas identity or pressure basis is unsafe; Gas molar flow, equivalents, or residence time is physically inconsistent | Gas use     |
| UO-09<br>Multistage closure                  | For multistage designs, are stream-introduction stages, cumulative flows, individual reactor volumes and times, interstage operations, and total process residence time consistent?                     | Stage sequence changes the chemistry; Stage flow or residence-time closure is physically impossible                 | Multi-stage |

| Criterion                                     | Exact evaluation question                                                                                                                                                                                                   | Critical-error conditions                                                                                                               | Applies |
|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|---------|
| UO-10<br>Inventory feasibility                | Does every selected equipment item exist, and are flow, volume, material, ID, temperature, pressure, wavelength, and connectivity within declared limits?                                                                   | Invented or unavailable essential equipment; Operating point exceeds a hard equipment limit                                             | Always  |
| UO-11<br>Physical and transport plausibility  | Are phase behavior, mixing, pressure drop, heat transfer, mass transfer, and transport assumptions physically plausible and supported at the proposed operating point?                                                      | Unmitigated boiling, precipitation, plugging, runaway, or pressure failure; Transport assumption makes the design physically impossible | Always  |
| UO-12<br>Hazards and controls                 | Are chemistry-specific thermal, pressure, gas, peroxide, azide, catalyst, solvent, and incompatibility hazards recognized and converted into concrete controls?                                                             | Major hazard omitted; Proposed execution lacks a necessary containment, purge, interlock, relief, or compatibility control              | Always  |
| UO-13<br>Operating procedure and work-up      | Are feed preparation, leak testing, priming, startup, steady-state qualification, collection, normal and emergency shutdown, depressurization, quench, separation, work-up, and waste handling reproducible where required? | Missing procedure creates an immediate execution hazard; The experiment cannot be reconstructed from the delivered instructions         | Always  |
| UO-14<br>Evidence and uncertainty calibration | Are source facts, calculations, assumptions, analogies, performance predictions, uncertainty, validation needs, confidence, and SCREEN/BLOCK/EXECUTE disposition represented honestly and proportionately?                  | Invented evidence or unsupported performance guarantee; Execution recommended despite an acknowledged unresolved critical defect        | Always  |

NOT\_APPLICABLE was permitted only when the criterion's applicability rule was false. Missing required information was rated 0 or 1 rather than excluded. In the frozen rubric, gas bookkeeping (UO-08) and multistage closure (UO-09) were the only criteria that could routinely become non-applicable. Critical-error flags were recorded separately and did not impose an arbitrary cap on the numerical score.

For candidate  $i$ , criterion  $k$ , and judge  $j$ , the criterion consensus was:

$$C_{i,k} = \frac{1}{J} \sum_{j=1}^J s_{i,k,j}$$

where  $s(i,k,j)$  is the judge's integer rating from 0 to 4. If  $A(i)$  is the set of applicable criteria, the candidate score was:

$$S_i = \frac{1}{|A_i|} \sum_{k \in A_i} \frac{C_{i,k}}{4}$$

The resulting candidate score ranged from 0 to 1. For each model, architecture, and repeat, the three chemistry candidate scores were first averaged to obtain a repeat-level campaign score:

$$R_{m,a,r} = \frac{1}{3} \sum_{q=1}^3 S_{m,a,r,q}$$

The reported model-architecture value was the mean of the three repeat-level scores, and the error bar was their sample standard deviation ( $n = 3$ ,  $\text{ddof} = 1$ ). It was not a confidence interval. The matched architecture effect was calculated before aggregation:

$$\Delta_{m,q,r} = S_{\text{FlowPilot},m,q,r} - S_{\text{one-shot},m,q,r}$$

A positive value therefore favored FlowPilot for that matched case-repeat pair. Judge disagreement was retained separately and was not used as a substitute for generation-repeat variability.

### Case study 1: Hydrogenolysis

*Benchmark title: Three-phase N-diphenylmethyl hydrogenolysis*

Source of the hydrogenolysis benchmark case: Tu et al., reference 1. Source exclusion is a retrieval control, not proof of absence from pretraining.

#### **RAW BATCH PROTOCOL:**

*N-Diphenylmethylazetidin-3-ol (DMAOL; MW 239.32 g/mol) was prepared as a 3.8 wt% solution in methanol. For molarity conversion, explicitly use and label an assumed solution density of 0.792 g/mL, giving approximately 0.126 M DMAOL. A 200 mL charge was contacted with hydrogen over 0.5 g of 7 wt% Pd(OH)2/Al2O3 catalyst in a stirred batch reactor at 60 °C and 21 bar for 5 h. Conversion to 3-azetidinol was 38.1%. Translate this into one conservative continuous-flow packed-bed screening design. Acetic acid is optional and must not be introduced unless explicitly justified as a screening variable. Report feed preparation, gas-liquid mixing, packed-bed operation, temperature, pressure basis, liquid flow, hydrogen inlet/STP flow, pressure-corrected in-channel hydrogen flow, hydrogen equivalents, nominal empty-bed space time, liquid-holdup/contact residence-time basis, separation, collection, startup, shutdown, nitrogen purge, and hydrogen/methanol/pressure safety controls.*

#### **OBJECTIVE:**

*Produce one executable, inventory-constrained first gas-liquid-solid flow screen without using the held-out source answer.*

#### **HARD CONSTRAINTS:**

- Use only the authoritative inventory supplied below.
- Hydrogen is the required reducing reagent and must be delivered through the certified H2 MFC.
- Use a single DMAOL/methanol liquid feed and the declared gas-liquid mixer before the Pd(OH)2/Al2O3 packed bed.
- Operate at 60 °C and the available 21 bar BPR setpoint; explicitly state whether gas compression uses 21 bar absolute or 21 bar gauge.
- Report hydrogen flow at inlet/STP and pressure-corrected in-channel conditions, with the conversion calculation.
- Report H2 equivalents from gas molar flow divided by DMAOL molar flow.
- Distinguish nominal empty-bed space time from liquid-holdup/contact residence time; do not silently use total gas-plus-liquid flow as liquid reaction time.
- Use the nitrogen line for startup leak/purge verification and shutdown displacement of residual hydrogen.
- Return SCREEN unless the inventory cannot support a safe design.

AVAILABLE INVENTORY (JSON):

```
{
  "BPR_available": [21.0],
  "collectors": [
    {
      "equipment_id": "collector_h2_250",
      "max_pressure_bar": 1.0,
      "name": "Grounded vented liquid collection vessel",
      "type": "collection vessel",
      "volume_mL": 250.0
    }
  ],
  "connectors": [
    {
      "equipment_id": "connector_h2_pbr",
      "material": "stainless steel",
      "max_pressure_bar": 40.0,
      "name": "Hydrogen-rated packed-bed connector set",
      "supported_ID_mm": [4.35],
      "type": "packed-bed union"
    }
  ],
  "degassers": [],
  "filters": [],
  "gas_hardware": [
    {
      "equipment_id": "gas_h2_cylinder",
      "gas": "H2",
      "max_pressure_bar": 40.0,
      "name": "Certified hydrogen cylinder and regulator",
      "type": "gas_supply"
    },
    {
      "equipment_id": "mfc_h2_100",
      "gas": "H2",
      "max_flow_sccm": 100.0,
      "max_pressure_bar": 40.0,
      "min_flow_sccm": 1.0,
      "name": "Hydrogen mass-flow controller",
      "type": "MFC"
    },
    {
      "equipment_id": "gas_n2_cylinder",
      "gas": "N2",
      "max_pressure_bar": 40.0,
      "name": "Nitrogen purge cylinder and regulator",
      "type": "gas_supply"
    }
  ],
  "light_sources": [],
  "mixers": [
    {
      "equipment_id": "mixer_h2_t",
      "material": "stainless steel",
      "max_inputs": 2,
      "max_pressure_bar": 40.0,
      "max_total_flow_mL_min": 5.0,
      "min_total_flow_mL_min": 0.1,
      "name": "Hydrogen-service T-mixer",
      "supported_ID_mm": [4.35],
      "type": "T-mixer"
    }
  ],
  "pressure_controllers": [
    {
      "equipment_id": "bpr_h2_21",
      "max_pressure_bar": 21.0,
      "min_pressure_bar": 21.0,
      "name": "Hydrogen-service BPR",
      "setpoints_bar": [21.0],
      "type": "BPR"
    }
  ],
  "pumps": [
    {
      "compatible_materials": ["methanol", "DMAOL"],
      "equipment_id": "pump_h2_pilot",
      "max_flow_rate_mL_min": 2.0,
      "max_pressure_bar": 40.0,
      "min_flow_rate_mL_min": 0.1,
      "name": "Pressure-rated plunger"
    }
  ]
}
```

```

pump", "type": "plunger"}], "reactor_trains": [], "reactors": [{"ID_mm": 4.35, "allowed_temperatures_C":
[60.0], "configuration": "single 200 mm gas-liquid-solid packed
tube", "equipment_id": "reactor_h2_pbr_300", "material": "stainless
steel", "max_pressure_bar": 40.0, "max_temperature_C": 100.0, "min_pressure_bar": 0.0, "min_temperature_
C": 20.0, "name": "Pd(OH)2/Al2O3 micro-packed bed", "system": "hydrogenation skid", "type": "packed-
bed", "volume_mL": 3.0}], "safety_accessories": [{"capabilities":
["backflow_prevention"], "compatible_hazards":
["hydrogen"], "equipment_id": "safety_h2_check_valve", "max_pressure_bar": 40.0, "name": "Hydrogen
non-return check valve", "type": "check valve"}, {"capabilities":
["vented_separator"], "compatible_hazards":
["hydrogen"], "equipment_id": "safety_h2_vent", "max_pressure_bar": 25.0, "name": "Dedicated hydrogen
separator safe vent", "type": "safe vent"}], "schema_version": "flowpilot_lab_inventory_v3.0", "separators":
[{"equipment_id": "separator_h2_gl", "max_flow_mL_min": 10.0, "max_pressure_bar": 25.0, "name": "Vent
ed gas-liquid separator", "supported_phases": ["gas", "liquid"], "type": "gas-liquid
separator"}], "strict_assignment": true, "temperature_controllers": [{"allowed_temperatures_C":
[60.0], "equipment_id": "bath_h2_60", "max_temperature_C": 90.0, "min_temperature_C": 20.0, "name": "Pa
cked-bed water bath", "type": "water bath"}], "tubing":
[{"ID_mm": 4.35, "equipment_id": "tube_h2_pilot_ss", "length_m": 0.2, "material": "stainless
steel", "max_pressure_bar": 40.0, "max_temperature_C": 100.0, "name": "Hydrogen-service packed
tube", "transparent": false}]}
CHEMIST-CONFIRMED CHEMISTRY IDENTITY:
{"confirmed": true, "mechanism_type": "heterogeneous catalytic
hydrogenolysis", "reaction_class": "hydrogenolysis", "reaction_name": "N-diphenylmethyl
hydrogenolysis"}

```

## Case study 2: Photochemical oxidation

*Benchmark title: Photocatalytic aerobic oxidation of Fmoc-L-methionine*

Source of the photochemical benchmark case: Thomson et al., reference 2. The reaction name is not the article title. The paper's supporting information includes Fmoc-L-methionine oxidation; the campaign's extracted numerical facts have not been revalidated page by page in this document revision.

### RAW BATCH PROTOCOL:

*Fmoc-L-methionine (3.7 mmol, 1.0 equiv) and 4,7-diphenyl-2,1,3-benzothiadiazole (10 mol%) were dissolved in acetonitrile at 0.10 M in a round-bottom flask. The solution was exposed to air and irradiated with a 1.5 W LED centered near 420 nm at 21 C for 5 min. Fmoc-methionine sulfoxide was formed in 99% yield. Translate this into one conservative continuous-flow photochemical screening design. Report feed*

*preparation, air delivery and oxygen-equivalent basis, gas-liquid mixing, pressure basis, photoreactor assignment, wavelength, temperature, liquid flow, inlet/STP and pressure-corrected in-channel gas flow, liquid contact residence time, reactor volume, separation, collection, startup, shutdown, and oxygen/acetonitrile/light safety controls.*

**OBJECTIVE:**

*Produce one executable, inventory-constrained first gas-liquid photochemical flow screen without using the held-out source answer.*

**HARD CONSTRAINTS:**

- Use only the authoritative inventory supplied below.*
- Use one premixed Fmoc-L-methionine/photocatalyst/acetonitrile liquid feed and the declared air MFC.*
- Use the declared gas-liquid mixer before the 420 nm photoreactor.*
- Operate at the available 3 bar BPR setpoint and state whether pressure is absolute or gauge.*
- Report air at inlet/STP and pressure-corrected in-channel conditions and calculate oxygen equivalents from 21 mol% oxygen.*
- Use liquid flow and the 1.0 mL reactor volume to define liquid contact residence time; do not use total gas-plus-liquid volumetric flow as reaction time.*
- No inline degasser is available or required because air is a reagent.*
- All selected equipment must be named by its inventory equipment\_id.*
- Return SCREEN unless the inventory cannot support a safe design.*

**AVAILABLE INVENTORY (JSON):**

```
{ "BPR_available": [3.0], "collectors":  
[ { "equipment_id": "collector_photo_100", "max_pressure_bar": 1.0, "name": "Amber vented collection  
vessel", "type": "collection vessel", "volume_mL": 100.0 } ], "connectors":  
[ { "equipment_id": "connector_photo_set", "material": "PFA", "max_pressure_bar": 10.0, "name": "PFA  
photochemistry connector set", "supported_ID_mm": [1.0], "type": "photoreactor union set" } ], "degassers":  
[], "filters": [], "gas_hardware":  
[ { "equipment_id": "gas_air_clean", "gas": "air", "max_pressure_bar": 10.0, "name": "Clean dry air supply and  
regulator", "type": "gas_supply" },  
{ "equipment_id": "mfc_air_10", "gas": "air", "max_flow_sccm": 10.0, "max_pressure_bar": 10.0, "min_flow_  
sccm": 0.1, "name": "Air mass-flow controller", "type": "MFC" } ], "light_sources":  
[ { "allowed_temperatures_C":  
[21.0, 25.0], "compatible_reactor": "coil", "equipment_id": "light_photo_420", "max_temperature_C": 40.0, "  
min_temperature_C": 15.0, "name": "420 nm photochemistry  
LED", "power_W": 1.5, "wavelength_nm": 420.0 } ], "mixers":  
[ { "equipment_id": "mixer_photo_t", "material": "PFA", "max_inputs": 2, "max_pressure_bar": 10.0, "max_tot
```

```

al_flow_mL_min":2.0,"min_total_flow_mL_min":0.01,"name":"PFA gas-liquid T-
mixer","supported_ID_mm":[1.0],"type":"T-mixer"}], "pressure_controllers":
[{"equipment_id":"bpr_photo_3","max_pressure_bar":3.0,"min_pressure_bar":3.0,"name":"Photochemist
ry BPR","setpoints_bar":[3.0],"type":"BPR"}], "pumps":[{"compatible_materials":["acetonitrile","Fmoc-
L-methionine","4,7-diphenyl-2,1,3-
benzothiadiazole"],"equipment_id":"pump_photo_liquid","max_flow_rate_mL_min":1.0,"max_pressure_
bar":10.0,"min_flow_rate_mL_min":0.01,"name":"Photochemistry HPLC
pump","type":"HPLC"}], "reactor_trains":[], "reactors":[{"ID_mm":1.0,"allowed_temperatures_C":
[21.0,25.0],"configuration":"irradiated PFA
coil","equipment_id":"reactor_photo_100","light_source":"light_photo_420","material":"PFA","max_pre
ssure_bar":10.0,"max_temperature_C":40.0,"min_pressure_bar":0.0,"min_temperature_C":15.0,"name":"
One milliliter PFA photocoil","system":"420 nm photochemistry module","type":"photochemical
coil","volume_mL":1.0,"wavelength_nm":420.0}], "safety_accessories":[{"capabilities":
["shield_or_containment","vented_separator"],"compatible_hazards":
["high_intensity_light","oxygen_enriched_organic_stream","acetonitrile_vapor"],"equipment_id":"safety
_photo_enclosure","max_pressure_bar":1.0,"name":"Interlocked photochemical enclosure and
vent","type":"interlocked enclosure"}], "schema_version":"flowpilot_lab_inventory_v3.0","separators":
[{"equipment_id":"separator_photo_gl","max_flow_mL_min":5.0,"max_pressure_bar":5.0,"name":"Vent
ed gas-liquid separator","supported_phases":["gas","liquid"],"type":"gas-liquid
separator"}], "strict_assignment":true,"temperature_controllers":[{"allowed_temperatures_C":
[21.0,25.0],"equipment_id":"controller_photo_thermostat","max_temperature_C":40.0,"min_temperature
_C":15.0,"name":"Photoreactor thermostat","type":"recirculating thermostat"}], "tubing":
[{"ID_mm":1.0,"equipment_id":"tubing_photo_pfa_10","length_m":1.273,"material":"PFA","max_press
ure_bar":10.0,"max_temperature_C":60.0,"name":"PFA photoreactor coil tubing","transparent":true}}]
CHEMIST-CONFIRMED CHEMISTRY IDENTITY:
{"confirmed":true,"mechanism_type":"photocatalytic aerobic
oxidation","reaction_class":"oxidation","reaction_name":"photocatalytic aerobic oxidation of Fmoc-L-
methionine"}

```

### Case study 3: CuAAC

*Benchmark title: Cu/C-catalyzed azide-alkyne cycloaddition*

Source of the CuAAC benchmark case: Fuchs et al., reference 3. Frozen case inputs and inventory constraints are retained unchanged.

*RAW BATCH PROTOCOL:*



*Benzyl azide (0.50 mmol, 1.0 equiv) and phenylacetylene (0.55 mmol, 1.1 equiv) were combined with Cu/C catalyst (10 mol% Cu) in acetone (2.0 mL; benzyl azide concentration 0.25 M). The homogeneous liquid reaction mixture was heated under microwave batch conditions at 150 °C for 3 min. The corresponding 1,4-disubstituted 1,2,3-triazole was isolated in 94% yield. Translate this batch procedure into one conservative continuous-flow screening design. Preserve the Cu/C catalyst strategy and explicitly report feed preparation, reactor topology, temperature, liquid flow, pressure, reactor/contact volume, residence-time basis, collection, work-up, and azide/pressure safety controls.*

**OBJECTIVE:**

*Produce one executable, inventory-constrained first flow screen without using the held-out source answer.*

**HARD CONSTRAINTS:**

- Use only the authoritative inventory supplied below.
- Use a single premixed liquid feed through the Cu/C packed-bed cartridge.
- Do not invent a gas stream, photochemical equipment, or unlisted hardware.
- Report catalyst-contact residence time and make reactor volume, liquid flow, and residence time mathematically consistent.
- Return SCREEN unless the inventory cannot support a safe design.

**AVAILABLE INVENTORY (JSON):**

```
{ "BPR_available": [20.0], "collectors":  
[ { "equipment_id": "collector_amber_100", "max_pressure_bar": 1.0, "name": "Vented collection  
vessel", "type": "collection vessel", "volume_mL": 100.0 }, { "connectors":  
[ { "equipment_id": "connector_xcube_cart", "material": "stainless  
steel", "max_pressure_bar": 40.0, "name": "X-Cube cartridge connector", "supported_ID_mm":  
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[], "pressure_controllers":  
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bed", "volume_mL": 0.3 }, { "safety_accessories": [ { "capabilities":  
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```
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*CHEMIST-CONFIRMED CHEMISTRY IDENTITY:*

```
{"confirmed": true, "mechanism_type": "copper-catalyzed
cycloaddition", "reaction_class": "cuaac", "reaction_name": "copper-catalyzed azide-alkyne cycloaddition"}
```

### Internal FlowPilot module-ablation screen

The internal screen used Qwen3.8-27B as the fixed generator and the same three chemistry cases. Fifteen architecture conditions were tested, giving 45 condition-case outcomes. Each cell was generated once; SD therefore describes between-case variation, not generation-repeat uncertainty. Disabled specialists received no LLM call or default-score penalty, and active council weights were renormalized. **Table S10** now lists all 15 conditions from the archived module summary. Main-text **Figure 4d** displays a selected subset. The archive's executable-outcome fraction is retained as metadata and should not be interpreted as independent experimental implementability or as a fair cross-architecture safety rating.

**Table S10.** All 15 internal module-ablation conditions, restored from the archived 45-outcome summary. Mean score and sample SD summarize three chemistries with one generation per cell. Exec. is the archive's executable-status fraction, not experimental success or a safety verdict. LLM calls are means per outcome; flags are totals over the three cases.

| Condition                       | Mean  | SD    | Exec. | Calls | Flags |
|---------------------------------|-------|-------|-------|-------|-------|
| Without preselection refinement | 0.935 | 0.028 | 100%  | 11.0  | 0     |
| One-pass council                | 0.932 | 0.019 | 100%  | 9.7   | 0     |
| Full FlowPilot                  | 0.928 | 0.027 | 100%  | 13.3  | 0     |
| Without winner revision         | 0.926 | 0.042 | 100%  | 13.0  | 0     |
| Without DFMEA                   | 0.926 | 0.042 | 100%  | 12.7  | 0     |
| Without Kinetics agent          | 0.924 | 0.057 | 100%  | 12.0  | 0     |
| Deterministic selection         | 0.920 | 0.041 | 100%  | 12.7  | 0     |
| Candidate budget 4              | 0.915 | 0.049 | 100%  | 18.7  | 0     |
| Without Safety agent            | 0.915 | 0.060 | 100%  | 12.3  | 0     |
| Without Fluidics agent          | 0.913 | 0.042 | 100%  | 10.7  | 0     |

| Condition               | Mean  | SD    | Exec. | Calls | Flags |
|-------------------------|-------|-------|-------|-------|-------|
| Candidate budget 1      | 0.907 | 0.048 | 100%  | 12.3  | 0     |
| Without Skeptic audit   | 0.904 | 0.039 | 100%  | 13.3  | 0     |
| Without Chemistry agent | 0.902 | 0.054 | 100%  | 12.0  | 0     |
| One-shot                | 0.743 | 0.167 | 0%    | 1.0   | 14    |
| No council              | 0.670 | 0.137 | 0%    | 3.0   | 19    |

The screen supports descriptive module attribution. It should not be interpreted as a statistically powered ranking of every internal configuration. Notably, removal of an individual specialist did not necessarily reduce the aggregate score for this small case set, whereas eliminating the complete council caused a substantial loss of executable design quality.

**Table S11.** Matched one-shot and FlowPilot architecture scores by generator model.

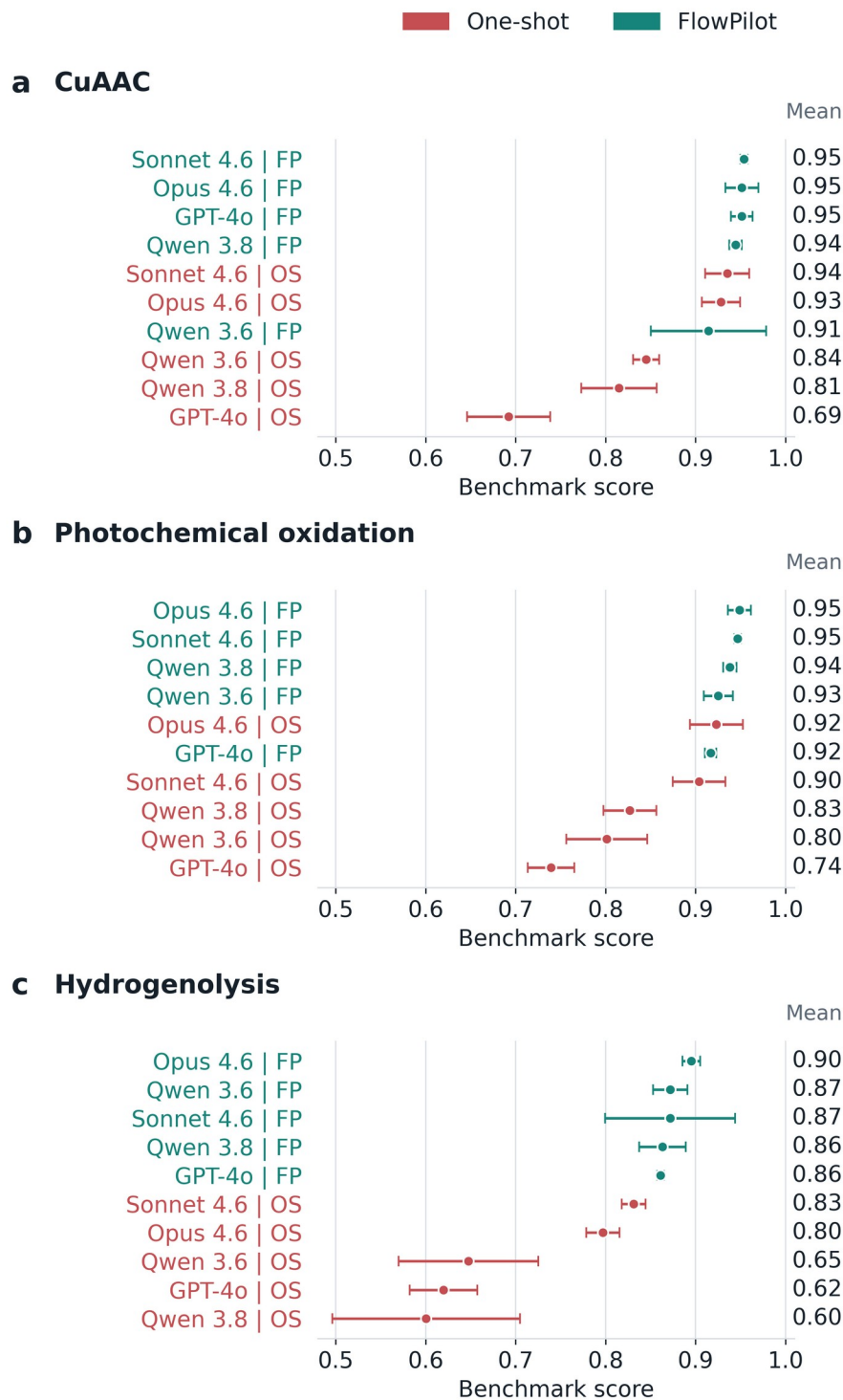
| Generator model   | One-shot        | FlowPilot       | Mean difference |
|-------------------|-----------------|-----------------|-----------------|
| Qwen3.6-27B       | 0.765 +/- 0.014 | 0.904 +/- 0.029 | +0.139          |
| Qwen3.8-27B       | 0.747 +/- 0.019 | 0.915 +/- 0.011 | +0.168          |
| GPT-4o            | 0.684 +/- 0.011 | 0.910 +/- 0.002 | +0.226          |
| Claude Sonnet 4.6 | 0.890 +/- 0.014 | 0.924 +/- 0.024 | +0.034          |
| Claude Opus 4.6   | 0.883 +/- 0.015 | 0.932 +/- 0.010 | +0.049          |

Values are mean +/- sample SD across three repeat-level campaign means. The difference column is FlowPilot minus one-shot.

Across the 45 retained outcomes per architecture:

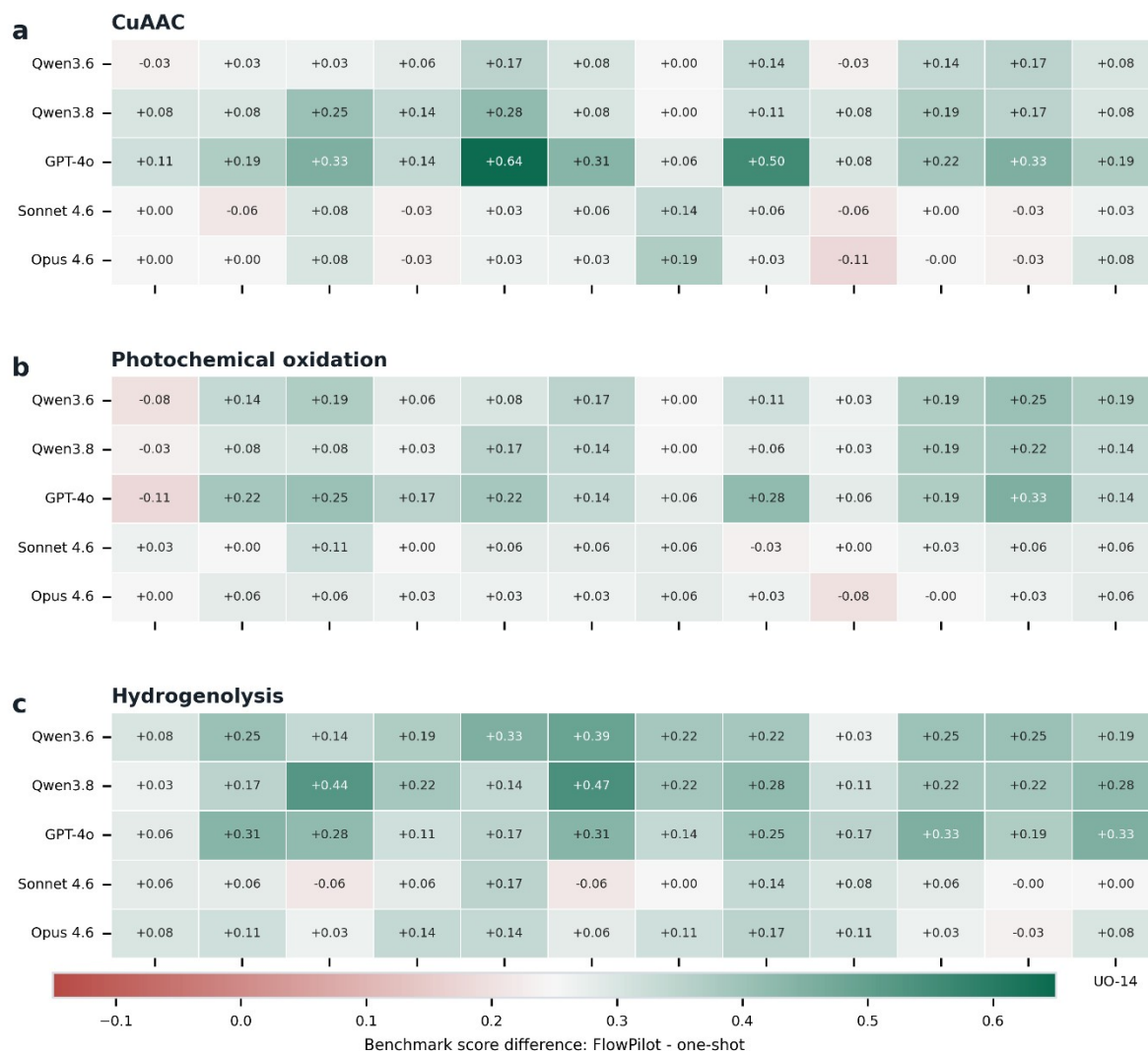
**Table S12.** Aggregate architecture-level score and critical-error summary.

| Architecture | Mean candidate score | Campaigns without critical flags | Raw judge critical flags |
|--------------|----------------------|----------------------------------|--------------------------|
| One-shot     | 0.794                | 18/45                            | 98                       |
| FlowPilot    | 0.917                | 43/45                            | 4                        |



Mean +/- sample SD; 3 repeats. FP: FlowPilot; OS: one-shot. Both Qwen models: 27B.

**Figure S7.** Case-specific benchmark scores for (a) CuAAC, (b) photochemical oxidation and (c) hydrogenolysis. Points and horizontal error bars show means and sample SDs across three generation repeats. Red denotes one-shot (OS); teal denotes FlowPilot (FP). Both Qwen models have 27B parameters. All 30 SDs are nonzero; small whiskers may be obscured by markers. Values beside the plots are rounded means.



**Figure S8.** Mean paired FlowPilot-minus-one-shot score difference for each applicable universal criterion, separated by chemistry. Positive values favor FlowPilot. UO-08 and UO-09 are omitted because their applicability is not uniform across the three chemistries.

### a CuAAC

|                      | Qwen 3.6 |   |    | Qwen 3.8 |   |    | GPT-4o |   |    | Sonnet 4.6 |   |    | Opus 4.6 |   |    |
|----------------------|----------|---|----|----------|---|----|--------|---|----|------------|---|----|----------|---|----|
|                      | OS       |   | FP | OS       |   | FP | OS     |   | FP | OS         |   | FP | OS       |   | FP |
| UO-02 Materials      |          |   | 1  |          |   |    | 1      |   |    |            |   |    |          |   |    |
| UO-03 Feeds          |          |   | 1  |          |   |    |        |   |    |            |   |    |          |   |    |
| UO-04 Stages         |          |   |    |          |   |    |        |   |    |            |   |    |          |   |    |
| UO-05 Topology       |          |   |    | 1        |   |    | 1      | 2 | 1  |            |   |    |          |   |    |
| UO-06 Liquid balance |          |   |    |          |   |    |        |   |    |            |   |    |          |   |    |
| UO-07 Residence time |          |   |    |          |   |    |        |   |    |            |   |    |          |   |    |
| UO-08 Gas            | -        | - | -  | -        | - | -  | -      | - | -  | -          | - | -  | -        | - | -  |
| UO-10 Inventory      |          |   |    |          |   |    | 1      | 1 | 1  |            |   |    |          |   |    |
| UO-11 Transport      |          |   | 1  |          |   |    | 1      |   | 1  |            |   |    |          |   |    |
| UO-12 Safety         |          |   |    |          | 1 |    | 1      |   | 1  |            |   |    |          |   |    |
| UO-13 Procedure      |          |   |    |          | 1 |    | 1      | 1 | 1  |            |   |    |          |   |    |
| Repeat               | 1        | 2 | 3  | 1        | 2 | 3  | 1      | 2 | 3  | 1          | 2 | 3  | 1        | 2 | 3  |

### b Photochemical oxidation

|                      | Qwen 3.6 |   |    | Qwen 3.8 |   |    | GPT-4o |   |    | Sonnet 4.6 |   |    | Opus 4.6 |   |    |
|----------------------|----------|---|----|----------|---|----|--------|---|----|------------|---|----|----------|---|----|
|                      | OS       |   | FP | OS       |   | FP | OS     |   | FP | OS         |   | FP | OS       |   | FP |
| UO-02 Materials      |          |   |    |          |   |    |        |   |    |            |   |    |          |   |    |
| UO-03 Feeds          | 1        |   |    |          |   |    |        |   |    | 1          |   |    |          |   |    |
| UO-04 Stages         |          |   |    |          |   |    |        |   |    |            |   |    |          |   |    |
| UO-05 Topology       |          |   |    |          |   |    |        |   |    |            |   |    |          |   |    |
| UO-06 Liquid balance |          |   |    |          |   |    |        |   |    |            |   |    |          |   |    |
| UO-07 Residence time |          |   |    |          |   |    |        |   |    |            |   |    |          |   |    |
| UO-08 Gas            | 1        |   |    | 2        |   |    | 1      | 2 |    | 1          |   |    |          |   |    |
| UO-10 Inventory      |          |   |    |          |   |    |        |   |    |            |   |    |          |   |    |
| UO-11 Transport      |          |   |    |          |   |    |        |   |    |            |   |    |          |   |    |
| UO-12 Safety         |          |   |    |          |   |    |        |   |    |            |   |    |          |   |    |
| UO-13 Procedure      | 1        |   |    | 1        | 1 |    | 1      |   | 1  |            |   |    |          |   |    |
| Repeat               | 1        | 2 | 3  | 1        | 2 | 3  | 1      | 2 | 3  | 1          | 2 | 3  | 1        | 2 | 3  |

### c Hydrogenolysis

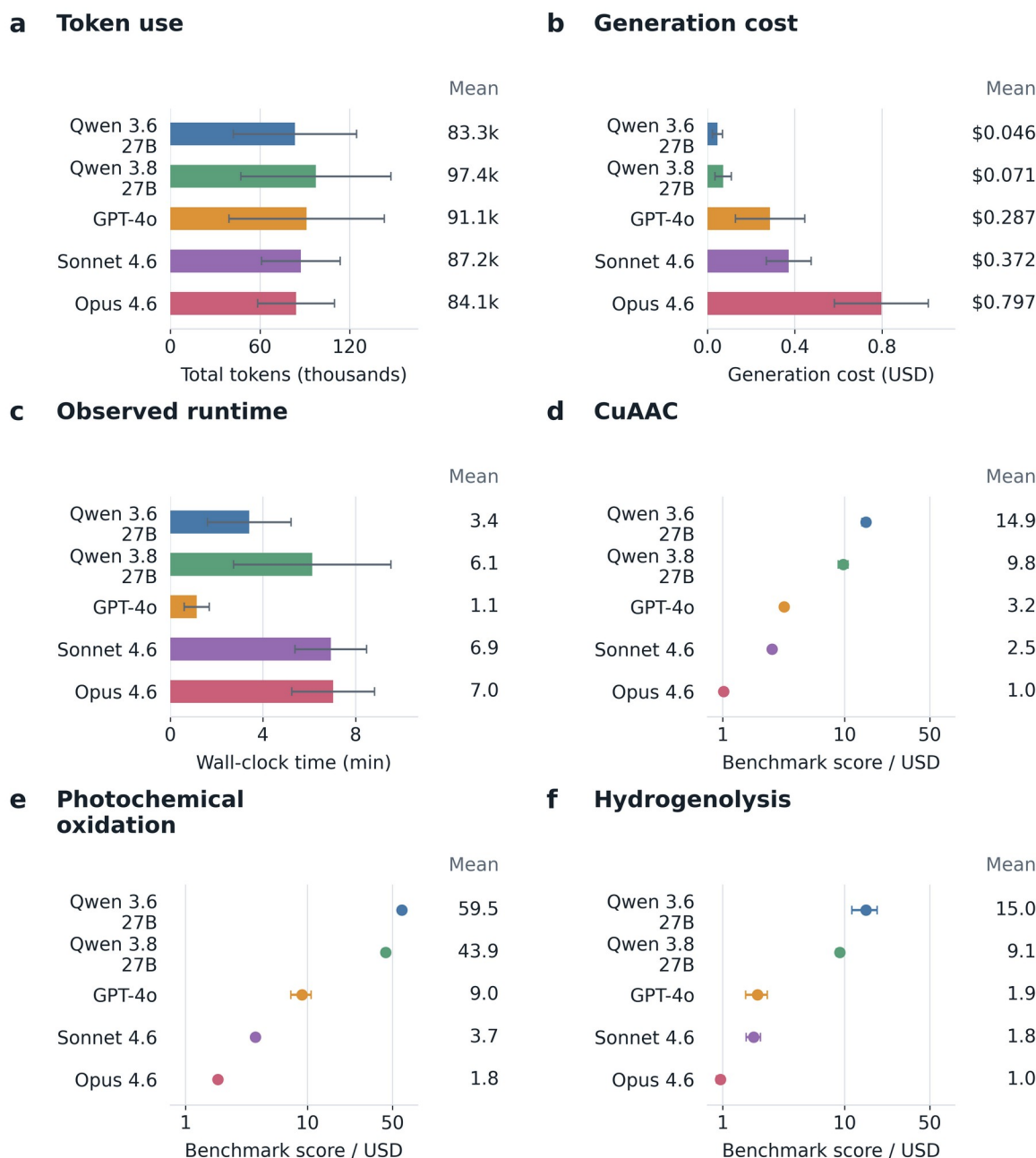
|                      | Qwen 3.6 |   |    | Qwen 3.8 |   |    | GPT-4o |   |    | Sonnet 4.6 |   |    | Opus 4.6 |   |    |
|----------------------|----------|---|----|----------|---|----|--------|---|----|------------|---|----|----------|---|----|
|                      | OS       |   | FP | OS       |   | FP | OS     |   | FP | OS         |   | FP | OS       |   | FP |
| UO-02 Materials      | 1        | 1 |    |          |   |    | 1      | 1 |    |            |   |    |          |   |    |
| UO-03 Feeds          |          | 1 | 1  | 2        | 3 |    | 1      | 1 | 1  |            |   |    |          |   |    |
| UO-04 Stages         |          |   |    |          |   |    |        |   |    |            |   |    |          |   |    |
| UO-05 Topology       | 1        | 2 | 1  |          |   |    | 1      |   |    | 1          | 1 |    | 1        | 1 |    |
| UO-06 Liquid balance | 1        | 2 |    |          | 1 |    | 1      |   |    |            |   |    |          |   |    |
| UO-07 Residence time |          | 2 |    |          |   |    |        | 1 |    |            |   |    |          | 2 |    |
| UO-08 Gas            |          | 1 | 1  | 2        | 3 | 2  | 2      | 2 | 1  |            |   |    | 1        | 1 |    |
| UO-10 Inventory      |          |   | 1  |          | 1 |    |        | 1 |    | 1          |   |    |          |   |    |
| UO-11 Transport      |          |   |    |          |   |    | 1      | 1 |    |            | 1 |    | 1        |   |    |
| UO-12 Safety         | 1        |   | 1  |          |   |    |        | 1 |    |            |   |    |          |   |    |
| UO-13 Procedure      |          | 1 | 1  |          | 1 |    | 1      | 1 |    |            |   |    |          |   |    |
| Repeat               | 1        | 2 | 3  | 1        | 2 | 3  | 1      | 2 | 3  | 1          | 2 | 3  | 1        | 2 | 3  |

Not applicable No flag 1 judge 2 judges 3 judges

OS: one-shot; FP: FlowPilot. Cells show judges flagging a critical issue, not error counts.

No critical flags: one-shot 18/45 campaigns; FlowPilot 43/45. Both Qwen models: 27B.

**Figure S9.** Critical flags for (a) CuAAC, (b) photochemical oxidation and (c) hydrogenolysis. Each model has adjacent one-shot (OS) and FlowPilot (FP) blocks with repeats 1-3. Rows identify criteria. Cell numbers count judges flagging a critical issue, not distinct physical errors. Pale cells have no flag; gray cells with a dash are not applicable. All 90 campaigns and 990 campaign-criterion cells are retained. No critical flag does not establish experimental success.



a-c: mean +/- sample SD; 9 campaigns per model (3 cases x 3 repeats).

d-f: mean +/- sample SD; 3 repeats per case; log scale. Runtime depends on the environment.

**Figure S10.** FlowPilot resource use and quality per generation cost. (a) Tokens, (b) estimated generation cost from the stored token-price schedule and (c) observed runtime: mean +/- sample SD across nine campaigns per model. (d-f) Mean benchmark-score/ USD ratios for CuAAC, photochemical oxidation and hydrogenolysis, respectively, with sample SDs across three repeats, on logarithmic axes. Ratios are calculated per campaign before averaging. Judge-model costs are excluded. Runtime is environment-dependent; token pricing is not a measurement of total local deployment cost.

## 4 Deterministic engineering, inventory realization, and refinement

FlowPilot separates model-proposed independent variables from quantities that can be recomputed. Flow rates, gas compression, molar equivalents, reactor volumes, stage times, pressure-drop indicators, heat-transfer indicators, and equipment assignments are recalculated after candidate selection and again after inventory realization. A numerical value becomes a run instruction only if the recomputed proposal, stage table, stream table, inventory allocation, process graph, and final contract agree within the implemented tolerances.

### 4.1 Residence-time and gas-flow conventions

Liquid molar flow is obtained from liquid concentration and volumetric flow. For a single liquid phase, nominal residence time closes as  $\tau = V/Q$ . Gas feeds are stored as MFC setpoints at the declared inlet/STP reference. The corresponding in-channel volumetric flow is calculated by ideal-gas scaling,  $Q_{g,ch} = Q_{g,STP}(T_{ch}/T_{STP})(P_{STP}/P_{abs})$ . FlowPilot reports both the inlet/STP apparent time,  $V/(Q_L + Q_{g,STP})$ , and the pressure-corrected in-channel apparent time,  $V/(Q_L + Q_{g,ch})$ , when those definitions are applicable. Gas equivalents are calculated from gas molar flow and limiting-liquid molar flow; they are not inferred from a gas-to-liquid volumetric ratio. In packed beds or poorly characterized multiphase systems, contact time is not equated automatically with empty-bed space time: measured phase holdup or residence-time-distribution data are requested where required.

**Table S13.** Deterministic quantities, definitions, and closure checks.

| Quantity              | Implemented definition or basis                         | Required closure                                                     |
|-----------------------|---------------------------------------------------------|----------------------------------------------------------------------|
| Limiting molar flow   | $n_{lim} = C_{lim} Q_L$                                 | Concentration, liquid flow, and stream composition agree             |
| Liquid residence time | $\tau_L = V_L/Q_L$                                      | Reactor liquid volume equals $Q_L \tau_L$                            |
| In-channel gas flow   | $Q_{g,ch} = Q_{g,STP}(T_{ch}/T_{STP})(P_{STP}/P_{abs})$ | Temperature and absolute-pressure bases are explicit                 |
| Gas equivalents       | $n_{gas}/n_{lim}$ from standard gas molar volume        | Gas identity and gas fraction are included                           |
| Gas holdup estimate   | $\epsilon_{g} = Q_{g,ch}/(Q_L + Q_{g,ch})$              | Used only with the declared two-phase approximation                  |
| Multistage total      | $\tau_{total} = \sum(\tau_i)$                           | Each stage uses cumulative local inlet flow and assigned volume      |
| Pressure drop         | Single- or two-phase model selected by phase regime     | Calculated drop remains within pump, tubing, reactor, and BPR limits |
| Thermal conductance   | $UA = U A_{wall}$ ; $Da_{th} = Q_{gen}/Q_{rem}$         | Assumptions, geometry, and thermal margin are reported               |



## 4.2 Multistage calculation and stream introduction

Each reactive stage receives a separate stage identifier, inlet composition, cumulative liquid and gas flow, assigned reactor volume, temperature, pressure, and residence-time basis. A feed introduced between reactors contributes only to downstream cumulative flow. Quench and work-up operations are represented in topology but are not counted as reactive stages. If only a total target time is available, allocation uses declared stage times or stage batch-time weights and otherwise applies an explicit equal-allocation fallback. Final validation checks both the stage sum and each local volume-flow-time identity.

### Heat-transfer screening

The heat-transfer module reports wall area, surface-to-volume ratio, overall thermal conductance  $UA$ , estimated heat generation and removal rates, a thermal Damkohler-like ratio, and a screening margin. The implemented relations are  $A_{wall} = \pi \cdot d \cdot L$ ,  $Q_{rem} = U \cdot A \cdot \Delta T_{lm}$ , and  $Da_{th} = Q_{gen} / Q_{rem}$ . Reaction enthalpy, overall heat-transfer coefficient, and driving-temperature difference can be estimated from reaction and reactor classes when measurements are unavailable. These values are screening assumptions rather than calorimetric evidence; an apparently favorable estimate does not replace reaction calorimetry for scale-up or strongly exothermic chemistry.

### Inventory normalization and executable topology

Laboratory inventories may be entered as structured JSON or extracted from PDF, Word, PowerPoint, spreadsheet, and text sources. Extracted content is normalized into typed equipment classes and reviewed before use. Each item carries a stable `equipment_id` and available capability fields. Free-text statements such as 'no inline degasser' become explicit unavailable capabilities or hard constraints rather than prompt-only notes. During design realization, every required operation is assigned to an inventory item. The allocator checks type, material, inner diameter, volume, flow range, temperature, pressure, wavelength, phase compatibility, and connectivity where declared. Under strict assignment, absent or invented hardware cannot be published in the executable topology.

**Table S14.** Inventory classes and deterministically checked capabilities.

| Inventory class           | Representative capability fields                                       | Design role                              |
|---------------------------|------------------------------------------------------------------------|------------------------------------------|
| Pumps and MFCs            | Minimum/maximum flow, pressure, compatible fluids or gas identity      | Quantified feed delivery                 |
| Mixers/connectors         | Inputs, material, pressure, supported ID, connectivity                 | Stream joining and interstage connection |
| Reactors/trains           | Type, volume, ID, material, temperature, pressure, light compatibility | Reactive residence volume                |
| Light/temperature control | Wavelength, power, allowed temperature setpoints                       | Photochemical and thermal control        |
| BPRs/tubing               | Setpoints or range, pressure rating, ID, length, material              | Pressure and hydraulic realization       |
| Separators/collectors     | Supported phases, flow, pressure, volume, venting                      | Depressurization and product collection  |

| Inventory class    | Representative capability fields            | Design role                                             |
|--------------------|---------------------------------------------|---------------------------------------------------------|
| Safety accessories | Hazard compatibility and control capability | Purge, check valve, shielding, containment, and venting |

A process diagram is generated from the same inventory-assigned topology stored in the final contract. If assignment or numerical closure fails, FlowPilot may render a requirements topology for diagnosis, but it does not label that diagram executable and does not publish numerical run instructions. This distinction prevents a polished intermediate graph from being mistaken for a laboratory-ready process.

### Incremental experimental refinement

Closed-loop refinement accepts one or more ExperimentRecord objects containing actual operating conditions, measured response, and operational observations. Measured evidence has higher authority than model inference. The module diagnoses underconversion, selectivity loss, solubility or fouling, pressure instability, and gas-liquid instability; it then proposes bounded changes to residence time, liquid and gas flow, temperature, concentration, pressure, tubing, filtration, or safety controls. For multiple experiments, the campaign preserves every observation, identifies the best observed anchor, and may fit a simple apparent response-versus-residence-time model. Extrapolated targets are reported as screening estimates with acceptance criteria, not guaranteed yields. Every cycle creates a new version while retaining the preceding design and measurements.

**Table S15.** Experimental feedback fields and their role in refinement.

| Evidence group      | Recorded fields                                                          | Refinement use                                        |
|---------------------|--------------------------------------------------------------------------|-------------------------------------------------------|
| Timing and geometry | Residence-time basis, inlet and channel time, flow, volume, tubing ID    | Reconcile actual exposure and throughput              |
| Gas delivery        | Gas identity, MFC/STP flow, channel flow, equivalents, pressure          | Recompute gas supply and apparent time bases          |
| Chemistry response  | Yield, conversion, selectivity, product and starting-material fractions  | Anchor response trend and target acceptance           |
| Operating state     | Temperature, concentration, wavelength, light power                      | Separate kinetic, photonic, and concentration changes |
| Run observations    | Pressure drift, clogging, precipitation, phase stability, impurity notes | Trigger feasibility and safety interventions          |

## 5 Computational provenance and interpretation boundaries

### Model routing and run records

The user interface exposes separate model routes for upstream chemistry interpretation and downstream proposal/council execution. The selected provider, model identifier, and local OpenAI-compatible endpoint mapping are applied for the duration of one serialized run and restored afterward. Model routing is written to pipeline-runtime provenance. Because hosted model aliases and local checkpoints can change, a

reproducible report should retain the exact model identifier, endpoint or provider, generation parameters, software revision, dependency environment, frozen DesignInputPackage, inventory profile, raw model events, deterministic audit, and canonical final result.

This ESI distinguishes the historical benchmark configuration from the current software. The archived worked example reports liquid-only geometric-volume time, inlet/STP apparent time and pressure-corrected apparent time as different fields. Current GUI defaults must not be used to retroactively relabel these measurements or regenerate archived benchmark scores. The held-out photochemical reference was machine-extracted and not page-verified during the campaign; internally inconsistent published-flow fields were explicitly withheld from numerical reference scoring.

**Table S16.** Minimum records required for audit and computational replay.

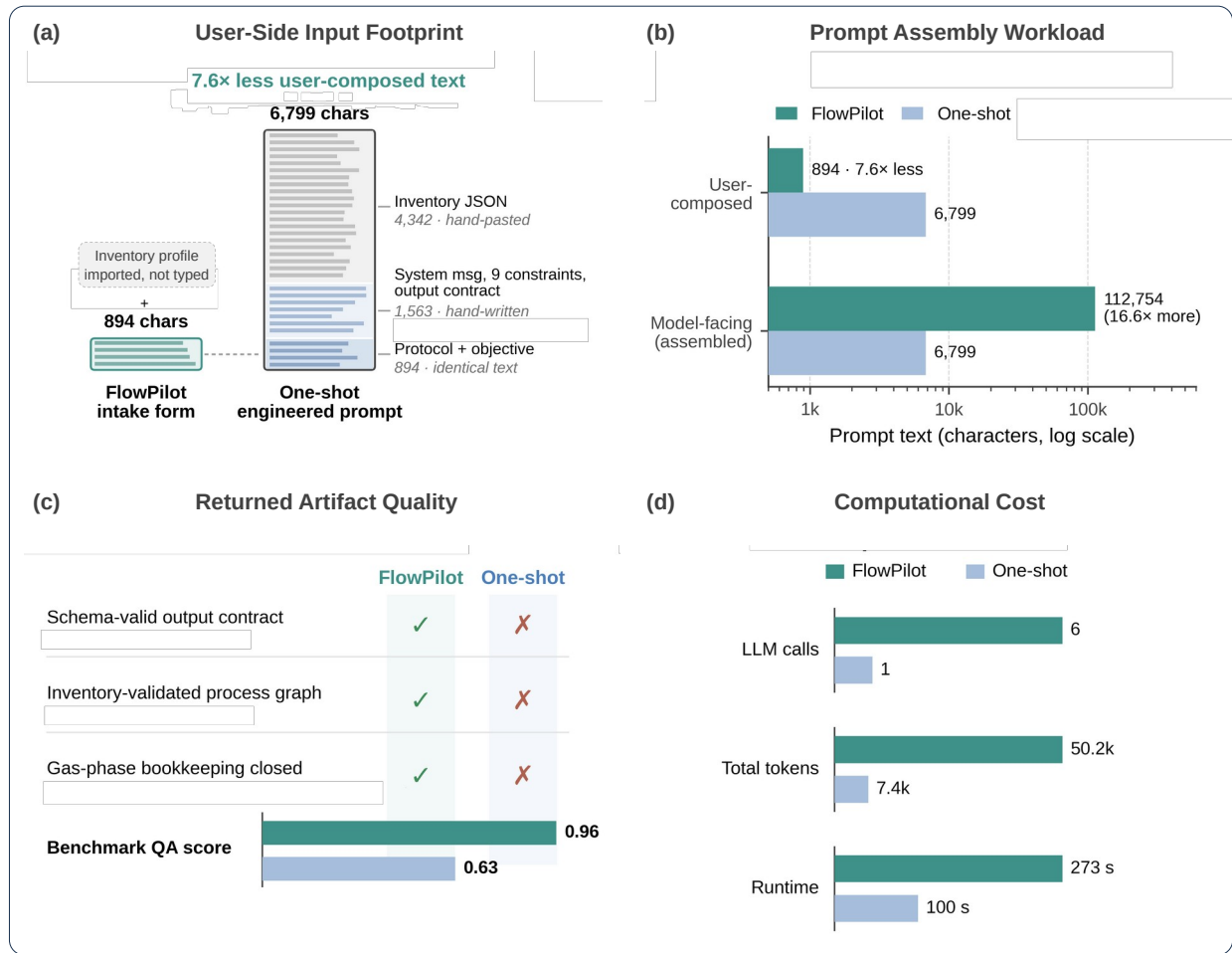
| Record               | Purpose                                          | Typical stored object                                           |
|----------------------|--------------------------------------------------|-----------------------------------------------------------------|
| Frozen input         | Reconstruct chemist-provided facts and authority | DesignInputPackage and inventory profile                        |
| Model configuration  | Identify stochastic generators and routes        | Provider, model ID, endpoint, temperature, seed where supported |
| Prompts/events       | Audit module inputs and raw responses            | Timestamped LLM event log and token telemetry                   |
| Deterministic state  | Verify calculations and gates                    | Calculation sheet, validation report, allocation, topology      |
| Canonical output     | Identify the published design                    | FinalDesignContract, content hash, rendered artifacts           |
| Software environment | Support replay of code-owned behavior            | Git revision, Python/dependency versions, configuration         |

## 6 Worked user-interaction example: FlowPilot versus one-shot prompting

This section preserves the exact archived inputs and provider responses for a matched Claude Opus 4.6 photochemical-oxidation run (repeat 01), together with the separately stored final structured design. It illustrates this benchmark's input-assembly and artifact-handling arrangements, not an inherent inability of other LLM interfaces to accept uploaded inventories or structured prompts. The comparison is descriptive, concerns a single run, and is not a usability trial or a token-matched experiment.

The example distinguishes user-facing input, internal computation, and returned artifacts. The recorded text counts are 894 characters for protocol plus objective and 6,799 for the complete one-shot prompt; the 4,342-character inventory is additional shared input, not absent from the FlowPilot task. FlowPilot used six calls and 50,175 tokens, versus one call and 7,430 tokens for one-shot. Its structured final design passed the archived deterministic checks and retained SCREEN disposition. However, its generated narrative was not synchronized to that final design and contained different conditions and unsupported assertions. The structured result and narrative must therefore be evaluated separately; neither is evidence of wet-lab validation.

Character counts quoted here are those recorded in the benchmark artifacts; the verbatim texts below are reproduced with display line-wrapping only (no text omitted or altered). **Figure S11** summarizes the comparison graphically; **Table S17** lists the underlying quantities.



**Figure S11. Historical prompt-footprint comparison for one matched Claude Opus 4.6 run.** (a) User-facing protocol plus objective versus the complete one-shot prompt; the structured inventory is additional shared input. (b) Six FlowPilot internal prompts total 112,754 model-facing characters. (c) Archived heuristic QA diagnostics (0.96 versus 0.63), distinct from the independent multi-LLM outcome score. Validation claims apply to the final structured artifact, not the unsynchronized generated narrative. (d) Recorded computation: 50,175 tokens and 273 s versus 7,430 tokens and 100 s. Exact inputs and outputs are reproduced in the following named subsections and companion files. This is a single-run artifact comparison, not an experimental or human-usability validation.

**Figure S11** separates user-composed text from system-assembled prompts. The numerical QA values in that historical figure (0.96 and 0.63) are legacy heuristic scores, not the equal-weight multi-LLM benchmark scores in Section 3. Their weights were 0.10 formal validity, 0.25 engineering integrity, 0.15 process completeness, 0.15 safety adequacy, 0.10 evidence provenance, 0.20 decision assurance and 0.05 actionability calibration. Trace-dependent components make that legacy metric unsuitable as architecture-neutral evidence of superiority. For this exact pair, the independent fixed-criteria scores were 0.935898

(FlowPilot) and 0.916667 (one-shot). The legacy scores are retained for provenance, not substituted for those benchmark results.

**Table S17.** Summary of the matched run.

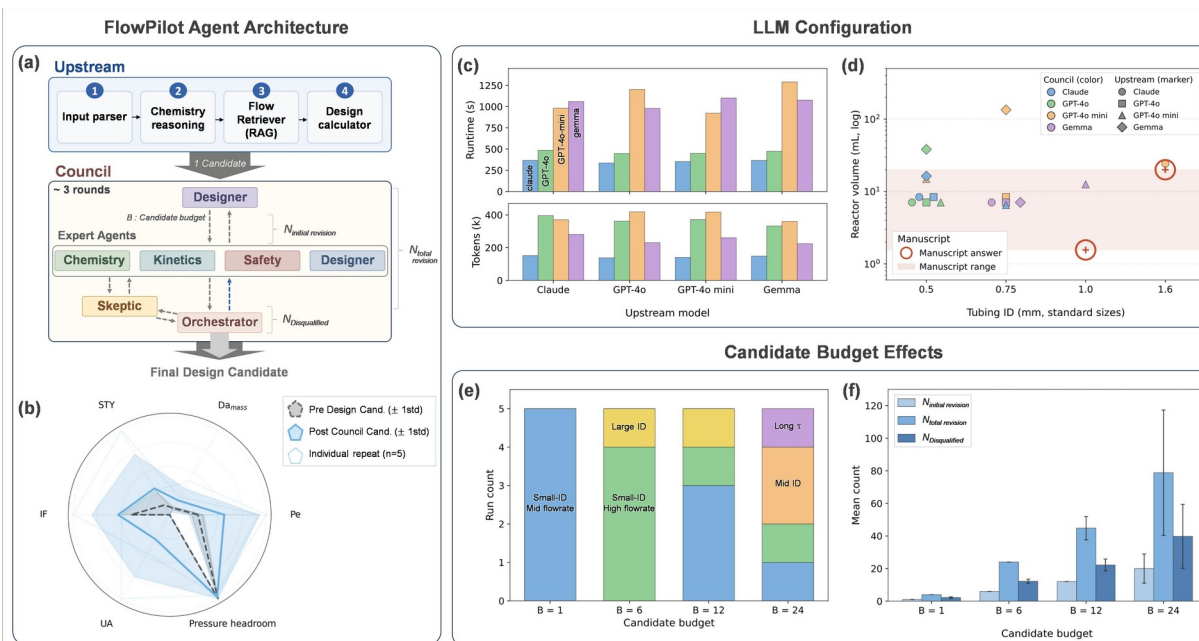
| Criteria                                      | FlowPilot                                                                                                                                                   | One-shot prompting                                                                                                                                                                                                                                                                                  |
|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>User-composed input</b>                    | 894 characters (protocol + objective); inventory imported as a structured file, constraints assembled automatically                                         | 6,799 characters in the archived system/user prompt: protocol, objective, constraints, inventory and output contract. This count describes the tested interface, not a universal typing requirement.                                                                                                |
| <b>Model-facing prompt text</b>               | 112,754 characters across six staged internal prompts                                                                                                       | 6,799 characters in a single prompt                                                                                                                                                                                                                                                                 |
| <b>LLM calls</b>                              | 6                                                                                                                                                           | 1                                                                                                                                                                                                                                                                                                   |
| <b>Input / output tokens</b>                  | 37,552 / 12,623 (total 50,175)                                                                                                                              | 2,095 / 5,335 (total 7,430)                                                                                                                                                                                                                                                                         |
| <b>Runtime</b>                                | 273 s                                                                                                                                                       | 100 s                                                                                                                                                                                                                                                                                               |
| <b>Returned artifact</b>                      | Inventory-assigned final SCREEN contract and <b>Figure S13</b> topology. The separately generated report is unsynchronized and is not executable authority. | Raw JSON response; six schema-validation errors; no inventory-validated process graph; two conflicting pressure-corrected gas-flow values (0.136 and 0.167 mL min <sup>-1</sup> ); O <sub>2</sub> basis 0.235 mol O <sub>2</sub> per mol substrate, below the stated 0.5 stoichiometric requirement |
| Legacy heuristic QA (not the multi-LLM score) | 0.96                                                                                                                                                        | 0.63                                                                                                                                                                                                                                                                                                |

## 6.1 Council model matrix and candidate-budget sensitivity

The council sensitivity experiment used the isoxazole 1,3-dipolar cycloaddition as a fixed test reaction. Upstream chemistry interpretation and downstream council execution were crossed in a 4 x 4 model matrix. The comparison records wall-clock runtime, token consumption, final reactor volume, tubing internal diameter, and deterministic validation status for every pairing (**Figure S12** and **Table S18**). These data are architecture-diagnostic rather than a general ranking of current foundation models.

Council vocabulary is used as follows. Need revision denotes a candidate with a correctable deficiency that is returned for a bounded edit. New versions are deterministic-recomputed descendants generated from that candidate. The combined pool contains the original candidates and all revised descendants after recalculation. Disqualified candidates fail a hard feasibility or critical-safety gate and cannot advance to final selection.

Candidate budgets  $B = 1, 6, 12$ , and  $24$  were evaluated in five repeats. Increasing the budget broadened the design families and increased revision activity, but the effect was non-monotonic and was accompanied by more disqualifications (**Table S19**). The five-repeat radar-area score changed from  $0.31 \pm 0.03$  before council review to  $0.48 \pm 0.23$  after review; this replaces the earlier single-run statement based on  $0.29$  to  $0.83$  (**Table S20**). Radar normalization and capping rules are reported explicitly in **Table S21**.



**Figure S12. Council architecture, model-matrix sensitivity, and candidate-budget analysis for the isoxazole 1,3-dipolar cycloaddition test reaction.** (a) Upstream pipeline and council workflow. (b) Pre- and post-council normalized engineering radar profiles; thin traces show five repeats. (c) Runtime and token use across the 4 x 4 upstream-council matrix. (d) Final reactor volume and tubing internal diameter relative to the manuscript operating region. (e) Design-family counts across candidate budgets  $B = 1, 6, 12$ , and  $24$ . (f) Revision and disqualification activity across those budgets. Complete numerical values and metric definitions are provided in **Tables S18-S21**.

**Table S18.** Runtime, token use, reactor volume, tubing internal diameter, and validation outcome for the 4 x 4 upstream-council model matrix.

| Upstream  | Council   | Runtime (s) | Tokens  | V_R (mL) | ID (mm) | Validated | Disposition     |
|-----------|-----------|-------------|---------|----------|---------|-----------|-----------------|
| claude    | claude    | 366.2       | 151,638 | 8.394    | 0.50    | yes       | validated       |
| claude    | gpt4o     | 485.8       | 395,734 | 7.069    | 0.50    | yes       | validated       |
| claude    | gpt4omini | 980.2       | 370,385 | 24.127   | 1.60    | yes       | validated       |
| claude    | gemma     | 1059.6      | 280,567 | 7.069    | 0.75    | yes       | validated       |
| gpt4o     | claude    | 335.5       | 137,685 | 8.394    | 0.50    | yes       | validated       |
| gpt4o     | gpt4o     | 447.0       | 361,807 | 7.069    | 0.50    | yes       | validated       |
| gpt4o     | gpt4omini | 1201.7      | 418,656 | 8.394    | 0.75    | yes       | validated       |
| gpt4o     | gemma     | 978.1       | 230,658 | 7.069    | 0.75    | yes       | validated       |
| gpt4omini | claude    | 14.0        | 4,945   | 20.106   | 1.60    | no        | screen required |
| gpt4omini | gpt4o     | 10.7        | 4,882   | 20.106   | 1.60    | no        | screen required |
| gpt4omini | gpt4omini | 10.6        | 4,816   | 20.106   | 1.60    | no        | screen required |
| gpt4omini | gemma     | 13.5        | 4,997   | 20.106   | 1.60    | no        | screen required |
| gemma     | claude    | 21.7        | 6,195   | 20.106   | 1.60    | no        | screen required |
| gemma     | gpt4o     | 18.9        | 6,155   | 20.106   | 1.60    | no        | screen required |
| gemma     | gpt4omini | 23.7        | 6,121   | 20.106   | 1.60    | no        | screen required |
| gemma     | gemma     | 21.8        | 6,269   | 20.106   | 1.60    | no        | screen required |

**Table S19.** Design-family counts and council revision activity by candidate budget, mean +/- sample SD where repeated (n = 5).

| B | Observed design families (count) | Changed     | New versions | Combined pool | Disqualified | n |
|---|----------------------------------|-------------|--------------|---------------|--------------|---|
| 1 | small-ID & mid flowrate: 5       | 1.0 +/- 0.0 | 3.0 +/- 0.0  | 4.0 +/- 0.0   | 2.2 +/- 0.4  | 5 |

| <b>B</b> | <b>Observed design families (count)</b>                                                  | <b>Changed</b> | <b>New versions</b> | <b>Combined pool</b> | <b>Disqualified</b> | <b>n</b> |
|----------|------------------------------------------------------------------------------------------|----------------|---------------------|----------------------|---------------------|----------|
| 6        | small-ID & high flowrate: 4; large-ID only: 1                                            | 6.0 +/- 0.0    | 18.0 +/- 0.0        | 24.0 +/- 0.0         | 12.2 +/- 1.3        | 5        |
| 12       | small-ID & mid flowrate: 3; small-ID & high flowrate: 1; large-ID only: 1                | 12.0 +/- 0.0   | 32.8 +/- 7.2        | 44.8 +/- 7.2         | 22.2 +/- 3.6        | 5        |
| 24       | small-ID & mid flowrate: 1; small-ID & high flowrate: 1; mid-ID balanced: 2; long tau: 1 | 20.0 +/- 8.9   | 58.8 +/- 29.5       | 82.8 +/- 29.5        | 39.8 +/- 19.6       | 5        |

**Table S20.** Five-repeat normalized radar metrics before and after council review, mean +/- sample SD (n = 5).

| <b>Metric</b>                 | <b>Pre-council</b> | <b>Post-council</b> | <b>n</b> |
|-------------------------------|--------------------|---------------------|----------|
| Peclet score                  | 0.293 +/- 0.062    | 0.565 +/- 0.366     | 5        |
| Mass-transfer Damkohler score | 0.079 +/- 0.015    | 0.172 +/- 0.142     | 5        |
| Space-time-yield score        | 0.115 +/- 0.157    | 0.315 +/- 0.409     | 5        |
| Intensification-factor score  | 0.400 +/- 0.091    | 0.533 +/- 0.274     | 5        |
| UA score                      | 0.000 +/- 0.000    | 0.293 +/- 0.444     | 5        |
| Pressure-headroom score       | 1.000 +/- 0.000    | 0.983 +/- 0.024     | 5        |
| Radar-area score              | 0.314 +/- 0.028    | 0.477 +/- 0.229     | 5        |



**Table S21.** Radar metric definitions and transformations used for the council analysis.

| Metric                                               | Definition used for the normalized radar score                                              |
|------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Peclet number (Pe)                                   | Pe is normalized with an upper cap of 100; values at or above the cap score 1.              |
| Mass-transfer Damkohler number (Da <sub>mass</sub> ) | Da <sub>mass</sub> is inverted so that lower mass-transfer limitation gives a higher score. |
| Space-time yield (STY)                               | STY is transformed by log-min-max scaling over the frozen comparison range.                 |
| Overall thermal conductance (UA)                     | UA is transformed by log-min-max scaling over the frozen comparison range.                  |
| Intensification factor (IF)                          | IF is normalized with an upper cap of 6.                                                    |
| Pressure headroom                                    | headroom = 1 - deltaP / pump <sub>max</sub> , then bounded to the plotted score interval.   |
| Radar-area score                                     | Area of the polygon formed by the six normalized axes, reported on a 0-1 scale.             |

## 6.2 FlowPilot user-facing intake (complete, verbatim)

The archived protocol plus objective contains 894 characters. The inventory is a separate 4,342-character input. The following transcription and the companion flowpilot\_user\_input.txt preserve the source text; line wrapping is a display operation.

Fmoc-L-methionine (3.7 mmol, 1.0 equiv) and 4,7-diphenyl-2,1,3-benzothiadiazole (10 mol%) were dissolved in acetonitrile at 0.10 M in a round-bottom flask. The solution was exposed to air and irradiated with a 1.5 W LED centered near 420 nm at 21 C for 5 min. Fmoc-methionine sulfoxide was formed in 99% yield. Translate this into one conservative continuous-flow photochemical screening design. Report feed preparation, air delivery and oxygen-equivalent basis, gas-liquid mixing, pressure basis, photoreactor assignment, wavelength, temperature, liquid flow, inlet/STP and pressure-corrected in-channel gas flow, liquid contact residence time, reactor volume, separation, collection, startup, shutdown, and oxygen/acetonitrile/light safety controls.

### OBJECTIVE:

Produce one executable, inventory-constrained first gas-liquid photochemical flow screen without using the held-out source answer.

## 6.3 One-shot engineered prompt (complete, verbatim)

The one-shot call used claude-opus-4-6, temperature 0.2 and recorded seed 1007535521. Its system and user messages contain 6,799 characters in total, with 2,095 recorded input tokens. The two messages below are restored from the archived prompt files, without inserting spaces inside JSON keys. Their system/user

labels are editorial and not included in the character count. The protocol, objective and inventory are shared with FlowPilot.

[SYSTEM MESSAGE]

You are a general chemistry assistant. Translate a batch chemistry protocol into a practical continuous-flow experiment.

[USER MESSAGE]

FROZEN DESIGN INPUT:

RAW BATCH PROTOCOL:

Fmoc-L-methionine (3.7 mmol, 1.0 equiv) and 4,7-diphenyl-2,1,3-benzothiadiazole (10 mol%) were dissolved in acetonitrile at 0.10 M in a round-bottom flask. The solution was exposed to air and irradiated with a 1.5 W LED centered near 420 nm at 21 C for 5 min. Fmoc-methionine sulfoxide was formed in 99% yield. Translate this into one conservative continuous-flow photochemical screening design. Report feed preparation, air delivery and oxygen-equivalent basis, gas-liquid mixing, pressure basis, photoreactor assignment, wavelength, temperature, liquid flow, inlet/STP and pressure-corrected in-channel gas flow, liquid contact residence time, reactor volume, separation, collection, startup, shutdown, and oxygen/acetonitrile/light safety controls.

OBJECTIVE:

Produce one executable, inventory-constrained first gas-liquid photochemical flow screen without using the held-out source answer.

HARD CONSTRAINTS:

- Use only the authoritative inventory supplied below.
- Use one premixed Fmoc-L-methionine/photocatalyst/acetonitrile liquid feed and the declared air MFC.
- Use the declared gas-liquid mixer before the 420 nm photoreactor.
- Operate at the available 3 bar BPR setpoint and state whether pressure is absolute or gauge.
- Report air at inlet/STP and pressure-corrected in-channel conditions and calculate oxygen equivalents from 21 mol% oxygen.
- Use liquid flow and the 1.0 mL reactor volume to define liquid contact residence time; do not use total gas-plus-liquid volumetric flow as reaction time.
- No inline degasser is available or required because air is a reagent.
- All selected equipment must be named by its inventory equipment\_id.
- Return SCREEN unless the inventory cannot support a safe design.

AVAILABLE INVENTORY (authoritative JSON):

```
{"BPR_available":3.0,"collectors":
[{"equipment_id":"collector_photo_100","max_pressure_bar":1.0,"name":"Amber vented collection
vessel","type":"collection vessel","volume_mL":100.0}], "connectors":
[{"equipment_id":"connector_photo_set","material":"PFA","max_pressure_bar":10.0,"name":"PFA
photochemistry connector set","supported_ID_mm":1.0,"type":"photoreactor union set"}], "degassers":
[], "filters":[], "gas_hardware":
[{"equipment_id":"gas_air_clean","gas":"air","max_pressure_bar":10.0,"name":"Clean dry air supply and
regulator","type":"gas_supply"},
{"equipment_id":"mfc_air_10","gas":"air","max_flow_sccm":10.0,"max_pressure_bar":10.0,"min_flow_
sccm":0.1,"name":"Air mass-flow controller","type":"MFC"}], "light_sources":
[{"allowed_temperatures_C":
[21.0,25.0],"compatible_reactor":"coil","equipment_id":"light_photo_420","max_temperature_C":40.0,"
min_temperature_C":15.0,"name":"420 nm photochemistry
LED","power_W":1.5,"wavelength_nm":420.0}], "mixers":
[{"equipment_id":"mixer_photo_t","material":"PFA","max_inputs":2,"max_pressure_bar":10.0,"max_tot
al_flow_mL_min":2.0,"min_total_flow_mL_min":0.01,"name":"PFA gas-liquid T-
```

```

mixer", "supported_ID_mm": [1.0], "type": "T-mixer"}], "pressure_controllers":
[{"equipment_id": "bpr_photo_3", "max_pressure_bar": 3.0, "min_pressure_bar": 3.0, "name": "Photochemist
ry BPR", "setpoints_bar": [3.0], "type": "BPR"}], "pumps": [{"compatible_materials": ["acetonitrile", "Fmoc-
L-methionine", "4,7-diphenyl-2,1,3-
benzothiadiazole"], "equipment_id": "pump_photo_liquid", "max_flow_rate_mL_min": 1.0, "max_pressure_
bar": 10.0, "min_flow_rate_mL_min": 0.01, "name": "Photochemistry HPLC
pump", "type": "HPLC"}], "reactor_trains": [], "reactors": [{"ID_mm": 1.0, "allowed_temperatures_C":
[21.0, 25.0], "configuration": "irradiated PFA
coil", "equipment_id": "reactor_photo_100", "light_source": "light_photo_420", "material": "PFA", "max_pre
ssure_bar": 10.0, "max_temperature_C": 40.0, "min_pressure_bar": 0.0, "min_temperature_C": 15.0, "name": "
One milliliter PFA photocoil", "system": "420 nm photochemistry module", "type": "photochemical
coil", "volume_mL": 1.0, "wavelength_nm": 420.0}], "safety_accessories": [{"capabilities":
["shield_or_containment", "vented_separator"], "compatible_hazards":
["high_intensity_light", "oxygen_enriched_organic_stream", "acetonitrile_vapor"], "equipment_id": "safety
_photo_enclosure", "max_pressure_bar": 1.0, "name": "Interlocked photochemical enclosure and
vent", "type": "interlocked enclosure"}], "schema_version": "flowpilot_lab_inventory_v3.0", "separators":
[{"equipment_id": "separator_photo_gl", "max_flow_mL_min": 5.0, "max_pressure_bar": 5.0, "name": "Vent
ed gas-liquid separator", "supported_phases": ["gas", "liquid"], "type": "gas-liquid
separator"}], "strict_assignment": true, "temperature_controllers": [{"allowed_temperatures_C":
[21.0, 25.0], "equipment_id": "controller_photo_thermostat", "max_temperature_C": 40.0, "min_temperature
_C": 15.0, "name": "Photoreactor thermostat", "type": "recirculating thermostat"}], "tubing":
[{"ID_mm": 1.0, "equipment_id": "tubing_photo_pfa_10", "length_m": 1.273, "material": "PFA", "max_press
ure_bar": 10.0, "max_temperature_C": 60.0, "name": "PFA photoreactor coil tubing", "transparent": true}]]

```

#### CHEMIST-CONFIRMED CHEMISTRY IDENTITY:

```

{"confirmed": true, "mechanism_type": "photocatalytic aerobic
oxidation", "reaction_class": "oxidation", "reaction_name": "photocatalytic aerobic oxidation of Fmoc-L-
methionine"}

```

Propose one complete flow design with numeric operating conditions, stream assignments, safety controls, and a brief rationale. Return one JSON object and no markdown with exactly these top-level fields: recommended\_disposition, disposition\_rationale, and proposal. The proposal value must be a JSON object, not a JSON-encoded string.

The proposal object uses these fields:

```

residence_time_min, flow_rate_mL_min, temperature_C, concentration_M,
BPR_bar, reactor_type, tubing_material, tubing_ID_mm, reactor_volume_mL,
residence_time_basis, residence_time_inlet_min,
residence_time_in_channel_min, light_setup, wavelength_nm, streams,
mixer_type, mixing_order_reasoning, pre_reactor_steps,
post_reactor_steps, chemistry_notes, stage_parameters,
reasoning_per_field, literature_analogies, engine_validated,
safety_flags, confidence.

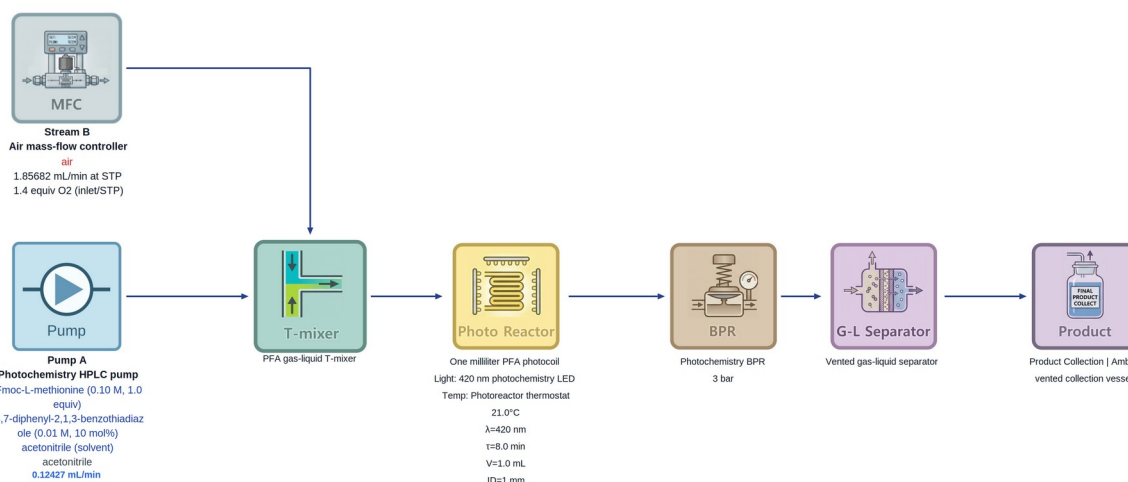
```

Each stream is an object with stream\_label, pump\_role, contents, solvent, phase, flow\_rate\_mL\_min, concentration\_M, molar\_equiv, gas\_flow\_sccm, and gas\_flow\_actual\_mL\_min where applicable. Distinguish inlet/STP gas flow from pressure-corrected in-channel gas flow. Ensure reactor\_volume\_mL is consistent with flow and residence time. State assumptions rather than inventing missing facts.

recommended\_disposition must be exactly EXECUTE, SCREEN, or BLOCK. Use BLOCK when the hard constraints make a safe design physically or operationally impossible. Use SCREEN for a feasible first experiment that still requires experimental validation. Use EXECUTE only for a directly executable design supported by sufficient evidence.

## 6.4 FlowPilot returned design

The final structured FlowPilot artifact contains the inventory-realized SCREEN design shown in **Figure S13** and summarized in **Table S22**. The 8,274-character generated report reproduced afterward is a distinct archived LLM narrative and does not agree with those final parameters. It is preserved as historical evidence, not endorsed as an executable procedure. The 17,149-character one-shot response is reproduced in the following subsection.



**Figure S13.** Horizontal redraw of the archived inventory-assigned topology for the Claude Opus 4.6 FlowPilot photochemical-oxidation run, repeat 01. The diagram represents the final structured design, not the unreconciled LLM narrative reproduced below. Its 1.0 mL reactor, 0.12427 mL/min liquid flow and 1.856819 mL/min air feed at inlet/STP correspond to a nominal  $V_R/Q_L$  time of 8.046994 min and 1.4 oxygen equivalents. This nominal time is not a measured liquid contact time. The recorded disposition is SCREEN, not experimentally validated execution. The exact one-shot response is preserved in the following subsection and the companion files. The layout and generic pump symbol have been updated for readability; recorded equipment, connectivity and operating conditions are unchanged.

**Table S22.** Final structured parameters and contemporaneous generated narrative for the same archived FlowPilot run.

| Field                         | Final structured record     | Raw LLM narrative               |
|-------------------------------|-----------------------------|---------------------------------|
| Reactor geometric volume (mL) | 1.0                         | 1.6094                          |
| Liquid flow (mL/min)          | 0.12427                     | 0.2                             |
| Reported residence time (min) | 8.046994; nominal $V_R/Q_L$ | 5.0; claimed liquid-holdup time |

| Field                     | Final structured record                          | Raw LLM narrative                                    |
|---------------------------|--------------------------------------------------|------------------------------------------------------|
| Air at inlet/STP (mL/min) | 1.856819                                         | 0.4483                                               |
| O2 equivalents            | 1.4                                              | Claims 1.0; arithmetic gives 0.210                   |
| Temperature / BPR         | 21 C / 3 bar gauge                               | 21 C / 3 bar gauge                                   |
| Validation status         | Archived final checks: ready; disposition SCREEN | NOT engine-validated; LOW confidence                 |
| Authority                 | Final structured contract                        | Historical raw narrative; not operating instructions |

### Archived raw narrative follows unchanged:

# Flow Chemistry Protocol: Photocatalytic Aerobic Oxidation of Fmoc-L-Methionine to Fmoc-Methionine Sulfoxide

---

#### ## (1) Proposed Flow Setup

**\*\*Reaction:\*\*** Singlet-oxygen-mediated thioether → sulfoxide oxidation using 4,7-diphenyl-2,1,3-benzothiadiazole as a triplet photosensitizer, with air as the O<sub>2</sub> source.

**\*\*Configuration:\*\*** Single-stage gas–liquid segmented (Taylor) flow photoreactor.

#### ### Streams

| Stream       | Contents                                                                                                                 | Phase  | Flow Rate                                        | Equipment            |
|--------------|--------------------------------------------------------------------------------------------------------------------------|--------|--------------------------------------------------|----------------------|
| <b>**A**</b> | (Liquid)   Fmoc-L-methionine (0.10 M, 1.0 equiv) + 4,7-diphenyl-2,1,3-benzothiadiazole (0.01 M, 10 mol%) in acetonitrile | Liquid | 0.2 mL/min                                       | Syringe/HPLC pump    |
| <b>**B**</b> | (Gas)   Clean dry air (21 mol% O <sub>2</sub> )                                                                          | Gas    | 0.4483 sccm (≈0.12 mL/min at operating pressure) | Mass flow controller |

**\*\*Mixing:\*\*** Streams A and B meet at a T-mixer (1.0 mm ID) to establish segmented gas–liquid (Taylor) flow before entering the irradiated zone.

#### ### Reactor

- **\*\*Type:\*\*** PFA coil reactor, 1.0 mm ID
- **\*\*Total reactor volume:\*\*** 1.6094 mL (liquid holdup ~1.0 mL; gas holdup fraction = 0.3786)
- **\*\*Residence time:\*\*** 5.0 min (defined as liquid-phase reactor volume ÷ liquid flow rate)
- **\*\*Temperature:\*\*** 21.0 °C (thermostatted)
- **\*\*Light source:\*\*** 420 nm LED module, 1.5 W, wrapped around the PFA coil
- **\*\*Back-pressure regulator:\*\*** 3.0 bar (gauge), giving ~4.0 bar absolute

#### ### Downstream

1. Gas–liquid separator to vent spent air safely
2. Collect liquid product in an amber vessel (protect from ambient light)

3. Discard the first ~3 reactor volumes (~3.0 mL) as startup waste before collecting steady-state product

### ### Preparation Steps

1. Dissolve Fmoc-L-methionine (0.10 M) and photocatalyst (0.01 M) in acetonitrile in an **\*\*amber bottle\*\*** to prevent premature photoactivation.
2. Prime the liquid pump and establish air flow through the MFC.
3. Establish stable segmented flow through the reactor **\*\*before\*\*** turning on the 420 nm LED.
4. Confirm stable back-pressure at 3.0 bar before beginning collection.

### ### Shutdown Sequence

Turn off the LED first → stop MFC → stop liquid pump → flush reactor with 5 reactor volumes of neat acetonitrile.

---

## ## (2) Rationale for Conditions

### ### Residence Time (5.0 min)

The batch protocol completes in 5 min. This was adopted directly as a conservative first screen. The automated design engine initially calculated a much shorter  $\tau$  (~0.25 min, representing a 20× intensification factor), but flagged severe O<sub>2</sub> mass-transfer limitation (transfer sufficiency only 0.19×) and poor axial dispersion ( $Pe \approx 2$ ) at that short time. The 5.0 min residence time provides a safe starting point. Literature analogies for related photocatalytic oxidations in flow show 16–384× batch-to-flow acceleration, suggesting that once baseline conversion is confirmed, residence time can likely be shortened substantially.

### ### Temperature (21.0 °C)

Matches the batch protocol. The chemistry plan identifies this reaction as temperature-insensitive. The thermal Damköhler number ( $Da_{th} = 0.003974$ ) confirms negligible heat accumulation — the reactor can remove heat ~250× faster than it is generated ( $UA = 1.931231$  W/K vs. heat generation of 0.077 W). No thermal runaway risk.

### ### Concentration (0.1 M)

Directly from the batch protocol. Consistent with literature analogies for photocatalytic sulfoxidations in acetonitrile.

### ### Tubing and Reactor Geometry (PFA, 1.0 mm ID)

PFA is transparent at 420 nm, enabling irradiation through the reactor wall. The 1.0 mm ID provides short optical path length (~0.5 mm average for an annular irradiation geometry), which is a major advantage over a batch round-bottom flask where photon penetration is limited. The small diameter also promotes well-defined Taylor flow slugs with thin liquid films, enhancing gas–liquid mass transfer (calculated  $kLa = 0.153$  s<sup>-1</sup>).

### ### Gas Flow and Pressure (0.4483 sccm air, 3.0 bar BPR)

At 0.2 mL/min liquid flow and 0.10 M substrate, the substrate molar flow is 0.020 mmol/min. The air flow delivers 0.020 mmol/min O<sub>2</sub> (1.0 equiv at the inlet). The O<sub>2</sub> transfer sufficiency is calculated at 11.3679×, meaning the gas–liquid mass transfer capacity exceeds the O<sub>2</sub> consumption requirement by more than 11-fold — O<sub>2</sub> delivery is not limiting under these conditions. The 3.0 bar BPR increases dissolved O<sub>2</sub> concentration via Henry's law (~4× vs. atmospheric) and stabilizes the gas–liquid flow pattern by preventing uncontrolled bubble expansion.

#### ### Wavelength (420 nm)

Matches the absorption maximum of 4,7-diphenyl-2,1,3-benzothiadiazole (~400–420 nm) and the batch protocol specification.

#### ### Catalyst Loading (10 mol%)

High by typical photocatalysis standards, but directly from the batch protocol. No reason to change for an initial screen.

---

### ## (3) Key Differences from the Batch Protocol

| Parameter                   | Batch                                                               | Flow                                                                                                      |
|-----------------------------|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| **Reactor geometry**        | Round-bottom flask (long optical path, poor photon penetration)     | 1.0 mm ID PFA coil (short path, uniform irradiation)                                                      |
| **Gas–liquid contact**      | Headspace exposure or balloon; limited interfacial area             | Segmented Taylor flow; high kLa (0.153 s <sup>-1</sup> ), thin liquid films around gas slugs              |
| **O <sub>2</sub> delivery** | Diffusion-limited from headspace                                    | Continuous co-flow; 11.4× excess transfer capacity                                                        |
| **Pressure**                | Atmospheric                                                         | 3.0 bar gauge (4.0 bar abs) — increases dissolved O <sub>2</sub> ~4×                                      |
| **Light exposure control**  | Entire flask irradiated; difficult to control dose precisely        | Only the reactor coil is irradiated; residence time defines photon dose exactly                           |
| **Over-oxidation control**  | Controlled by reaction time (manual quench by stopping irradiation) | Controlled by residence time (product exits irradiated zone automatically)                                |
| **Scale**                   | Fixed batch size                                                    | Continuous; throughput = 1.2 mmol/h at current settings, scalable by numbering up or increasing flow rate |

The primary flow advantages are (a) dramatically improved gas–liquid mass transfer and (b) uniform, short-path-length irradiation. Both address the two identified batch limitations.

---

### ## (4) Warnings and Flags from Engineering Validation

### This design is NOT engine-validated — screening is required

The automated design engine could not converge on a validated final design. The safety report states:

> **"ENGINE\_FALLBACK: screen required — no usable candidates after filtering"**

**\*\*Confidence level: LOW.\*\***

This means the 5.0 min / 0.2 mL/min conditions represent a conservative starting point, not an optimized design. The engine generated five screen candidates (S1–S5) exploring residence times of 0.25–0.5 min, tubing IDs of 0.5–1.6 mm, and temperatures of 21–41 °C, but none passed all engineering gates simultaneously.

#### ### What to do in practice

1. **\*\*Run the baseline first\*\*** (5.0 min, 21 °C, 0.1 M, 0.4483 sccm air, 3.0 bar BPR) and measure conversion and sulfoxide selectivity (vs. sulfone over-oxidation).
2. If conversion is high (>90%), systematically reduce residence time by increasing liquid flow rate (e.g., 0.4, 0.8, 1.0 mL/min →  $\tau$  = 2.5, 1.25, 1.0 min) to find the minimum  $\tau$  for acceptable yield.
3. If conversion is low, increase air flow (MFC allows up to 10 sccm) or modestly increase temperature (31–41 °C per screen candidates S2/S3).

#### ### Specific risks to monitor

- **\*\*Over-oxidation to sulfone:\*\*** The main selectivity concern. The product sulfoxide can react with additional  $\text{O}_2$ . If sulfone is detected, reduce residence time or lower  $\text{O}_2$  equivalents.
- **\*\*Photocatalyst precipitation:\*\*** At 0.01 M in acetonitrile, the benzothiadiazole catalyst should be fully soluble, but verify before running. Any precipitation will clog the 1.0 mm ID tubing.
- **\*\*Pressure drop:\*\*** Calculated two-phase pressure drop is negligible (0.009 bar), well within the 3.0 bar BPR capacity. Not a concern.
- **\*\*Feed solution stability:\*\*** The premixed substrate + catalyst solution must be kept in the dark (amber bottle/foil wrap). Exposure to ambient light could initiate premature reaction and consume dissolved  $\text{O}_2$ , leading to inconsistent results.
- **\*\*BPR pressure basis:\*\*** The design assumes 3.0 bar gauge (4.0 bar absolute). Confirm whether your BPR is rated in gauge or absolute pressure, as this affects gas compression calculations and dissolved  $\text{O}_2$  concentration.

### 6.5 One-shot raw response (complete, verbatim)

The 17,149-character one-shot provider response below is preserved exactly, including its code fence and errors. The archived FlowProposal parser reported six type-validation errors. The response contains conflicting channel-gas values (0.136 and 0.167 mL/min) and reports 0.235 oxygen equivalents. These are findings about the raw response, not a claim that every parsed benchmark field was scored identically. The final-outcome judge score and the legacy parser/QA diagnostics are distinct records.

```
```json
{
  "recommended_disposition": "SCREEN",
  "disposition_rationale": "The batch protocol is well-defined with mild conditions (room temperature, visible light, air as oxidant, short reaction time). All required equipment is available in inventory. The translation to flow is straightforward: a single liquid feed mixed with air in a T-mixer, irradiated in a 1.0 mL PFA photocoil at 420 nm, followed by gas-liquid separation and collection. This is a conservative first screen requiring experimental validation of conversion under segmented gas-liquid flow conditions.",
  "proposal": {
    "residence_time_min": 5.0,
    "flow_rate_mL_min": 0.20,
    "temperature_C": 21.0,
    "concentration_M": 0.10,
    "BPR_bar": 3.0,
    "reactor_type": "photochemical coil",
    "tubing_material": "PFA",
```



```

"tubing_ID_mm": 1.0,
"reactor_volume_mL": 1.0,
"residence_time_basis": "Liquid contact residence time = reactor_volume_mL /
liquid_flow_rate_mL_min. The gas slugs reduce the liquid holdup fraction, so the true liquid residence
time in the irradiated zone will be longer than 5.0 min. This 5.0 min value is a conservative lower bound
matching the batch irradiation time. The actual liquid-phase residence time depends on the gas-liquid slug
pattern and liquid holdup fraction (typically 0.4-0.6 for Taylor flow), which would give an effective
liquid residence time of roughly 8-12 min. This provides a safety margin for the first screen.",
"residence_time_inlet_min": 5.0,
"residence_time_in_channel_min": 5.0,
"light_setup": {
  "equipment_id": "light_photo_420",
  "name": "420 nm photochemistry LED",
  "power_W": 1.5,
  "wavelength_nm": 420,
  "notes": "Identical to batch: 1.5 W LED at 420 nm. The PFA coil is transparent and wrapped around
or positioned adjacent to the LED. Shorter optical path (1.0 mm ID) compared to batch flask should
improve photon delivery efficiency."
},
"wavelength_nm": 420,
"streams": [
  {
    "stream_label": "liquid_feed",
    "pump_role": "liquid reagent delivery",
    "equipment_id": "pump_photo_liquid",
    "contents": "Fmoc-L-methionine (0.10 M) and 4,7-diphenyl-2,1,3-benzothiadiazole (0.010 M, 10
mol%) dissolved in acetonitrile",
    "solvent": "acetonitrile",
    "phase": "liquid",
    "flow_rate_mL_min": 0.20,
    "concentration_M": 0.10,
    "catalyst_concentration_M": 0.010,
    "molar_equiv": {
      "Fmoc_L_methionine": 1.0,
      "photocatalyst": 0.10
    },
    "gas_flow_sccm": null,
    "gas_flow_actual_mL_min": null,
    "feed_preparation": "Dissolve Fmoc-L-methionine (1.375 g, 3.7 mmol) and 4,7-diphenyl-2,1,3-
benzothiadiazole (106.6 mg, 0.37 mmol) in acetonitrile to a final volume of 37.0 mL in an amber glass
bottle. Sonicate briefly if needed to ensure complete dissolution. This provides approximately 185 min of
run time at 0.20 mL/min, sufficient for startup, steady-state collection, and shutdown. Prepare feed under
subdued lighting to avoid premature photocatalytic activation."
  },
  {
    "stream_label": "air_feed",
    "pump_role": "gas reagent delivery (oxidant)",
    "equipment_id_supply": "gas_air_clean",
    "equipment_id_mfc": "mfc_air_10",
    "contents": "clean dry air (approximately 21 mol% O2, 79 mol% N2)",
    "solvent": null,
    "phase": "gas",

```

```

"flow_rate_mL_min": null,
"concentration_M": null,
"molar_equiv": null,
"gas_flow_sccm": 0.50,
"gas_flow_inlet_STP_mL_min": 0.50,
"gas_flow_actual_mL_min": 0.167,
"gas_flow_calculations": {
  "STP_conditions": "0 °C (273.15 K), 1.01325 bar",
  "operating_pressure_bar": 3.0,
  "operating_temperature_C": 21.0,
  "pressure_type": "gauge; absolute operating pressure = 3.0 bar gauge + 1.0 bar atmospheric = 4.0
bar absolute",
  "correction_factor": " $V_{\text{actual}} = V_{\text{STP}} \times (P_{\text{STP}} / P_{\text{actual}}) \times (T_{\text{actual}} / T_{\text{STP}}) = 0.50 \times$ 
 $(1.01325 / 4.0) \times (294.15 / 273.15) = 0.50 \times 0.2533 \times 1.0769 = 0.136 \text{ mL/min}$ ",
  "gas_flow_actual_corrected_mL_min": 0.136,
  "gas_to_liquid_ratio_actual": 0.68,
  "notes": "At 0.50 sccm air and 0.20 mL/min liquid, the gas-to-liquid volumetric ratio at operating
conditions is approximately 0.68, producing well-defined Taylor flow (segmented slug flow) in the 1.0
mm ID tubing. This regime maximizes gas-liquid interfacial area and ensures efficient oxygen mass
transfer."
},
"oxygen_equivalent_basis": {
  "air_flow_sccm": 0.50,
  "O2_fraction_in_air": 0.21,
  "O2_flow_sccm": 0.105,
  "O2_molar_flow_umol_per_min": 4.69,
  "calculation": " $n_{\text{O}_2} = (P \times V_{\text{dot}}) / (R \times T) = (1.01325\text{e}5 \text{ Pa} \times 1.05\text{e-}7 \text{ L/min} \times 1\text{e-}3 \text{ m}^3/\text{L}) /$ 
 $(8.314 \text{ J}/(\text{mol}\cdot\text{K}) \times 273.15 \text{ K}) \times 1\text{e}6 = 4.69 \text{ }\mu\text{mol/min}$  at STP",
  "substrate_molar_flow_umol_per_min": 20.0,
  "O2_to_substrate_ratio": 0.235,
  "stoichiometric_requirement": "The sulfoxidation requires 0.5 equiv O2 per substrate (2 electrons
per thioether, O2 is 4-electron oxidant in pairs). At 0.235 mol O2 per mol substrate, oxygen delivery is
approximately 0.47 stoichiometric equivalents per pass. Given the 5 min contact time and efficient Taylor
flow mass transfer, this should be sufficient for high conversion as demonstrated in the batch (which used
ambient air exposure). If conversion is low, air flow can be increased up to 10 sccm (MFC maximum).",
  "excess_note": "Conservative starting point. The batch protocol used open-flask air exposure which
provides effectively unlimited O2. In flow, the segmented regime provides high interfacial area. Air flow
can be increased in subsequent screens if needed."
}
},
],
"mixer_type": {
  "equipment_id": "mixer_photo_t",
  "name": "PFA gas-liquid T-mixer",
  "type": "T-mixer",
  "total_flow_at_operating_conditions_mL_min": 0.336,
  "within_mixer_limits": true,
  "mixer_limits_mL_min": "0.01 to 2.0"
},
"mixing_order_reasoning": "The liquid feed (premixed substrate + photocatalyst in acetonitrile) and air
are combined at the PFA T-mixer before entering the photoreactor. This generates a segmented gas-liquid
(Taylor) flow pattern that maximizes O2 mass transfer into the liquid slugs during irradiation. Premixing

```

the liquid avoids any solubility or precipitation issues at the mixer. The T-mixer is placed immediately upstream of the photoreactor coil to minimize non-irradiated volume.",

```
"pre_reactor_steps": [
  {
    "step": 1,
    "description": "Liquid feed from pump_photo_liquid and air from mfc_air_10 (supplied by
gas_air_clean) are combined at mixer_photo_t to generate segmented gas-liquid Taylor flow.",
    "equipment_ids": ["pump_photo_liquid", "gas_air_clean", "mfc_air_10", "mixer_photo_t"],
    "connections": "PFA tubing (tubing_photo_pfa_10) with connector_photo_set unions, 1.0 mm ID
throughout."
  }
],
"post_reactor_steps": [
  {
    "step": 1,
    "description": "The gas-liquid stream exits the photoreactor through bpr_photo_3 (3.0 bar gauge
back-pressure regulator) to maintain constant pressure in the reactor and ensure stable Taylor flow.",
    "equipment_id": "bpr_photo_3"
  },
  {
    "step": 2,
    "description": "After the BPR, the depressurized stream enters separator_photo_gl (vented gas-liquid
separator) to remove the gas phase (spent air). The gas vent is routed to the enclosure exhaust.",
    "equipment_id": "separator_photo_gl"
  },
  {
    "step": 3,
    "description": "The liquid product stream is collected in collector_photo_100 (amber vented
collection vessel, 100 mL). The amber glass protects the product from further photochemical reaction.
The vessel vent is connected to the enclosure exhaust to handle residual acetonitrile vapor.",
    "equipment_id": "collector_photo_100"
  }
],
"stage_parameters": {
  "reactor": {
    "equipment_id": "reactor_photo_100",
    "name": "One milliliter PFA photocoil",
    "volume_mL": 1.0,
    "ID_mm": 1.0,
    "material": "PFA",
    "configuration": "irradiated PFA coil",
    "light_source": "light_photo_420",
    "wavelength_nm": 420,
    "temperature_C": 21.0,
    "pressure_bar_gauge": 3.0,
    "pressure_bar_absolute": 4.0
  },
  "temperature_control": {
    "equipment_id": "controller_photo_thermostat",
    "setpoint_C": 21.0,
    "notes": "Recirculating thermostat maintains reactor at 21 °C, matching batch conditions. Also
removes heat from LED."
  }
}
```

```

    },
    "tubing": {
      "equipment_id": "tubing_photo_pfa_10",
      "ID_mm": 1.0,
      "material": "PFA",
      "length_m": 1.273,
      "calculated_volume_mL": 1.0,
      "volume_check": " $\pi \times (0.5 \text{ mm})^2 \times 1273 \text{ mm} = \pi \times 0.25 \times 1273 = 999.7 \text{ mm}^3 \approx 1.0 \text{ mL}$ . Consistent
with declared reactor volume."
    },
    "connectors": {
      "equipment_id": "connector_photo_set",
      "material": "PFA",
      "supported_ID_mm": 1.0
    },
    "safety_enclosure": {
      "equipment_id": "safety_photo_enclosure",
      "capabilities": ["shield_or_containment", "vented_separator"],
      "compatible_hazards": ["high_intensity_light", "oxygen_enriched_organic_stream",
"acetonitrile_vapor"],
      "notes": "All wetted components (mixer, reactor, BPR, separator, collector) are housed inside the
interlocked photochemical enclosure. The enclosure provides: (1) light shielding from 420 nm LED, (2)
containment and ventilation for acetonitrile vapor, (3) containment in case of tubing failure releasing O2-
enriched acetonitrile."
    }
  },
  "chemistry_notes": "This is a photocatalytic aerobic sulfoxidation. 4,7-Diphenyl-2,1,3-
benzothiadiazole absorbs at 420 nm and generates singlet oxygen ( $^1\text{O}_2$ ) or acts via electron-transfer
mechanism to oxidize the thioether of Fmoc-L-methionine to the corresponding sulfoxide. The reaction is
highly selective (sulfoxide over sulfone) under these mild conditions. Air serves as the terminal oxidant.
The batch protocol achieved 99% yield in 5 min at room temperature, indicating a fast and clean reaction.
In flow, the enhanced mass transfer of O2 in Taylor flow and the short optical path in the 1.0 mm ID
tubing should maintain or improve efficiency. The 0.10 M concentration matches batch conditions. No
competing side reactions are expected at this temperature and irradiation time.",
  "startup_procedure": [
    "1. Verify all connections are secure and the system is leak-tested with air at 4.0 bar absolute (3.0 bar
gauge).",
    "2. Ensure safety_photo_enclosure is closed and interlocked. Verify ventilation is active.",
    "3. Set controller_photo_thermostat to 21 °C and allow to equilibrate.",
    "4. With LED OFF, start pump_photo_liquid at 0.20 mL/min with pure acetonitrile (no substrate) to
fill and wet the system.",
    "5. Start mfc_air_10 at 0.50 sccm. Verify stable segmented flow visually (if viewport available) or by
stable pressure reading.",
    "6. Allow at least 3 reactor volumes (3.0 mL liquid, ~15 min) of acetonitrile + air to flush through the
system and establish steady-state pressure at bpr_photo_3.",
    "7. Switch liquid feed from pure acetonitrile to the prepared substrate/catalyst solution.",
    "8. Turn on light_photo_420 (420 nm, 1.5 W).",
    "9. Allow at least 3 liquid residence times (15 min) for steady state before beginning product
collection.",
    "10. Divert the outlet to collector_photo_100 and begin timed collection."
  ],
  "shutdown_procedure": [

```

```

"1. Turn off light_photo_420.",
"2. Switch liquid feed from substrate solution back to pure acetonitrile.",
"3. Continue pumping acetonitrile at 0.20 mL/min with air at 0.50 sccm for at least 3 reactor volumes (15 min) to flush substrate from the system.",
"4. Stop mfc_air_10.",
"5. Continue pumping acetonitrile for 2 additional reactor volumes (10 min) to purge residual air from the system.",
"6. Stop pump_photo_liquid.",
"7. Depressurize system by allowing flow through bpr_photo_3 until pressure equilibrates to ambient.",
"8. Turn off controller_photo_thermostat.",
"9. Cap or seal collector_photo_100 (amber vessel) and label with run parameters.",
"10. Open safety_photo_enclosure only after LED is confirmed off and system is depressurized."
],
"safety_flags": {
  "oxygen_enriched_organic_stream": {
    "risk": "Air mixed with acetonitrile under pressure creates a potentially flammable mixture. Acetonitrile has a flash point of 2 °C and LEL of 3.0 vol% in air.",
    "mitigations": [
      "Operating at 21 °C with no external heating above flash point.",
      "Using air (21% O2) rather than pure oxygen significantly reduces risk.",
      "Gas-to-liquid ratio is low (0.68 at operating conditions); the system is predominantly liquid-filled.",
      "All components rated to 10 bar; operating at 4.0 bar absolute provides >2× safety margin.",
      "safety_photo_enclosure provides containment and ventilation in case of leak.",
      "No ignition sources inside enclosure (LED is low-power, 1.5 W, non-thermal).",
      "Gas vent from separator_photo_gl is routed through enclosure exhaust to fume hood."
    ]
  },
  "acetonitrile_vapor": {
    "risk": "Acetonitrile is toxic (TLV-TWA 40 ppm). Vapor may be released at separator and collector.",
    "mitigations": [
      "safety_photo_enclosure is ventilated and interlocked.",
      "collector_photo_100 is vented through enclosure exhaust.",
      "separator_photo_gl gas vent is routed to exhaust.",
      "Total acetonitrile throughput is low (~0.20 mL/min)."
    ]
  },
  "high_intensity_light": {
    "risk": "420 nm LED (1.5 W) can cause eye damage.",
    "mitigations": [
      "safety_photo_enclosure shields operator from direct and reflected light.",
      "Interlock prevents LED operation when enclosure is open.",
      "Appropriate laser/LED safety eyewear available as secondary protection."
    ]
  },
  "pressure": {
    "risk": "System operates at 4.0 bar absolute (3.0 bar gauge).",
    "mitigations": [
      "All components rated to ≥5 bar (most to 10 bar).",
      "bpr_photo_3 provides passive pressure relief."
    ]
  }
}

```

```

    "collector_photo_100 is rated to 1.0 bar and is downstream of BPR (at ambient pressure).",
    "Leak test performed during startup with air before introducing solvent."
  ]
}
},
"reasoning_per_field": {
  "residence_time_min": "Set to 5.0 min to match the batch irradiation time. This is the liquid contact
residence time calculated as reactor_volume / liquid_flow_rate = 1.0 mL / 0.20 mL/min = 5.0 min. The
actual liquid residence time will be longer due to gas slugs reducing liquid holdup, providing a
conservative margin.",
  "flow_rate_mL_min": "0.20 mL/min liquid flow gives exactly 5.0 min residence time in the 1.0 mL
reactor. This is well within the pump range (0.01-1.0 mL/min) and provides a comfortable operating point
for stable flow.",
  "temperature_C": "21 °C matches the batch protocol exactly and is an allowed setpoint for both the
reactor and thermostat.",
  "concentration_M": "0.10 M matches the batch protocol exactly.",
  "BPR_bar": "3.0 bar gauge (4.0 bar absolute) is the only available BPR setpoint. This pressurization
serves to: (1) keep air dissolved/compressed for better mass transfer, (2) suppress bubble coalescence for
stable Taylor flow, (3) prevent acetonitrile boiling (not a concern at 21 °C but good practice).",
  "gas_flow_sccm": "0.50 sccm air provides approximately 0.47 stoichiometric equivalents of O2 per
pass relative to substrate. This is a conservative starting point; the MFC can deliver up to 10 sccm if more
O2 is needed. The gas-to-liquid ratio of ~0.68 at operating conditions is appropriate for Taylor flow.",
  "reactor_volume_mL": "1.0 mL as specified by reactor_photo_100. Verified by tubing dimensions:  $\pi \times (0.5 \text{ mm})^2 \times 1273 \text{ mm} \approx 1.0 \text{ mL}$ .",
  "wavelength_nm": "420 nm matches the batch protocol and the absorption maximum of 4,7-
diphenyl-2,1,3-benzothiadiazole photocatalyst.",
  "mixer_type": "PFA T-mixer is the only available gas-liquid mixer. Total flow (0.336 mL/min at
operating conditions) is within the mixer's range (0.01-2.0 mL/min).",
},
"literature_analogies": "Photocatalytic aerobic oxidations in flow using organic photocatalysts and
segmented gas-liquid flow are well-precedented. Similar sulfoxidations have been demonstrated in PFA
coil reactors with visible light LEDs. The use of air rather than pure O2 is safer and sufficient for this fast
reaction. Taylor flow in millimeter-scale tubing provides excellent gas-liquid mass transfer with
interfacial areas of 1000-5000 m2/m3, far exceeding batch stirred vessels.",
"engine_validated": false,
"confidence": 0.88
}
}
...

```

## 7 GUI operation, inventory management, and auditable examples

This section complements the architecture schematics with actual browser captures and archived operational evidence. The interface was exercised on 9 September 2026 using the existing web application; no screen was generated by an image model, no API response was mocked, and no design-generation request was submitted for this documentation exercise. Protocol entry, answer saving and inventory validation were performed live. Result and council views reopen the recorded run identified below. These are demonstrations of software interaction, not new chemical experiments or additions to the benchmark sample size.

The GUI example is the two-stage DPDTC-mediated amidation of 3-methyl-4-nitrobenzoic acid with benzylamine, with a thioester intermediate. Its input specifies 30 min at 95 C for activation, 5-10 min cooling, and another 30 min at 95 C after amine addition. This example is separate from the three benchmark chemistries in Section 3. The original run used claude-opus-4-6 upstream and claude-sonnet-4-6 downstream with 12 candidates. The captured current interface and the archived scientific-design policy are identified in the companion records; they must not be assumed to be identical to every historical benchmark configuration.

### 7.1 Batch entry, follow-up questions, and saved answers

**Figures S14 and S15** show the user-facing input process. The chemist enters the protocol, binds the saved laboratory profile, and reviews the questions selected by the versioned question bank. The screenshot session used deterministic question selection with LLM-assisted extraction switched off, allowing the interface sequence to be reproduced without a new provider call. This check establishes repeatability of this question-selection mode; it does not establish reproducibility of model-generated chemistry or numerical designs.

Three fresh requests with the same protocol, profile and extraction mode returned identical pending question IDs and question-set hashes. The pending IDs were Q-OBJ-001, Q-CHEM-001, Q-HIST-001, Q-HYP-001. The user supplied the archived objective and chemistry confirmation, supplied the archived hypotheses, and marked absent experimental history unavailable. The answer-saving endpoint returned `ready_for_design = true`. Complete questions, answers and intermediate packages are retained in the companion evidence, including items not simultaneously visible inside the scrolling GUI panel.

**Table S23. Observed GUI steps and the records that preserve them.**

| User action          | Observed behavior                                                                                                        | Saved evidence                               |
|----------------------|--------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| Enter batch protocol | The protocol remains editable before analysis; model and laboratory selectors are separate controls.                     | intake_initial_api.json; <b>Figure S14a</b>  |
| Analyze and answer   | Stable question IDs are displayed; absent history can be explicitly unavailable.                                         | intake_recheck_1-3.json; <b>Figure S15</b>   |
| Save answers         | The demonstration reaches a frozen, ready input package. Readiness concerns input completeness, not chemical validation. | intake_answered_api.json; <b>Figure S14b</b> |
| Run or reopen        | A new run requires an explicit design command. This documentation session instead reopened an existing run.              | capture_manifest.json; archived result.json  |
| Inspect and export   | Stage conditions, topology, response bindings, council records and JSON are separate views of the recorded result.       | <b>Figures S17-S18; Table S25</b>            |



(a) Batch protocol input

01 / PROTOCOL

0 answers stored

Standardized intake

FlowPilot intake

Provide the batch protocol. I will extract stated facts and apply the versioned core and deterministic conditional question bank.

Initial batch protocol

A 4.0 mL vial equipped with a PTFE stir bar was charged with 3-methyl-4-nitrobenzoic acid (1 equiv, 0.5 mmol), N,N-dimethylpyridin-4-amine (DMAP) (6.1 mg, 0.1 equiv, 0.05 mmol), and DPDTC (130. mg, 1.05 equiv, 0.525 mmol) under ambient atmosphere without argon or nitrogen purging. 2-MeTHF (1.0 mL, 0.5 M) was added, and the vial was capped and sealed with Teflon tape. The reaction vial was placed in a pre-heated oil bath maintained at 95 deg C and stirred at 500-600 rpm for 30 min. The vial was removed from the oil bath and allowed to cool at room temperature for 5-10 min. Once cooled, the cap was removed and benzylamine (1.05 equiv, 0.525 mmol) and 2-MeTHF (0.25 mL, 0.4 M) were quickly added. The vial was capped, sealed with Teflon tape, and returned to the oil bath at 95 deg C, and stirred at 500-600 rpm for another 30 min before workup.

☐ LLM-assisted extraction

Analyze intake

(b) Answered package

02 / CLARIFY

✓ COMPLETE

Fixed + conditional questions

flowpilot\_questions\_v1.1

be804c8fa675

✓

Design input is frozen

All mandatory questions are answered or explicitly unavailable.

**Figure S14.** Actual FlowPilot interface: (a) entry of the complete two-stage batch protocol before analysis; (b) the completion state after saving the required answers. The panels are different moments in the same live intake demonstration. LLM-assisted extraction was disabled. Frozen means that the answered input package is ready for submission, not that a chemical design or yield has been validated. Full-screen originals are retained alongside these interface crops.

02 / CLARIFY

🔍 Fixed + conditional questions

🕒 4 PENDING

flowpilot\_questions\_v1.1

be804c8fa675

Q-0BJ-001
OBJECTIVE · CORE

What is the main goal for the first flow design? Examples: first safe screening design, reproduce batch yield, improve throughput, maximize conversion, minimize residence time, or test a specific hypothesis.

Identify a practical two-stage flow process that reduces the overall processing time while maintaining high final amide yield and stable continuous operation. FlowPilot should explore the residence time

Q-CHEM-001
CHEMISTRY IDENTITY · CORE

Please confirm the intended reaction transformation. State the transformation family and, where known, the substrate, product, bond formed, and bond broken.

This is a two-stage DPDTC-mediated amide formation through a thioester intermediate. In Stage 1, 3-methyl-4-nitrobenzoic acid is converted to the corresponding 2-pyridyl thioester using DPDTC and

Q-HIST-001
HISTORICAL DATA · CORE

Do you have historical experimental data or previous flow attempts for this reaction? If yes, provide a table with conditions and results.

Table/CSV/text with entry, T, concentration, pressure, flows, residence times, reactor volume, tubing ID, product/yield/conversion/selectivity, and notes; or mark unavailable.

Save answers

**Figure S15.** Actual follow-up question panel with stable IDs, question-bank version, question-set hash and entered answers. The visible examples concern objective and chemistry identity; other questions remain within the scrollable panel. Scrollbars and partially visible textarea contents are native interface behavior, not edited text. Complete input and answer records accompany the ESI. Three identical deterministic intake requests produced the same pending IDs and hash.

## 7.2 Inventory input, normalization, and laboratory constraints

The inventory workspace separates source material from a normalized, editable profile (**Figure S16**). It accepts uploaded laboratory documents and free-text equipment or constraint descriptions; the JSON editor provides a structured validation route. Users review equipment identifiers, quantities, units, capability limits and unavailable functions before saving a version or selecting Use in design. Source extraction is not itself laboratory confirmation. A schema-valid profile may still contain warnings or missing capabilities, and design-specific compatibility is checked separately.

The demonstrated structured route uses KHU Laboratory Inventory, version 4, schema `flowpilot_inventory_profile_v3.0`. Its seven reactor entries and three pump entries are inventory records, not a claim of seven reactor families or three physical pumps: one pump entry describes three Chemyx units. The profile combines source-derived data with subsequent chemist confirmations. The BPR named Noname has confirmed setpoints 2, 8 and 9 bar even though the original PDF did not list BPR settings. The retained source-warning text refers to that original omission and is not a complete description of the later pressure-controller entry.

During the interface check, uploading this JSON through the generic Extract inventory operation with LLM assistance disabled returned an empty inventory and validation warnings. Pasting the existing JSON into the editor and selecting Validate edits retained all seven reactor and three pump records. **Figure S16** therefore documents the structured validation route, not a successful PDF extraction experiment. The failed extraction response and screenshot are retained as a current input-path limitation; the source inventory was not overwritten or resaved during the capture session.

**Table S24. Examples of the KHU version-4 inventory data and their interpretation.**

| <b>Inventory item</b> | <b>Stored information</b>                                                                                                             | <b>Design consequence or qualification</b>                                                                                           |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Inline degassing      | Unavailable; inline degasser equipment explicitly forbidden. Offline pre-degassing with a sealed reservoir is an allowed alternative. | Omit the inline device; do not equate its absence with oxygen exclusion being unnecessary.                                           |
| Reactor records       | PFA: 2 mL; two 5 mL units at 1.016 mm ID; 10 mL; 5 mL at 0.762 mm ID. FEP: 10 and 20 mL.                                              | Assign actual volume/material/ID combinations rather than inventing an intermediate coil size.                                       |
| Temperature limits    | Listed PFA coils: maximum 80 C; listed FEP coils: maximum 50 C.                                                                       | These profile limits restrict this inventory selection; they are not universal material-property limits.                             |
| Pump records          | Chemyx Fusion 100 (three units), Vapourtec SF-10, Vapourtec R2C+; separate pressure and flow limits.                                  | Use the chosen pump and syringe configuration. The profile minimum is not automatically a globally suitable experimental feed rate.  |
| Pressure controller   | Chemist-confirmed BPR: setpoints 2, 8, 9 bar; maximum 10 bar.                                                                         | Preserve pressure-basis declarations and check the entire flow path, not just this device.                                           |
| Mixers and connectors | No dedicated mixer entry; one connector entry. The displayed design uses a generic T-mixer requiring verification.                    | An assumed passive accessory is not evidence that a suitable three-port mixer exists. A two-port union alone cannot merge two feeds. |
| Source caveats        | Conflicting inch/metric tubing labels; missing radiant power; original BPR omission.                                                  | Preserve caveats and provenance. A numerical unknown-value sentinel must not be treated as measured zero power.                      |

**(a) Inventory input**

**SOURCE**  
 **Import laboratory inventory**

**Profile name**  
KHU Laboratory Inventory

**Laboratory**  
Kyung Hee University (KHU)

  
**Choose PDF, Word, PowerPoint, spreadsheet, or JSON**  
khu\_inventory\_v4.json

**Additional equipment and constraints**  
No inline degasser. Offline pre-degassing and a sealed solvent reservoir are permitted; do not add an inline degasser to the process.

☐ LLM-assisted extraction

 **Extract inventory**

**Figure S16.** Inventory workspace, panel (a): actual upload controls and the separate field for additional equipment or constraints. The KHU JSON file is selected here. This capture shows available controls, not evidence that every supported document format has been validated. The structured JSON-validation result is shown on the following page.

**(b) Normalized profile and warnings**

**NORMALIZED PROFILE**

VALID

 **Schema preview**

|                      |                         |                         |
|----------------------|-------------------------|-------------------------|
| <b>7</b><br>REACTORS | <b>3</b><br>PUMPS       | <b>3</b><br>TUBING      |
| <b>8</b><br>LIGHTS   | <b>2</b><br>GAS DEVICES | <b>1</b><br>ACCESSORIES |

Reactors

KHU PFA coil 2 mL (2 mL), KHU PFA coil 5 mL - unit 1 (5 mL), KHU PFA coil 5 mL - unit 2 (5 mL), KHU PFA coil 10 mL (10 mL), KHU FEP coil 10 mL (10 mL), KHU FEP coil 20 mL (20 mL), KHU PFA coil 5 mL (0.03 inch ID) (5 mL)

Pumps

Chemyx Fusion 100 syringe pump (3 units), Vapourtec SF-10 E-series peristaltic pump, Vapourtec R-series R2C+ HPLC pump

Pressure

Noname

Constraints

9 declared

 No mixers are defined; a passive standard mixing fitting may be used as a disclosed screening assumption and must be verified before a run.

 Tubing metric ID labels conflict with their inch labels; converted inch values are used and documented in source\_caveats.

 BPR settings are absent from the source and remain empty.

 Manual photoreactor radiant power is absent; power\_W=0.0 is used only as a backward-compatible unknown-value sentinel.

 Validate edits

 Save profile

 Use in design

**Figure S16** (continued). Inventory workspace, panel (b): validated KHU profile after loading the structured JSON through the editor and selecting Validate edits. Equipment record counts, assigned names, warnings, and save/use controls remain visible. VALID denotes schema/profile validation, not complete hardware availability or laboratory safety. Warnings were not removed for presentation.

### 7.3 Stage-resolved output and run provenance

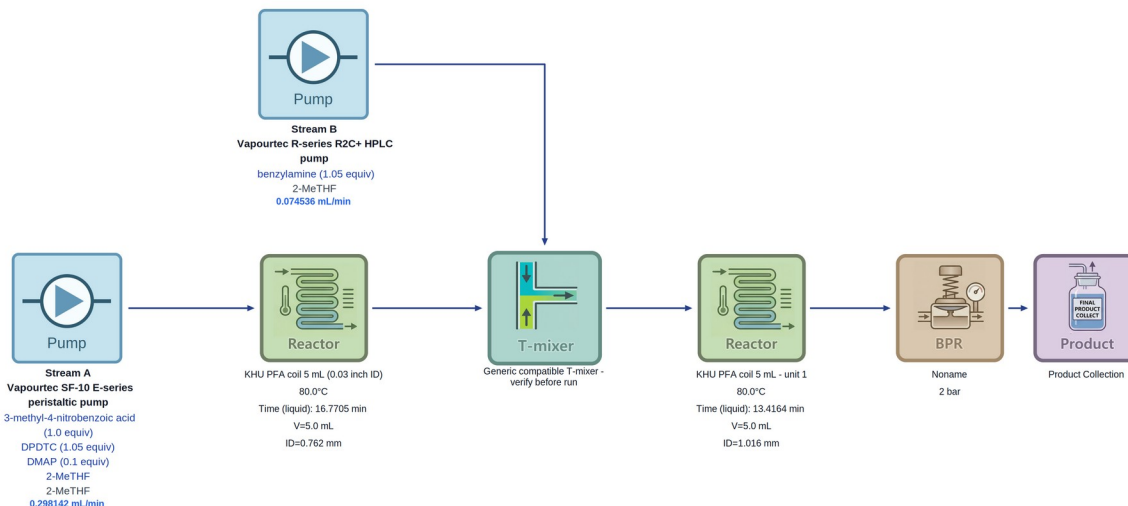
**Figure S17** pairs a horizontal redraw of the archived DPDTC topology with the original GUI process-summary screenshot. **Table S25** transcribes the final stage values at readable manuscript text size. The second liquid feed enters between the reactors, so the Stage 2 flow is the sum of the upstream liquid and the new amine feed. Both stages are liquid-only in this example. The generic gas-basis note visible in the GUI is not evidence of a gas feed here. For a gas-containing design, inlet/STP apparent time is a reporting convention and must not be presented as measured in-channel contact time.

**Table S25. Archived DPDTC screening parameters shown by the GUI; not a validated synthesis procedure.**

| Quantity                  | Stage 1                   | Stage 2                   |
|---------------------------|---------------------------|---------------------------|
| Reaction                  | Thioester formation       | Amide formation           |
| Reactor volume / material | 5 mL / PFA                | 5 mL / PFA                |
| Tubing ID                 | 0.762 mm                  | 1.016 mm                  |
| Cumulative liquid flow    | 0.298142 mL/min           | 0.372678 mL/min           |
| New feed at this stage    | Stream A: 0.298142 mL/min | Stream B: 0.074536 mL/min |
| Nominal liquid time       | 16.7705 min               | 13.4164 min               |
| Temperature               | 80 C                      | 80 C                      |
| Recorded pressure         | 2 bar gauge               | 2 bar gauge               |
| Gas feed                  | None                      | None                      |

The approximately 30.19 min combined nominal time is a proposed screen, not a measured conversion time. The source batch used 95 C; the selected profile restricts the chosen PFA coils to 80 C. The UI reports LOW confidence and requires verification of the generic interstage mixer. Neither a passing contract nor council agreement establishes that reduced temperature and shorter holds preserve yield. The pump pictograms are generic: equipment names and identifiers specify the peristaltic and HPLC pumps and take precedence over the icon's internal syringe-pump label.

## (a) Process topology



## (b) Stage and feed tables

Process summary CSV

Gas reference: 273.15 K, 1.01325 bar. Gas-stage time = reactor volume / (liquid flow + inlet/STP gas flow); liquid-stage time = volume / liquid flow.

| Stage / reactor                                                                                                                           | Volume mL | Liquid mL/min | Gas at STP mL/min | Time min          | Temperature °C | Pressure bar | Light nm |
|-------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------|-------------------|-------------------|----------------|--------------|----------|
| <b>Stage 1: Thioester formation (carboxylic acid activation with DPDTC/DMAP)</b><br>KHU PFA coil 5 mL (0.03 inch ID)<br>PFA - 0.762 mm ID | 5         | 0.298142      | 0                 | 16.7705<br>liquid | 80             | 2<br>gauge   | --       |
| <b>Stage 2: Aminolysis - amide bond formation</b><br>KHU PFA coil 5 mL - unit 1<br>PFA - 1.016 mm ID                                      | 5         | 0.372678      | 0                 | 13.4164<br>liquid | 80             | 2<br>gauge   | --       |

Feed streams

| Stream / enters     | Composition                                                                             | Phase  | Flow mL/min        | Feed concentration M | Reagent equiv | Delivery equipment                        |
|---------------------|-----------------------------------------------------------------------------------------|--------|--------------------|----------------------|---------------|-------------------------------------------|
| <b>A</b><br>Stage 1 | 3-methyl-4-nitrobenzoic acid (1.0 equiv); DPDTC (1.05 equiv); DMAP (0.1 equiv); 2-MeTHF | liquid | 0.298142<br>liquid | 0.5                  | 1             | Vapourtec SF-10 E-series peristaltic pump |
| <b>B</b><br>Stage 2 | benzylamine (1.05 equiv)                                                                | liquid | 0.074536<br>liquid | 2.1                  | 1.05          | Vapourtec R-series R2C+ HPLC pump         |

**Figure S17.** Archived-run output: (a) topology redrawn with a horizontal main process path, right-angle interstage feed and generic pump symbols; equipment names and numerical labels are retained from the archived graph; (b) the original GUI process-summary screenshot, including the interstage feed and cumulative Stage 2 flow. Both panels describe the same archived result. Numerical values are also transcribed in **Table S25**. The generic T-mixer remains a disclosed verification requirement. This is a software screening output, not a wet-lab validated protocol. Original interface captures and graph files remain in the companion evidence.



## 7.4 Examples of critical findings and evaluator limitations

**Table S26** gives traceable examples from retained one-shot and FlowPilot outcomes. They were selected to illustrate different failure mechanisms, not sampled to estimate a failure rate. A judge flag, a source-supported design inconsistency, and a deterministic software block are different observations. Multiple criteria can flag the same underlying defect. The archived benchmark scores and **Figure S9** flag counts are unchanged; the arithmetic recheck below is an explicit post hoc audit, not a silent rescore.

**Table S26. Concrete archived issues and one checked evaluator false positive.**

| Record / criterion                                                   | Observed issue                                                                                                                                                                                          | Interpretation                                                                                                                                                                                                                                       |
|----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| One-shot Opus 4.6; hydrogenolysis repeat 01; N2-D1FC65D8; UO-05      | The procedure places a vented gas-liquid separator before the sole pressure-defining BPR while describing a pressurized reactor. The original post-reactor instructions confirm this order.             | One judge flagged UO-05. The stated pressure-control and venting arrangement is incomplete as written; component pressure ratings alone do not resolve the topology.                                                                                 |
| One-shot Opus 4.6; hydrogenolysis repeat 02; N2-B6923FF7; UO-07      | The main parameter reports 20.27 min, while its own basis text gives 3.0 mL / 0.1 mL/min = 30.0 min empty-bed time and a different holdup time.                                                         | Two judges flagged UO-07. The claimed basis and serialized value disagree. Distinct time definitions must not be collapsed into one unlabeled value.                                                                                                 |
| FlowPilot Qwen3.6-27B; CuAAC repeat 03; N2-110E4388; UO-02/03/11     | The final preparation instruction charges 0.25 mmol Cu/C into a 10 mL feed even though the catalyst is also assigned to a packed-bed cartridge. The saved operating_procedure confirms the instruction. | One judge raised three criterion flags for one solids-assignment defect. The feed-preparation compiler and heterogeneous-catalyst placement must agree; passing geometry checks is insufficient.                                                     |
| FlowPilot Opus 4.6; oxidation repeat 01; worked example in Section 6 | The archived council summary reports no usable candidates and a calculator center-point fallback. The raw narrative and final structured conditions then differ ( <b>Table S22</b> ).                   | Operational fallback and artifact synchronization are separate limitations. A completed pipeline does not establish successful council selection or consistency of every report artifact.                                                            |
| FlowPilot GPT-4o; oxidation repeat 03; N2-E85D424E; UO-08            | One Qwen judge called 3.3158 sccm versus 3.3158 mL/min at STP a factor-of-1000 defect.                                                                                                                  | This specific allegation is incorrect. These volumetric units are equal at the same standard conditions. The arithmetic gives approximately 1 O <sub>2</sub> equivalent, not 0.001. The historical flag remains disclosed as a judge false positive. |

Unit audit for N2-E85D424E: oxygen feed =  $(3.3158 \text{ mL/min air} \times 0.21) / (22.414 \text{ mL/mmol}) = 0.0310662 \text{ mmol/min O}_2$ . Dividing by the recorded substrate flow of 0.031068 mmol/min gives 0.99994 equivalent. The same reference temperature and pressure must be used for both volume units. This checks the stated thousand-fold allegation only; it does not validate all other aspects of that design. It also explains why the heatmap should be described as judge-reported critical flags rather than confirmed error counts.

## 7.5 Recorded council discussion and disagreement

**Figure S18** shows the actual Council tab for the same DPDTC run. Four domain reviewers evaluate the candidate set, after which a Skeptic and Chief record cross-domain assessment and selection. **Table S27** reproduces exact saved excerpts showing a disagreement about candidate 6 and the eventual selection of candidate 1. These are explicit model-authored review outputs, not a reconstruction of hidden model reasoning and not a fabricated dialogue. The GUI groups domain reviews and synthesis into two display rounds; this must not be interpreted as two complete candidate-generation/revision cycles.

Recorded candidate discussion, before final inventory realization. These statements are model assessments, not final run conditions.

|                          |         |
|--------------------------|---------|
| ▼ Round 1                |         |
| 4 recorded contributions |         |
| ► DrChemistry            | WARNING |
| ► DrKinetics             | WARNING |
| ► DrFluidics             | WARNING |
| ► DrSafety               | WARNING |
| ▼ Round 2                |         |
| 2 recorded contributions |         |
| ► Skeptic                | WARNING |
| ► Chief                  | WARNING |

**Figure S18.** Actual Council tab with expandable domain, Skeptic and Chief records. WARNING states are retained. Display rounds organize the recorded contributions and are not evidence of independent physical validation or two full optimization cycles. Exact source excerpts and the complete machine-readable discussion accompany **Table S27**.

**Table S27. Exact archived council excerpts; statements are model assessments, not endorsed chemical facts.**

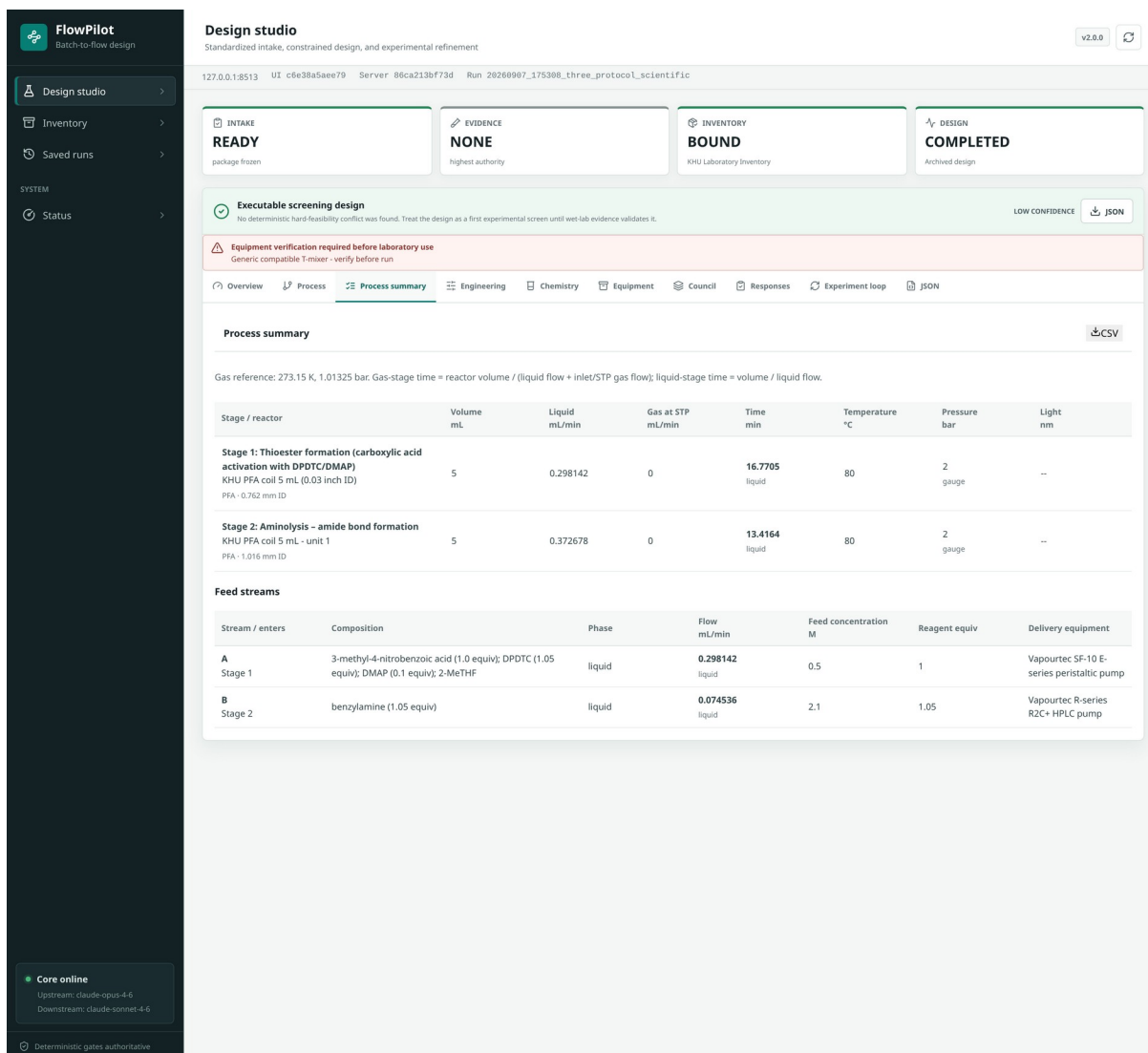
| Speaker and decision                                                           | Verbatim recorded assessment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DrChemistry<br>Candidate 6<br>reject                                           | S1 16.8 min / S2 6.7 min; total 23.5 min. S2 at 6.7 min is the shortest Stage 2 across all candidates. With uncharacterized kinetics, this risks incomplete amide coupling entering collection. Throughput gain does not justify this yield risk for a screening experiment.                                                                                                                                                                                                                                                                                                                                                       |
| DrKinetics<br>Candidate 6<br>acceptable                                        | $\tau_1=16.8$ min (10 mL coil), $\tau_2=6.7$ min; total 23.5 min. High feed throughput. Tests whether Stage 2 aminolysis can complete in <7 min at 80 °C. Vapourtec SF-10 at 0.596 mL/min well within range. Stage 2 exposure very short.                                                                                                                                                                                                                                                                                                                                                                                          |
| DrSafety<br>Candidate 6<br>acceptable                                          | S1 16.8 min in 10 mL coil / S2 6.7 min, total ~23 min. High feed throughput (0.596 mL/min S1). Very short S2 hold is the primary risk; tests whether aminolysis is fast once thioester is formed.                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Skeptic<br>Conflicting domain preferences<br>No hard veto on this disagreement | DrChemistry rejected candidates 6 and 12; DrKinetics and DrSafety accepted both as valid exploratory boundary points. DrFluidics accepted both. No hard violation was identified by any reviewer for either candidate. The rejections by DrChemistry are chemical-risk preferences, not hard violations. No veto is warranted on that basis alone.                                                                                                                                                                                                                                                                                 |
| Chief<br>Final candidate selection<br>Candidate 1                              | Candidate 1 delivers the highest feed throughput among candidates whose total residence time is at or below the 60-minute batch cycle (including cooling), at 30.2 min total with both stages halved symmetrically. It uses two available 5 mL PFA coils within all hard constraints, employs the SF-10 and R2C+ pumps well within their operating ranges, and the 2 bar BPR is available. The symmetric 16.8/13.4 min allocation explores whether both stages can be shortened simultaneously—the most direct test of the throughput objective—without arbitrarily assuming one stage is rate-limiting. No hard violations exist. |

The discussion exposes uncertainty rather than proving the selected timing is correct. The Chief's phrase '60-minute batch cycle (including cooling)' is itself inconsistent with the source's two 30 min heated holds plus 5-10 min cooling. The quoted statement is preserved as an audit example. The saved Skeptic record also assumes a connector can serve an interstage T-junction; the actual port count and compatible mixing hardware must be verified rather than inferred from agreement among agents. No laboratory validation follows from these recorded assessments.

## 7.6 Evidence package and capture reproducibility

The companion evidence package separates fresh GUI actions from archived generation artifacts. Full-screen originals and cropped screenshots are saved at device scale factor 2. The retained GUI screenshots are unaltered apart from cropping and page arrangement. The capture manifest records the API/build identity, request URLs and screenshot hashes. The live intake rechecks used the same saved protocol and KHU profile three times. No design job was submitted, no saved inventory version was overwritten, and no benchmark score was recalculated. **Figure S17(a)** is now an explicitly identified publication redraw; its original screenshot remains in that evidence package.

**Figure S19** retains the full application context: navigation between Design studio, Inventory and Saved runs; the build/run identifier; intake, evidence and inventory state; the screening disposition and equipment warning; and the result tabs. It locates the enlarged interface crops in the overall workflow. The server address shown is a local capture endpoint, not a public service URL.



**Figure S19.** Full application screenshot for the reopened DPDTC run. The sidebar exposes the input, inventory and archived-run workspaces, while the header identifies the build and run. The result retains LOW confidence and the equipment-verification warning despite showing a completed screening design. This image provides navigation context; detailed numerical values are in **Table S25** and full-resolution screenshots in the evidence package.

**Table S28. Companion records for the added operational examples.**

| <b>Record</b>                                             | <b>Purpose</b>                                                                                        |
|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| capture_manifest.json; screenshots/                       | Actual UI capture sequence, build identity and full-resolution images.                                |
| intake_initial_api.json; intake_answered_api.json         | Live question-selection and answer-saving responses.                                                  |
| intake_reproducibility_check.json                         | Three identical deterministic question sets and hashes.                                               |
| khu_inventory_v4.json;<br>inventory_validated_api.json    | Saved inventory source and structured validation result.                                              |
| inventory_import_api.json                                 | Preserved empty-profile result from the generic extraction path; not presented as successful parsing. |
| gui_archive_result.json;<br>gui_archive_final_design.json | Unmodified run result and final structured conditions.                                                |
| council_exact_excerpts.json;<br>council_full_record.json  | Verbatim excerpt provenance and complete discussion records.                                          |
| benchmark_examples/;<br>selected_judge_findings.csv       | Candidate results, selected judge outputs and identities for <b>Table S26</b> .                       |
| gas_flag_arithmetic_check.json;<br>SHA256SUMS.txt         | Transparent post hoc unit check and evidence-file checksums.                                          |

## 8 Connected-process case studies supporting main-text Figures 5 and 6

This section preserves the supplied batch protocols, frozen chemist responses, and archived September design proposals separately from the August architecture-comparison campaigns in Section 3. **Tables S31 and S34** summarize the new KHU experimental outcomes without treating shared response-set entries as replicates. Section 12 provides the full procedures, photographs, characterization, and implementation reconciliation. Proposed conditions remain proposals even where the laboratory implemented matching settings; no measured residence-time distribution or optimized operating point is inferred.

### 8.1 Photoredox Giese addition followed by sulfoxidation (main-text Figure 5)

The target sequence converts the alpha-silyl sulfide and acrylonitrile to a sulfide, then to sulfoxide 4a. The chemistry source is Park et al. [4]. Stage 1 is oxygen-free; pure oxygen is introduced only at the Stage 2 inlet in the revised flow proposal. The original batch protocol uses ambient air in the second step. The change to pure oxygen is a collaborator-requested constraint in the September 18 reruns, not an autonomous discovery attributed to the model.

#### Supplied batch protocol (verbatim collaborator transcription).

Step 1: Giese reaction A re-sealable pressure tube (13 × 100 mm) equipped with a magnetic stir bar was charged with (((4-methoxyphenyl)thio)methyl)trimethylsilane (1, 0.2 mmol, 1.0 equiv), acrylonitrile (2, 0.4 mmol, 2.0 equiv), and [Ir(dF(CF<sub>3</sub>)ppy)<sub>2</sub>(dtbpy)]PF<sub>6</sub> (0.001 mmol, 0.5 mol %) under an argon atmosphere. A degassed solvent mixture (2.0 mL, 0.1 M with respect to 1) of EtOH:pH 9 buffer (5:1, v/v) was added, and the resulting light greenish-yellow mixture, under vigorous magnetic stirring, was positioned 3 cm from a pair of 5 W blue LEDs ( $\lambda_{\text{max}} = 452 \text{ nm}$ ) and irradiated at room temperature (25 °C, maintained with a cooling fan to counteract heating from the LEDs) for 4 h

Step 2: Oxidation reaction After 4 h, the cap of the tube was removed to expose the reaction mixture to ambient air, providing atmospheric oxygen as the terminal oxidant, and stirring was continued under otherwise identical irradiation conditions for an additional 6 h.

**Table S29. Figure 5 response-set provenance and shared constraints.**

| Set | Design emphasis                                                  | Source / shared constraints                                                                                                                             |
|-----|------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | Integrated performance                                           | KHU revised V2, slide 5                                                                                                                                 |
| 2   | Conversion-focused                                               | KHU revised V2, slide 8                                                                                                                                 |
| 3   | Throughput / processing time                                     | KHU revised V2, slide 11                                                                                                                                |
| All | Oxygen exclusion in Stage 1; at least 2.0 equivalents in Stage 2 | Original fixed-ID responses below; September 18 rerun additionally requires pure O <sub>2</sub> . Inventory v7 adds MFC setting-grid information to v6. |

#### Fixed-ID responses (verbatim; hypotheses are not measured facts).

##### Response Set 1

Responses to Q-OBJ-001: Please enter here —> Develop a safe, reproducible, and inventory-compatible two-stage flow process that maximizes the final yield of sulfoxide 4a while avoiding unnecessarily long



residence times. Design the two stages as one connected process, because Stage 1 determines the composition entering Stage 2.

Responses to Q-HYP-001: Please enter here —> Stage 1 performance may depend on effective oxygen exclusion and photon delivery. Incomplete sulfide formation will limit the material available for subsequent oxidation. Stage 2 may depend on oxygen availability and gas–liquid transport. Use the batch reaction time as a reference rather than a fixed target, and determine the residence time based on flow performance. Balance photon delivery, oxygen feed, pressure, and stage-specific residence times based on the final sulfoxide yield and selectivity.

Responses to Q-GAS-002: Please enter here —> Use at least 2.0 equiv of oxygen relative to the limiting substrate; higher oxygen equivalents may be used if justified by the reaction and operating conditions.

Responses to Q-GAS-003: Please enter here —> Introduce oxygen only at the Stage 2 inlet. Stage 1 must remain oxygen-free.

### **Response Set 2**

Responses to Q-OBJ-001: Please enter here Prioritize high conversion through both stages and a high final yield of sulfoxide 4a. Choose the shortest residence times reasonably justified for this conversion-focused starting design rather than adding a large time excess by default. Keep reactor volumes and flow rates practical for flow screening.

Responses to Q-HYP-001: Please enter here —> Stage 1 requires sufficient photon delivery and residence time to generate enough sulfide for Stage 2. Use the batch reaction time as a reference rather than a fixed target, and determine the residence time based on flow performance. Stage 2 requires adequate oxygen delivery, but oxygen feed, pressure, gas fraction, and residence time are coupled. Residence time should be optimized based on the final sulfoxide yield and selectivity rather than increased by default.

Responses to Q-GAS-002: Please enter here —> Use at least 2.0 equiv of oxygen relative to the limiting substrate; higher oxygen equivalents may be used if justified by the reaction and operating conditions.

Responses to Q-GAS-003: Please enter here —> Introduce oxygen only at the Stage 2 inlet. Stage 1 must remain oxygen-free.

### **Response Set 3**

Responses to Q-OBJ-001: Please enter here —> Identify a practical two-stage flow design that balances final sulfoxide yield and selectivity with substrate throughput, total processing time, oxygen delivery, and stable operation. Prefer a compact, higher-throughput starting design where chemically justified.

Responses to Q-HYP-001: Please enter here —> Stage 1 may maintain sufficient sulfide formation at higher liquid throughput if photon delivery remains effective. Use the batch reaction time as a reference rather than a fixed target, and determine the residence time based on flow performance. In Stage 2, pressure and oxygen feed can alter oxygen availability, gas fraction, and contact time simultaneously. Evaluate the overall process based on substrate throughput, stage-specific residence times, and the final sulfoxide yield and selectivity. Shorter residence times may be explored, but should be validated experimentally.

Responses to Q-GAS-002: Please enter here —> Use at least 2.0 equiv of oxygen relative to the limiting substrate; higher oxygen equivalents may be used if justified by the reaction and operating conditions.

Responses to Q-GAS-003: Please enter here —> Introduce oxygen only at the Stage 2 inlet. Stage 1 must remain oxygen-free..

Archived execution: outputs/figure5\_pure\_oxygen\_physics\_20260918, attempt\_01 for each set; upstream Claude Opus 4.6 and downstream Claude Sonnet 4.6; scientific\_v2 policy with a candidate budget of 12. The run folders retain the input package, inventory snapshot, model-call records, council record, calculation checks and result JSON. This description does not imply bit-for-bit reproducibility from model names alone.

**Table S30. Figure 5 nominal stage conditions and component-feed targets from the archived final exports. Panel I: stages.**

| Set / stage | Coil / module                      | Q liquid / gas (mL min-1) | Time (min) | T / BPR         |
|-------------|------------------------------------|---------------------------|------------|-----------------|
| 1 / 1       | 2 mL PFA; Vapourtec UV-150; 450 nm | 0.02 / 0                  | 100        | 25 C / 7 bar(g) |
| 1 / 2       | 20 mL FEP; Manual 2; 448 nm        | 0.02 / 0.09               | 181.82     | 25 C / 7 bar(g) |
| 2 / 1       | 5 mL PFA; Vapourtec UV-150; 450 nm | 0.02 / 0                  | 250        | 25 C / 7 bar(g) |
| 2 / 2       | 20 mL FEP; Manual 2; 448 nm        | 0.02 / 0.09               | 181.82     | 25 C / 7 bar(g) |
| 3 / 1       | 2 mL PFA; Vapourtec UV-150; 450 nm | 0.02 / 0                  | 100        | 25 C / 7 bar(g) |
| 3 / 2       | 20 mL FEP; Manual 2; 448 nm        | 0.02 / 0.09               | 181.82     | 25 C / 7 bar(g) |

**Table S30 (continued). Panel II: identical nominal component feeds in all three sets. Liquid Feed A uses EtOH:pH 9 buffer (5:1, v/v). Its common volumetric flow is 0.0200 mL min-1; listing three components does not mean three liquid feeds. The gas feed is pure oxygen at inlet/STP.**

| Component                     | Feed / entry  | C (M)              | Molar flow (mmol min-1) | Equiv.  |
|-------------------------------|---------------|--------------------|-------------------------|---------|
| Alpha-silyl sulfide precursor | A / Stage 1   | 0.100              | 0.00200                 | 1.00    |
| Acrylonitrile                 | A / Stage 1   | 0.200              | 0.00400                 | 2.00    |
| Ir photocatalyst              | A / Stage 1   | 0.000500           | 0.0000100               | 0.00500 |
| Oxygen                        | Gas / Stage 2 | Not a liquid stock | 0.004016                | 2.008   |

Calculation basis:  $Q_{\text{gas,STP}} = 0.0900 \text{ mL min}^{-1}$  and  $STP = 273.15 \text{ K}$  and  $1 \text{ atm}$ . The archived calculator uses a gas molar volume of  $22.41272243 \text{ mL mmol}^{-1}$  (approximately 22.414 at STP). Thus  $n_{\text{dot}}(\text{O}_2) = 0.0900/22.41272243 = 0.004016 \text{ mmol min}^{-1}$  and  $\text{equiv}(\text{O}_2) = 0.004016/(0.100 \times 0.0200) = 2.008$ . The nominal Stage 2 inlet/STP index is  $20/(0.0200 + 0.0900) = 181.82 \text{ min}$ . This index is useful for consistent reporting but is not physical in-channel residence, gas holdup, oxygen uptake or a mass-transfer measurement. A dissolved-oxygen or conversion measurement is still needed to connect oxygen feed to reaction performance.

Equipment interpretation: Stage 1 uses an inventory-confirmed UV-150-compatible PFA coil and the UV-150 light; Stage 2 uses the Manual 2 module and its own 448 nm strip light. The UV-150 light is not detached for use with an unrelated manual module. The liquid pump is a Vapourtec E-series channel with BLUE tubing; gas delivery uses the FF-C00 MFC, a CV3301 gas-line check valve and the P713 mixing junction. No inline degasser is assigned. Offline deoxygenation and protection of the Stage 1 feed remain required by the oxygen-free protocol.

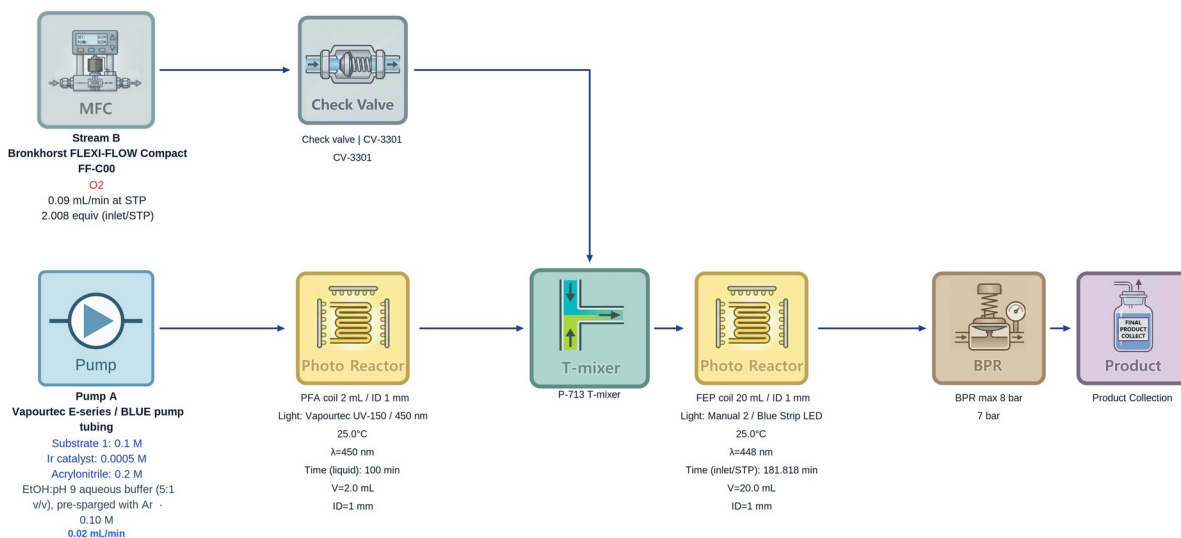
Unresolved experimental checks: the gas-line check valve addresses liquid entry into the gas branch, not gas migration into the upstream liquid reactor. The earlier collaborator incident is therefore not declared solved by its presence. Confirm valve orientation and pressure rating, liquid-branch isolation/protection, actual junction pressures, oxygen-compatible hardware, and startup/shutdown procedures before testing the archived 7 bar(g) proposal. The optional transient tool used illustrative dynamics and a 12 bar(g) simulated gas-source pressure; that value is not a laboratory setpoint and exceeds the recorded MFC inlet rating. Its outputs must not be treated as a calibrated incident prediction or safety clearance. Gas pressure traces and equipment response data were unavailable.

**Table S31. KHU-reported Figure 5 outcomes and actual settings. Sets 1/3 share a single entry in the supplied result table; this is not evidence of independent repeats. NR means not reported. Calculated times are identified separately from measurements.**

| Reported item                                      | Sets 1 and 3 (grouped) | Set 2                  |
|----------------------------------------------------|------------------------|------------------------|
| NMR yield of 4a (%)                                | 97                     | 98                     |
| Isolated yield / mass                              | 94% / 42.0 mg          | NR                     |
| R1 / R2 volume (mL)                                | 2 / 20                 | 5 / 20                 |
| Liquid feed (mL/min)                               | 0.020                  | 0.020                  |
| Oxygen controller setting (mL/min)                 | 0.090                  | 0.090                  |
| R1 nominal liquid time (min)                       | 100                    | 250                    |
| R2 inlet/STP index (min), calculated               | 181.82                 | 181.82                 |
| Reported temperature                               | 25 ± 1 °C, both stages | 25 ± 1 °C, both stages |
| Recorded system pressure (bar)                     | 5.1–5.4                | 5.2–5.6                |
| BPR cartridge rating                               | 7 bar                  | 7 bar                  |
| NMR internal standard                              | 1,3-Benzodioxole       | 1,3-Benzodioxole       |
| Replicate count / SD / sample window               | NR                     | NR                     |
| Substrate conversion / selectivity / intermediates | NR                     | NR                     |

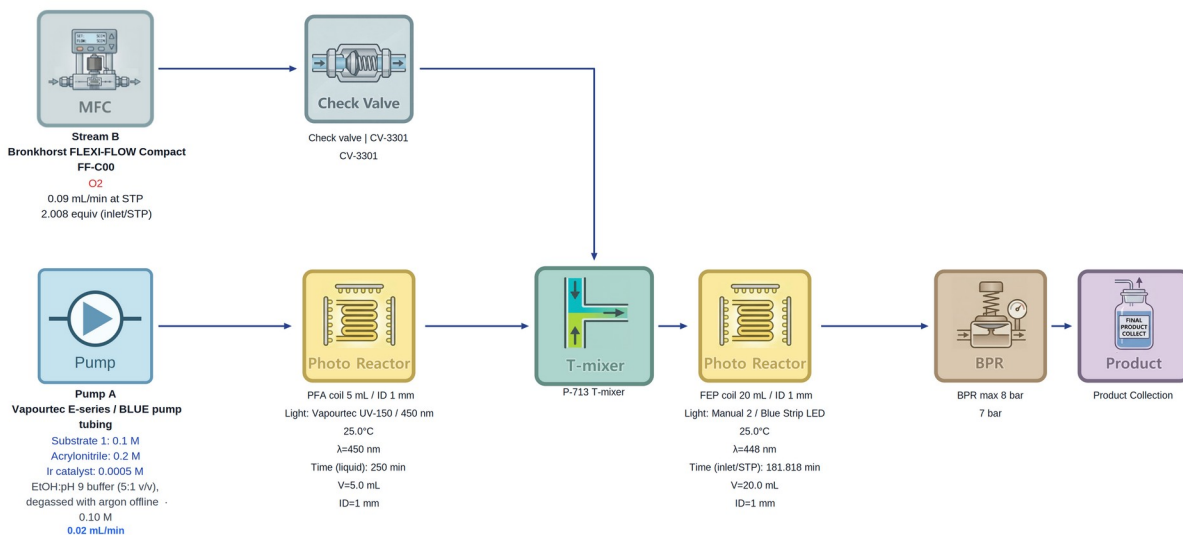
Report the observed operating point rather than copying proposed values when the experiment differs. Identical Set 1 and Set 3 settings do not create an independent repeat unless separate experiments are performed and recorded. Analytical yield, isolated yield, conversion and selectivity must not be interchanged. No wet-lab error bars should be added until the actual replicate structure is known.

**Figure S20. Panel (a): Figure 5, response Set 1.**



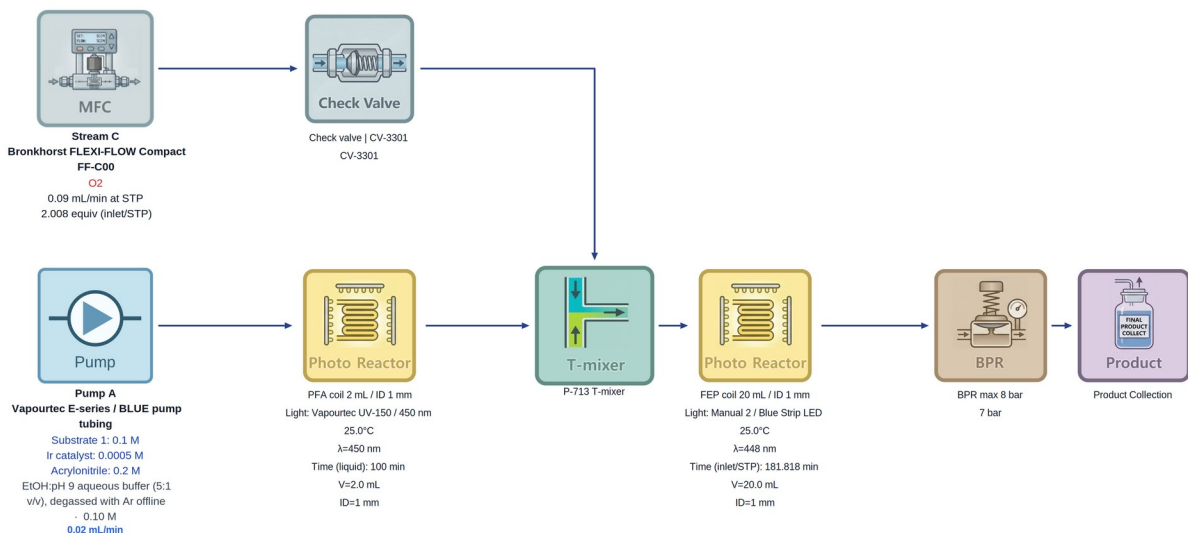
(a) Horizontal redraw of the archived topology for response Set 1. The Stage 2 time label in the archived topology is an inlet/STP index, not measured in-channel contact time. The branch-protection and pressure checks described above remain unresolved. Only the layout has changed; node labels, equipment icons and process connections are retained.

**Figure S20 (continued). Panel (b): Figure 5, response Set 2.**



(b) Horizontal redraw of the archived topology for response Set 2. The Stage 2 time label in the archived topology is an inlet/STP index, not measured in-channel contact time. The branch-protection and pressure checks described above remain unresolved. Only the layout has changed; node labels, equipment icons and process connections are retained.

**Figure S20 (continued). Panel (c): Figure 5, response Set 3.**



(c) Horizontal redraw of the archived topology for response Set 3. The Stage 2 time label in the archived topology is an inlet/STP index, not measured in-channel contact time. The branch-protection and pressure checks described above remain unresolved. Only the layout has changed; node labels, equipment icons and process connections are retained.

## 8.2 Telescoped DPDTC-mediated amidation (main-text Figure 6)

The supplied protocol forms N-benzyl-3-methyl-4-nitrobenzamide through a 2-pyridyl thioester; the chemistry source is Saunders et al. [5]. Batch timing, nominal concentrations and stage order are protocol facts. Flow residence times, the omission of interstage cooling, and the use of a premixed activation feed are proposed choices requiring validation.

### Supplied batch protocol (verbatim collaborator transcription).

A 4.0 mL vial equipped with a PTFE stir bar was charged with 3-methyl-4-nitrobenzoic acid (1 equiv, 0.5 mmol), N,N-dimethylpyridin-4-amine (DMAP) (6.1 mg, 0.1 equiv, 0.05 mmol), and DPDTC (130. mg, 1.05 equiv, 0.525 mmol) under ambient atmosphere without argon or nitrogen purging. 2-MeTHF (1.0 mL, 0.5 M) was added, and the vial was capped and sealed with Teflon tape. The reaction vial was placed in a pre-heated oil bath maintained at 95 °C and stirred at 500–600 rpm for 30 min. The vial was removed from the oil bath and allowed to cool at room temperature for 5–10 min. Once cooled, the cap was removed and benzylamine (1.05 equiv, 0.525 mmol) and 2-MeTHF (0.25 mL, 0.4 M) were quickly added. The vial was capped, sealed with Teflon tape, and returned to the oil bath at 95 °C, and stirred at 500–600 rpm for another 30 min before workup.

**Table S32. Figure 6 response-set provenance and common chemistry constraints.**

| Set | Design emphasis                               | Source / common requirements                                                                                             |
|-----|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| 1   | Integrated performance                        | KHU revised V2, slide 16                                                                                                 |
| 2   | Conversion-focused                            | KHU revised V2, slide 19                                                                                                 |
| 3   | Throughput / processing time                  | KHU revised V2, slide 22                                                                                                 |
| All | Sequential thioester formation then amidation | Amine enters after Stage 1. Inventory v6; component concentration and molar-flow reporting; system-aware pump selection. |

### Fixed-ID responses (verbatim).

#### Response Set 1

Responses to Q-OBJ-001: Please enter here —> Develop a safe, reproducible, and inventory-compatible two-stage telescoped flow process that maximizes the final amide yield while avoiding unnecessarily long residence times. Design the two stages as one connected process, because Stage 1 determines the composition entering Stage 2.

Responses to Q-CHEM-001: Please enter here —> This is a two-stage DPDTC-mediated amide formation through a thioester intermediate. In Stage 1, 3-methyl-4-nitrobenzoic acid is converted to the corresponding 2-pyridyl thioester using DPDTC and DMAP, without isolating the intermediate. In Stage 2, benzylamine is added to the Stage 1 effluent to form N-benzyl-3-methyl-4-nitrobenzamide. The amide C(O)–N bond is formed and the thioester C(O)–S bond is cleaved during Stage 2.

Responses to Q-HYP-001: Please enter here —> Stage 1 conversion will directly affect the amount of thioester available for Stage 2 amidation. The cooling step before benzylamine addition may be a batch-handling step rather than an essential reaction step and should be evaluated in flow. Use the batch temperature and reaction times as reference conditions rather than fixed targets, and determine the operating

conditions based on flow performance. The Stream A/Stream B molar ratio should match the intended stoichiometry, taking both concentration and flow rate into account.

### Response Set 2

Responses to Q-OBJ-001: Please enter here —> Prioritize high conversion through both stages and a high final amide yield. Choose the shortest residence times reasonably justified for this conversion-focused starting design rather than adding a large time excess by default. Keep reactor volumes and flow rates practical for flow screening.

Responses to Q-CHEM-001: Please enter here —> This is a two-stage DPDTC-mediated amide formation through a thioester intermediate. In Stage 1, 3-methyl-4-nitrobenzoic acid is converted to the corresponding 2-pyridyl thioester using DPDTC and DMAP, without isolating the intermediate. In Stage 2, benzylamine is added to the Stage 1 effluent to form N-benzyl-3-methyl-4-nitrobenzamide. The amide C(O)–N bond is formed and the thioester C(O)–S bond is cleaved during Stage 2.

Responses to Q-HYP-001: Please enter here —> Stage 1 requires sufficient conversion to generate enough thioester for Stage 2. Stage 2 requires sufficient reaction time for conversion to the final amide. Use the batch temperature and reaction times as reference conditions rather than fixed targets, and determine the operating conditions based on flow performance. Evaluate the overall process based on the final amide yield rather than either stage alone. The Stream A/Stream B molar ratio should reflect the intended stoichiometry based on both concentration and flow rate.

### Response Set 3

Responses to Q-OBJ-001: Please enter here —> Identify a practical two-stage flow design that balances final amide yield with substrate throughput, total processing time, and stable operation. Prefer a compact, higher-throughput starting design where chemically justified.

Responses to Q-CHEM-001: Please enter here —> This is a two-stage DPDTC-mediated amide formation through a thioester intermediate. In Stage 1, 3-methyl-4-nitrobenzoic acid is converted to the corresponding 2-pyridyl thioester using DPDTC and DMAP, without isolating the intermediate. In Stage 2, benzylamine is added to the Stage 1 effluent to form N-benzyl-3-methyl-4-nitrobenzamide. The amide C(O)–N bond is formed and the thioester C(O)–S bond is cleaved during Stage 2.

Responses to Q-HYP-001: Please enter here —> Stage 1 must maintain sufficient thioester formation as throughput is increased. Direct transfer from Stage 1 to benzylamine addition may avoid the batch cooling step if chemically and operationally acceptable. Use the batch temperature and reaction times as reference conditions rather than fixed targets. Evaluate the overall process based on substrate throughput, stage-specific residence times, and the final amide yield. Shorter residence times may be explored, but should be validated experimentally. The Stream A/Stream B molar ratio should match the intended stoichiometry based on both concentration and flow rate.

Archived numerical source: `outputs/khu_revised_six_20260915/presentation/figure6_set1` through `figure6_set3`. The recorded routing is Claude Opus 4.6 upstream and Claude Sonnet 4.6 downstream, using `scientific_v2` with a candidate budget of 12. The collaborator export dated 20260915\_163845 provides the displayed topologies and component tables. These are preserved existing outputs, not newly generated designs in this document revision. The original model records and finalized JSON, not a later figure redraw, define the case provenance.



**Table S33. Figure 6 stage conditions and component-feed targets. Panel I: final stage values; all coils are ETFE, 2.4 mm ID, operated at a proposed 95 C and 7 bar(g).**

| Set / stage | Volume (mL) | Total liquid Q (mL min <sup>-1</sup> ) | V/Q (min) | Interpretation     |
|-------------|-------------|----------------------------------------|-----------|--------------------|
| 1 / 1       | 5           | 0.16                                   | 31.25     | Activation; Feed A |
| 1 / 2       | 5           | 0.2                                    | 25        | Amidation; A + B   |
| 2 / 1       | 10          | 0.28                                   | 35.714    | Activation; Feed A |
| 2 / 2       | 5           | 0.35                                   | 14.286    | Amidation; A + B   |
| 3 / 1       | 10          | 0.28                                   | 35.714    | Activation; Feed A |
| 3 / 2       | 5           | 0.35                                   | 14.286    | Amidation; A + B   |

**Table S33 (continued). Panel II: nominal component concentrations and molar flows. Feed A contains acid, DPDTC and DMAP; Feed B contains benzylamine. Both use 2-MeTHF. Listed concentrations are preparation targets, not measured reactor concentrations.**

| Set / feed | Component                    | C (M) | Feed Q (mL min <sup>-1</sup> ) | Molar flow (mmol min <sup>-1</sup> ) |
|------------|------------------------------|-------|--------------------------------|--------------------------------------|
| 1 / A      | 3-methyl-4-nitrobenzoic acid | 0.5   | 0.16                           | 0.08                                 |
| 1 / A      | DPDTC                        | 0.525 | 0.16                           | 0.084                                |
| 1 / A      | DMAP                         | 0.05  | 0.16                           | 0.008                                |
| 1 / B      | benzylamine                  | 2.1   | 0.04                           | 0.084                                |
| 2 / A      | 3-methyl-4-nitrobenzoic acid | 0.5   | 0.28                           | 0.14                                 |
| 2 / A      | DPDTC                        | 0.525 | 0.28                           | 0.147                                |
| 2 / A      | DMAP                         | 0.05  | 0.28                           | 0.014                                |
| 2 / B      | benzylamine                  | 2.1   | 0.07                           | 0.147                                |
| 3 / A      | 3-methyl-4-nitrobenzoic acid | 0.5   | 0.28                           | 0.14                                 |
| 3 / A      | DPDTC                        | 0.525 | 0.28                           | 0.147                                |
| 3 / A      | DMAP                         | 0.05  | 0.28                           | 0.014                                |
| 3 / B      | Benzylamine                  | 2.1   | 0.07                           | 0.147                                |

Stoichiometric closure: for Set 1,  $n_{\text{dot}}(\text{acid}) = 0.500 \times 0.160 = 0.0800 \text{ mmol min}^{-1}$  and  $n_{\text{dot}}(\text{benzylamine}) = 2.10 \times 0.0400 = 0.0840 \text{ mmol min}^{-1}$ ; their ratio is 1.05.  $Q_{\text{Stage2}} = 0.160 + 0.0400 = 0.200 \text{ mL min}^{-1}$ , giving  $5/0.200 = 25.00 \text{ min}$ . Sets 2 and 3 preserve the same molar ratio at 0.280 and 0.0700 mL min<sup>-1</sup>. The fully mixed acid-equivalent concentration is 0.400 M in all sets. This is not a measured thioester concentration: at Stage 1 fractional conversion  $X_1$ , the maximum thioester concentration available after mixing is  $0.400 X_1 \text{ M}$  before considering side reactions.

Stock preparation and materials record: prepare and document separate Feed A and Feed B solutions at the target concentrations above, recording the actual final solution volumes, reagent assay/purity, preparation time and any precipitation or discoloration. The batch notation '0.4 M' refers to acid-equivalent

concentration after the total solvent volume reaches 1.25 mL; it is not the concentration of the separate benzylamine stock. The proposed 2.10 M benzylamine stock and premixed activation feed require solubility and stability checks. If acid activation proceeds in the reservoir, the true reaction history extends beyond the stated coil residence time. Do not assume zero premix reaction or quantitative thioester formation. Feed vessels, pump channels and sampling must be documented so that this possible confound can be evaluated.

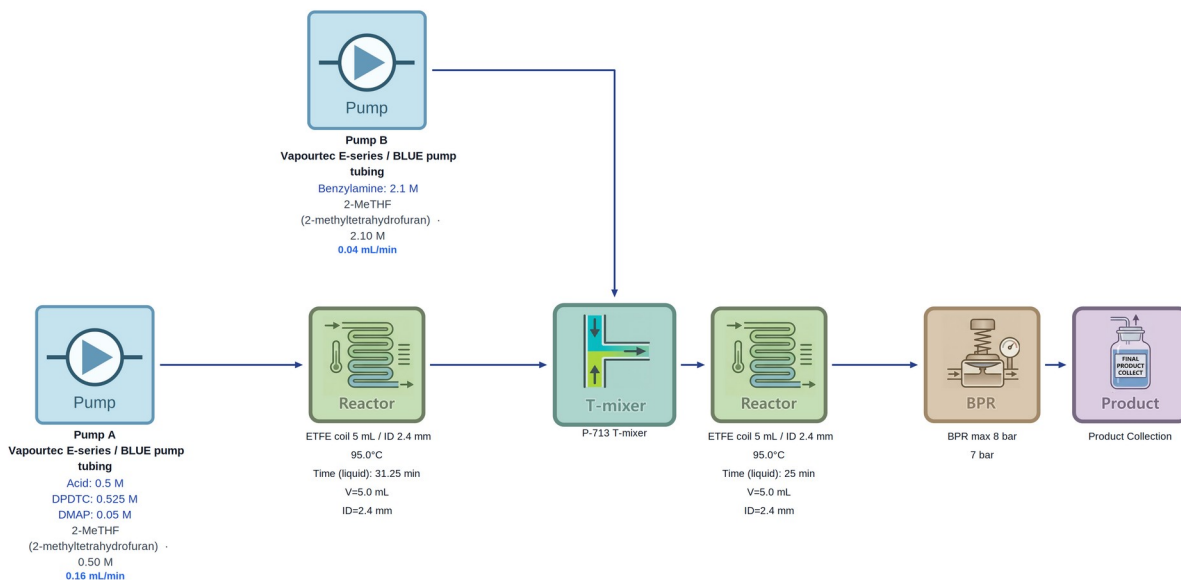
Process interpretation: choose compatible channels of the same Vapourtec E-series system where possible; the diagrams use the general Pump icon rather than implying that every pump is a syringe pump. No irradiation or gas feed is part of this thermal sequence. Downstream volumetric flow includes the benzylamine addition, so the two stage times cannot be calculated from a single unchanged inlet flow. Confirm temperature and pressure limits of every wetted component, operation above the solvent's normal boiling temperature, and whether direct hot interstage transfer is acceptable. Startup waste and offline workup are separate from the two nominal heated-coil residence times. Workup, isolation and analytical methods must be recorded from the actual experiment, not inferred from the topology.

**Table S34. KHU-reported Figure 6 outcomes and actual settings. Sets 2/3 share one entry in the supplied result table. Times are nominal V/Q calculations. NR means not reported; the table does not imply independent experimental replicates.**

| Reported item                              | Set 1                  | Sets 2 and 3 (grouped) |
|--------------------------------------------|------------------------|------------------------|
| NMR yield of 3b (%)                        | 86                     | 68                     |
| Isolated yield / mass                      | 84% / 227.0 mg         | NR                     |
| R1 / R2 volume (mL)                        | 5 / 5                  | 10 / 5                 |
| Feed A / B flow (mL/min)                   | 0.160 / 0.040          | 0.280 / 0.070          |
| R1 / R2 nominal time (min)                 | 31.25 / 25.00          | 35.7143 / 14.2857      |
| Temperature of both reactors               | 95 °C                  | 95 °C                  |
| Recorded system pressure (bar)             | 5.2–5.3                | 5.5–5.8                |
| BPR cartridge rating                       | 7 bar                  | 7 bar                  |
| Acid / DPDTC / DMAP in Feed A (M)          | 0.500 / 0.525 / 0.0500 | 0.500 / 0.525 / 0.0500 |
| Benzylamine in Feed B (M)                  | 2.10                   | 2.10                   |
| NMR internal standard                      | 1,3-Benzodioxole       | 1,3-Benzodioxole       |
| Replicate count / SD / sample window       | NR                     | NR                     |
| Stage 1 conversion / residual intermediate | NR                     | NR                     |

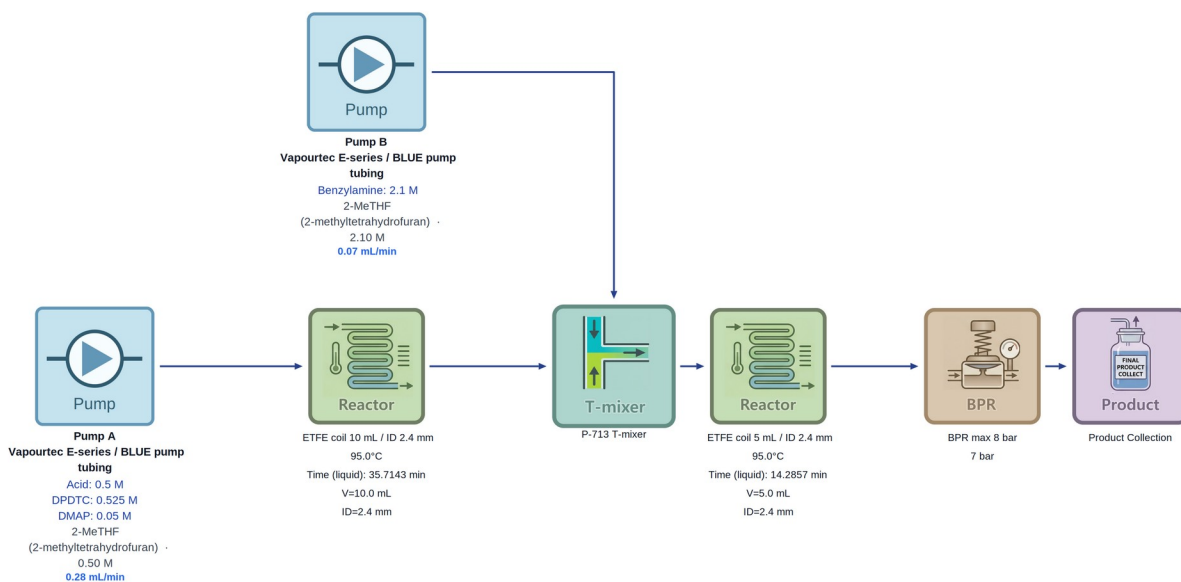
The laboratory record groups Sets 2 and 3 at the same operating point and gives 68% NMR yield for that grouped entry. It does not supply replicate identifiers or a distribution of yields. Set 1 gives 86% NMR and 84% isolated yield. Section 12 reports the supplied procedures and characterization; quantitative NMR calculation records, sample collection windows, and intermediate composition remain to be supplied.

**Figure S21. Panel (a): Figure 6, response Set 1.**



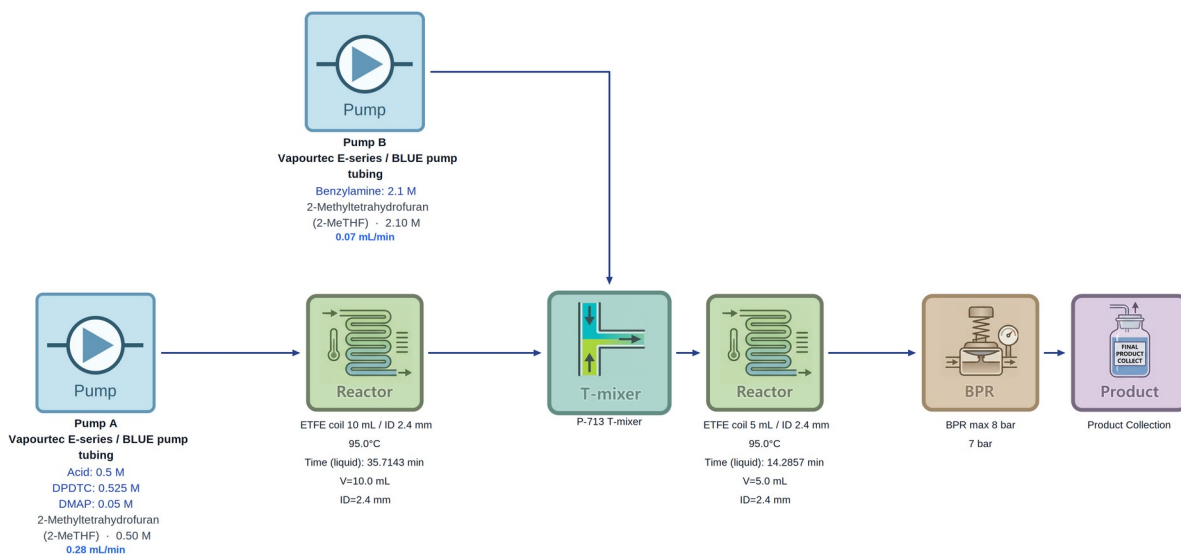
(a) Horizontal redraw of the archived topology for response Set 1. Stage residence times use the liquid flow entering each reactor; Stage 2 includes both liquid feeds. The diagrams are archived design proposals, not measured conversions or pressure validation. Only the layout has changed; node labels, equipment icons and process connections are retained.

**Figure S21 (continued). Panel (b): Figure 6, response Set 2.**



(b) Horizontal redraw of the archived topology for response Set 2. Stage residence times use the liquid flow entering each reactor; Stage 2 includes both liquid feeds. The diagrams are archived design proposals, not measured conversions or pressure validation. Only the layout has changed; node labels, equipment icons and process connections are retained.

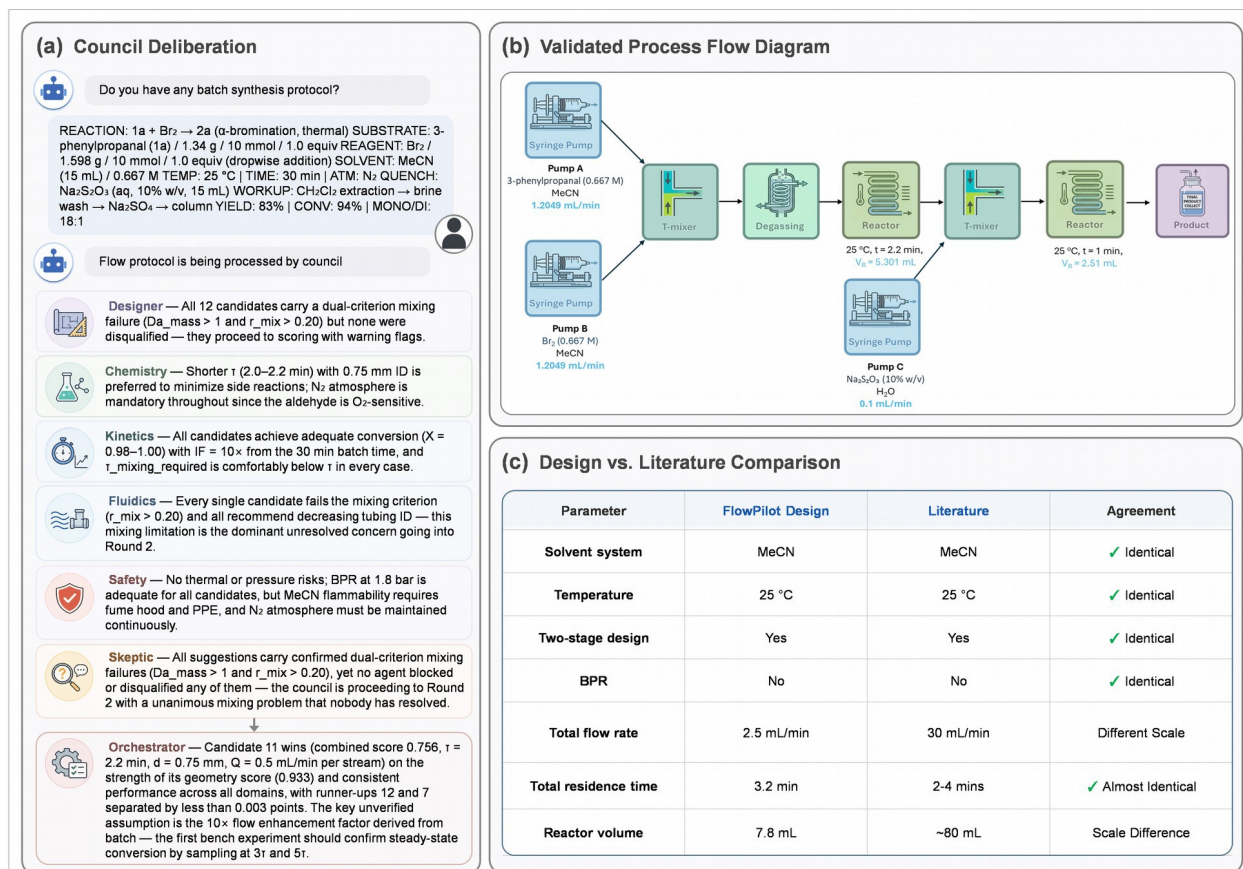
**Figure S21 (continued). Panel (c): Figure 6, response Set 3.**



(c) Horizontal redraw of the archived topology for response Set 3. Stage residence times use the liquid flow entering each reactor; Stage 2 includes both liquid feeds. The diagrams are archived design proposals, not measured conversions or pressure validation. Only the layout has changed; node labels, equipment icons and process connections are retained.

## 9 Archived preprint alpha-bromination case

**Figure S22** reproduces the original preprint **Figure 6** artwork without modification, including its original internal labels and values. This earlier aldehyde-bromination example, based on Lu et al. [6], is retained as a historical computational illustration and is not the DPDTC amidation now shown in main-text **Figure 6**. The original artwork records unresolved mixing/design concerns; its model confidence, timings and predicted outcomes are not new measured results or an approved procedure. Reproduction here must not be taken to validate those archived statements.



**Figure S22.** Unmodified **Figure 6** from the supplied FlowPilot preprint (preprint.pdf, page 14): historical alpha-bromination case and the associated model discussion and process comparison. The complete original artwork is preserved, not redrawn. Its chemistry source is Lu et al. [6]. Any predictions or unresolved qualifications within the image belong to that archived output and do not describe the new wet-lab amidation case. Source artwork extraction and image hashes are retained with this revision.

## 10 Major pipeline evolution and version boundaries

**Table S35** records major workflow changes using repository snapshots, dated reports and archived runs. Dates identify available evidence, not independently verified release or deployment dates; no release tags were found. The purpose is to explain why the workflow evolved, not to count minor bug fixes or claim that each change increased chemical yield. Earlier benchmark numbers remain attached to their frozen campaigns. In particular, a recorded August 20 campaign used two candidates, whereas the later scientific-policy case studies used twelve; later intake, inventory and backflow behavior must not be retroactively attributed to the earlier benchmark.

**Table S35. Major FlowPilot development milestones and their rationale. Snapshot identifiers and dated artifacts support chronology; they are not controlled before/after efficacy experiments.**

| Date / evidence                                      | Major addition                                                                 | Reason / interpretation                                                                        |
|------------------------------------------------------|--------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| 2026-03-31; snapshot 213ad6b3                        | Batch parsing, chemistry planning, literature retrieval and Streamlit diagrams | Ground batch-to-flow proposals in chemistry and literature.                                    |
| 2026-04-15 to 04-22; f08f178a, 491d436c, 3ecf7e44    | Deterministic engineering calculator and specialist council                    | Separate dependent numerical calculations from model judgment; record conflicting assessments. |
| 2026-04-24 onward; e596a073 and April 27 runs        | Model-pair and candidate-budget benchmark harness                              | Persist model calls and compare council configurations, costs and candidate diversity.         |
| 2026-06-26 release note; June 29 snapshot 02a03f89   | Closed-loop experiment refinement and experiment-loop interface                | Use measured outcomes to revise unsupported design assumptions.                                |
| 2026-06-29 run; June 30 snapshot 1d90c256            | Fixed-ID intake and frozen DesignInputPackage                                  | Record objectives, evidence, equipment and hypotheses before design.                           |
| 2026-07-06; snapshot 98d92241                        | Stronger inventory enforcement and gas/liquid reconciliation                   | Retain hardware limits, specified equivalents and numerical closure after changes.             |
| 2026-07-28 runs; July 30 snapshot 0530af8e           | Architecture-ablation study with saved cells                                   | Separate contributions of retrieval, engineering, council and inventory.                       |
| 2026-08-07 and 08-19 audits; ec9a48a7                | Canonical final-design contract                                                | Prevent disagreement among selected conditions, inventory realization, GUI and topology.       |
| 2026-08-19 to 08-25 campaigns; ec9a48a7, 6e49d9ac    | Repeated fixed-criterion architecture comparison                               | Measure judge scores, critical flags, variability and resource use under shared inputs.        |
| 2026-09-04; a72e7c1f                                 | React/FastAPI GUI and inlet/STP reporting                                      | Expose consistent stage-resolved outputs and reproducible intake behavior.                     |
| 2026-09-07 documents; September 14 snapshot 7a130a40 | Scientific design policy and inventory-bound candidate screening               | Keep stage order, objective, uncertain kinetics and selected conditions explicit.              |



| Date / evidence                                                                        | Major addition                                                                    | Reason / interpretation                                                                                       |
|----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| 2026-09-14 to 09-18 laboratory-inventory revisions; 7a130a40, 7c3998e7; v6/v7 profiles | System-aware pump/module compatibility, feed stoichiometry and setting increments | Translate collaborator clarifications into implementable equipment combinations and explicit component flows. |
| 2026-09-18 trial records; additions beyond b48a5929                                    | Branch-specific backflow checks and experimental transient-pressure tools         | Expose unprotected reverse paths and conditional startup risks; not a calibrated failure-probability model.   |

Evidence anchors: outputs/FlowPilot\_latest\_release\_2026-06-26.md;  
 outputs/thq\_intake\_claude\_gpt4o\_20260629/run.log;  
 outputs/gui\_visual\_audit\_20260807/final\_contract\_overhaul/validation\_summary.json;  
 outputs/flowpilot\_systemic\_fix\_20260819/SYSTEMIC\_FIX\_REPORT.md;  
 deliverables/manuscript\_benchmark\_visualizations\_20260825/METHODS\_AND\_SCOPE.md;  
 outputs/validation/20260904\_standardized\_intake\_stp\_report.md; docs/flowpilot\_2\_development.md;  
 inventory\_khu/KHU\_inventory\_20260915\_v6.json;  
 outputs/council\_backflow\_ab\_20260918\_111811/REPORT.md; and docs/council\_transient\_physics.md.  
 The transient-pressure extension was an experimental addition beyond snapshot b48a5929, using assumed dynamics without laboratory pressure traces.

A later rerun may evaluate a revised workflow, but it does not replace or improve a previously recorded benchmark result without a separately identified campaign. Computational replay, mocked integration checks, repeated model generations, and independent laboratory repeats are different evidence types and must remain separately labeled.

## 11 Related flow-chemistry and process-design agents

This section compares FlowPilot with six studies selected for their direct relevance to flow-chemistry or chemical-process agents. **Table S36** records the task, delivered output, quantitative engineering scope, equipment context and type of validation reported in the checked sources [7-12]. Journal articles, a conference paper and preprints are identified separately. The comparison is descriptive: it does not assign performance scores, infer absent capabilities from silence, or equate different benchmarks. Source versions were checked on 28 September 2026.

**Table S36.** Task-level comparison of FlowPilot with directly related flow-chemistry and chemical-process agents. Panel I: inputs, delivered outputs and reported validation. Bracketed numbers refer to the ESI reference list.

| System / source                                   | Input and purpose                                                              | Delivered output                                                                      | Reported evidence                                                                                        |
|---------------------------------------------------|--------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Chat-microreactor<br>Pan et al., 2025 [7]         | Literature-derived flow information and a microreactor design query.           | Extracted parameters, predicted flow patterns and knowledge-enhanced design guidance. | Flow-pattern prediction and a piperacillin design application.                                           |
| LLM-RDF<br>Ruan et al., 2024 [8]                  | Synthesis-development tasks and experimental-platform interactions.            | Procedures, equipment instructions, analyses, optimization and scale-up guidance.     | Aerobic oxidation development plus kinetics, photochemical optimization and reactor-design applications. |
| SapoMind<br>Wang et al., 2026 [9]                 | Lanolin saponification requirements and an experimental optimization workflow. | Recommended conditions, experimental designs and automated microreactor experiments.  | Continuous-flow saponification experiments, product-quality checks and process-greenness assessment.     |
| Text-to-Simulation<br>Tian et al., 2026 [10]      | Natural-language process specifications.                                       | Process topology, configured parameters and executable simulation flowsheets.         | Simona computational evaluation, including simulation convergence and design time.                       |
| CeProAgents<br>Yang et al., 2026 [11]<br>Preprint | Engineering documents, process diagrams and simulation parameters.             | Knowledge extraction, flowsheet interpretation/design and optimized parameters.       | CeProBench knowledge, concept and parameter tasks, including Aspen Plus optimization.                    |
| CAAF<br>Zhang, 2026 [12]<br>Preprint              | Explicit engineering requirements and formalized domain constraints.           | Checked candidates, constraint-conflict diagnostics and controlled revision.          | Constructed engineering benchmarks, including a pharmaceutical flow-reactor constraint problem.          |

| System / source            | Input and purpose                                                                        | Delivered output                                                                                           | Reported evidence                                                                                                                                                               |
|----------------------------|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FlowPilot<br>Present study | Batch protocol, fixed-ID follow-up answers, objective, history and laboratory inventory. | Connected topology, feed compositions/molar flows, stage conditions, equipment assignments and provenance. | Matched architecture benchmarks, archived run audits, <b>and</b> two connected-stage laboratory cases. Reported isolated yields: sulfoxide 4a, 94%; amide 3b, 84% (Section 12). |

**Table S36** (continued). Panel II: engineering and equipment scope, and the appropriate comparison with FlowPilot. These are scope distinctions, not proof that a capability is exclusive to either system.

| System / source                                   | Quantitative engineering                                                             | Equipment / process scope                                                            | Interpretation for comparison                                                                                                                                                                    |
|---------------------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Chat-microreactor<br>Pan et al., 2025 [7]         | Flow-property extraction and ensemble-learning flow-pattern prediction.              | Microreactor and multiphase-flow design context.                                     | The emphasis is local hydrodynamic prediction and design; FlowPilot reconciles a connected stream-and-stage proposal against a supplied inventory.                                               |
| LLM-RDF<br>Ruan et al., 2024 [8]                  | Stoichiometric calculations, kinetic analysis and reactor scale-up reasoning.        | Automated synthesis and analysis hardware; photoelectrochemical reactor design.      | Experimental development and reactor engineering overlap substantially. The relevant distinction is the specified batch-to-flow output contract and coupled stage/equipment records.             |
| SapoMind<br>Wang et al., 2026 [9]                 | Operating-condition optimization across quality and resource-use objectives.         | A continuous-flow microreactor for the target saponification chemistry.              | Reaction-specific optimization on an implemented platform complements initial connected-process construction under a laboratory inventory. Both need measured outcomes.                          |
| Text-to-Simulation<br>Tian et al., 2026 [10]      | Topology generation, parameter configuration, simulator feedback and search.         | Unit operations represented in process-simulation software.                          | Simulator executability and physical laboratory implementation are distinct. FlowPilot targets reaction-stage feeds and inventory-bound configurations rather than replacing process simulation. |
| CeProAgents<br>Yang et al., 2026 [11]<br>Preprint | Specialist cohorts combine discussion with structured workflows and simulator tools. | Industrial process diagrams and simulation configurations.                           | Process-development coordination is shared prior art. The comparison concerns task scale, inputs and delivered records, not agent coordination alone.                                            |
| CAAF<br>Zhang, 2026 [12]<br>Preprint              | Machine-readable invariants enforced by a deterministic assertion interface.         | Modeled reactor operating limits in the benchmark; not a wet-lab flow demonstration. | Deterministic validation is shared methodology. The distinction concerns the reaction-to-configuration workflow and laboratory evidence, not assertion checking itself.                          |

| System / source            | Quantitative engineering                                                                                                                    | Equipment / process scope                                                          | Interpretation for comparison                                                                                                                                                                    |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FlowPilot<br>Present study | Dependent quantities are recalculated and checked against declared conventions and limits; candidate assumptions undergo specialist review. | Selected laboratory items and coupled stream/stage assignments in a common record. | Numerical closure, physical implementability after review, and measured chemistry are separate outcomes. The record is an inspectable proposal, not an autonomous execution or safety guarantee. |

### 11.1 Interpretation and experimental boundary

A common input-output description does not establish identical capabilities. FlowPilot's proposed contribution is the reconciliation of reaction sequencing, feed composition, molar flow, stage-specific calculations and available equipment in a shared record. The scope comparison cannot establish priority for every feature or replace a controlled comparison against the cited systems. The architecture benchmark in Section 3 compares the recorded one-shot and FlowPilot configurations only; it should not be read as a ranking of these six external methods.

Laboratory implementation and chemical response are reported separately in Section 12. **Figure 5** concerns an oxygen-free/oxygen-fed photochemical sequence; **Figure 6** concerns telescoped activation and amidation. **Tables S31 and S34** summarize the measured outcomes, while **Tables S38–S39** reconcile proposals with the supplied operating records and feed calculations. These laboratory observations do not convert the literature comparison into a controlled head-to-head test of the six external systems.

## 12 Laboratory implementation, experimental procedures, and analytical data

This section documents the KHU implementation of the photoredox Giese addition/oxidation and DPDTC-mediated amidation cases. **Tables S31 and S34** report the measured outcomes, and **Tables S38–S39** connect the implemented equipment and nominal feeds to the design proposals. The photographs, spectra, experimental procedures, and grouped response-set entries are retained from the laboratory record. Grouped sets are not treated as independent measurements.

### 12.1 General materials, instrumentation, and analytical methods

All reagents were purchased from commercial suppliers (Acros Organics, Alfa Aesar, Tokyo Chemical Industry, and Sigma-Aldrich) and used without further purification unless otherwise noted. Selected compounds were purified by flash column chromatography on silica gel 60 (0.040–0.063 mm) prior to use. Analytical thin-layer chromatography (TLC) was performed on silica gel 60 F254 glass plates (Merck/Supelco, 20 × 20 cm). Visualization was achieved by UV irradiation (254 or 365 nm) or staining (anisaldehyde, ninhydrin, or KMnO<sub>4</sub>) followed by heating when appropriate.

Flow reactions were performed using a Vapourtec E-series system (part no. 50-1319) equipped with a Blue V3 pump tube (part no. 50-1301). For the photochemical reaction (**Figure 5**), the system was fitted with an organometallic reagent kit (part no. 50-1311). Reactor 1 was a UV–150 photochemical reactor equipped with an LED power supply (part no. 50-1453) and irradiated with a 450 nm LED (part no. 50-1448). The effluent then passed through a second coil reactor irradiated with a blue LED strip. Reactor 1 was maintained at 25 °C using an E-Series cooling module (part no. 50-1314), while reactor 2 was maintained at 25 °C using a water bath and fan cooling. O<sub>2</sub> gas was delivered via a Bronkhorst FLEXI-FLOW compact mass flow controller (FF-C00). For the amide coupling reaction (**Figure 6**), both reactor coils were held at 95 °C using a water bath. Both setups were fitted with a 7 bar cartridge back-pressure regulator.

All isolated products were characterized by <sup>1</sup>H and <sup>13</sup>C NMR spectroscopy and LRMS. Known products were identified by comparison with published NMR data. Copies of all <sup>1</sup>H, <sup>13</sup>C NMR spectra are provided in the Supporting Information. NMR spectra were recorded at ambient temperature on a JEOL JNM-ECZ500R FT-NMR spectrometer (<sup>1</sup>H, 500 MHz; <sup>13</sup>C{<sup>1</sup>H}, 126 MHz). Chemical shifts are reported relative to residual solvent peaks (CDCl<sub>3</sub> δ = 7.26 ppm for <sup>1</sup>H, 77.16 ppm for <sup>13</sup>C{<sup>1</sup>H}; DMSO-d<sub>6</sub>, δ = 2.50 ppm for <sup>1</sup>H, 39.52 ppm for <sup>13</sup>C{<sup>1</sup>H}). All <sup>13</sup>C NMR spectra were broadband <sup>1</sup>H-decoupled. Coupling constants (J) are given in Hertz (Hz). Signal multiplicities: br (broad), s (singlet), d (doublet), t (triplet), q (quartet), p (quintet), m (multiplet). Low-resolution mass spectra (LRMS) were recorded on an Advion expressionS CMS spectrometer using electrospray ionization (ESI) in positive-ion mode.

The 7 bar values identify the installed BPR cartridge, not the measured system pressure. **Tables S31 and S34** report the laboratory pressure ranges separately. The pressure reference (gauge or absolute), sensor position, and calibration were not specified. The reported 95 °C water-bath arrangement is retained pending laboratory confirmation.

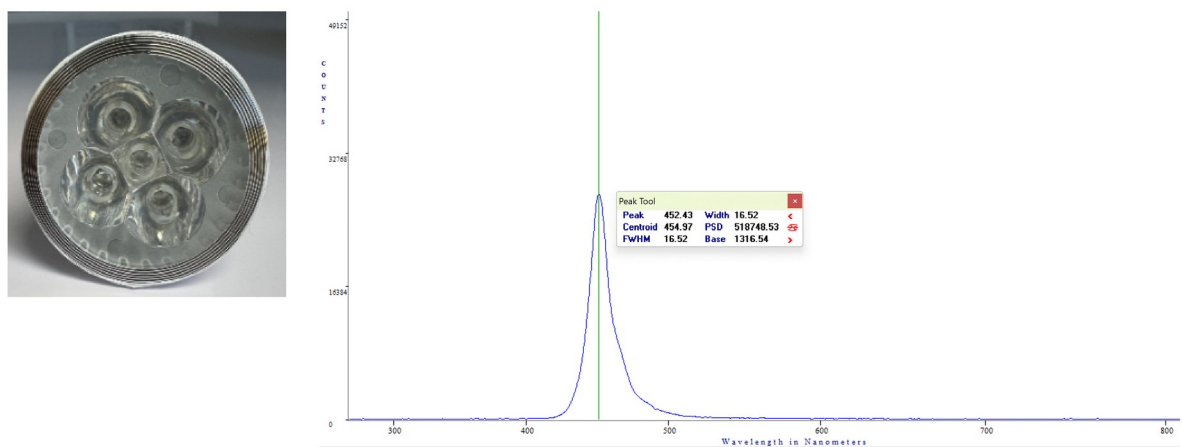
## 12.2 Irradiation sources and recorded specifications

The batch irradiation used two MR16 5 W blue LED spotlights. Flow Stage 1 used the Vapourtec 450 nm LED in the UV-150 module, and Stage 2 used a separate blue LED strip. **Figure S23** reproduces the supplied photographs and emission spectra. **Table S37** distinguishes electrical input from the radiant-power specification; these values are not photon fluxes at the reacting solution. No actinometric calibration or position-dependent irradiance measurement was supplied.

**Table S37.** Irradiation-source specifications transcribed from the KHU supplement. Peak wavelengths are those reported in the supplied specification/spectrum images, not an independently repeated calibration.

| Light source / use                | Electrical specification | Reported radiant power | Peak wavelength |
|-----------------------------------|--------------------------|------------------------|-----------------|
| MR16 spotlight / batch, two lamps | 5 W per lamp; 12 V       | Not reported           | 452.4 nm        |
| Vapourtec LED / flow R1           | 60 W input               | 24 W                   | 450.32 nm       |
| Blue strip / flow R2              | 10 W; 24 V               | Not reported           | 447.50 nm       |

**a Batch: two MR16 5 W lamps**

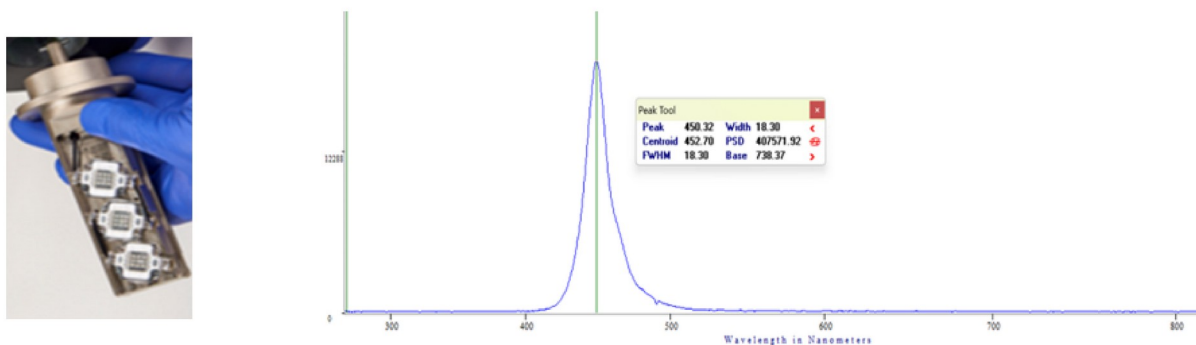


Original KHU lamp photograph and emission-spectrum image; specifications in Table S37.

**Figure S23a.** KHU photograph and emission spectrum of the MR16 5 W batch lamp. Artwork and spectral traces are reproduced from the supplied experimental supplement. **Table S37** gives the corresponding specifications; lamp wattage is not assumed to equal incident optical power at the reaction mixture.



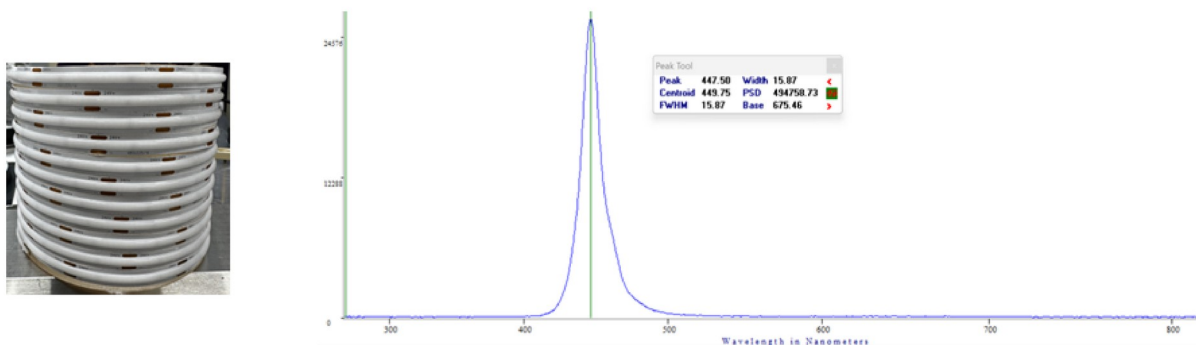
**b Flow R1: Vapourtec 450 nm LED**



Original KHU lamp photograph and emission-spectrum image; specifications in Table S37.

**Figure S23b.** KHU photograph and emission spectrum of the Vapourtec flow-stage-1 LED. Artwork and spectral traces are reproduced from the supplied experimental supplement. **Table S37** gives the corresponding specifications; lamp wattage is not assumed to equal incident optical power at the reaction mixture.

**c Flow R2: blue LED strip**



Original KHU lamp photograph and emission-spectrum image; specifications in Table S37.

**Figure S23c.** KHU photograph and emission spectrum of the blue strip LED for flow-stage-2 irradiation. Artwork and spectral traces are reproduced from the supplied experimental supplement. **Table S37** gives the corresponding specifications; lamp wattage is not assumed to equal incident optical power at the reaction mixture.

### 12.3 Photoredox Giese addition and oxidation: preparation and operation

The transformation converts (((4-methoxyphenyl)thio)methyl)trimethylsilane (1a) and acrylonitrile (2a) to sulfide 3a and then sulfoxide 4a, following the chemistry of Park et al. [4]. **Figure S24** reproduces KHU's supplied reaction/setup comparison and **Figure S25** shows the actual apparatus. The compared literature values belong to the cited publication, not to new FlowPilot experiments.

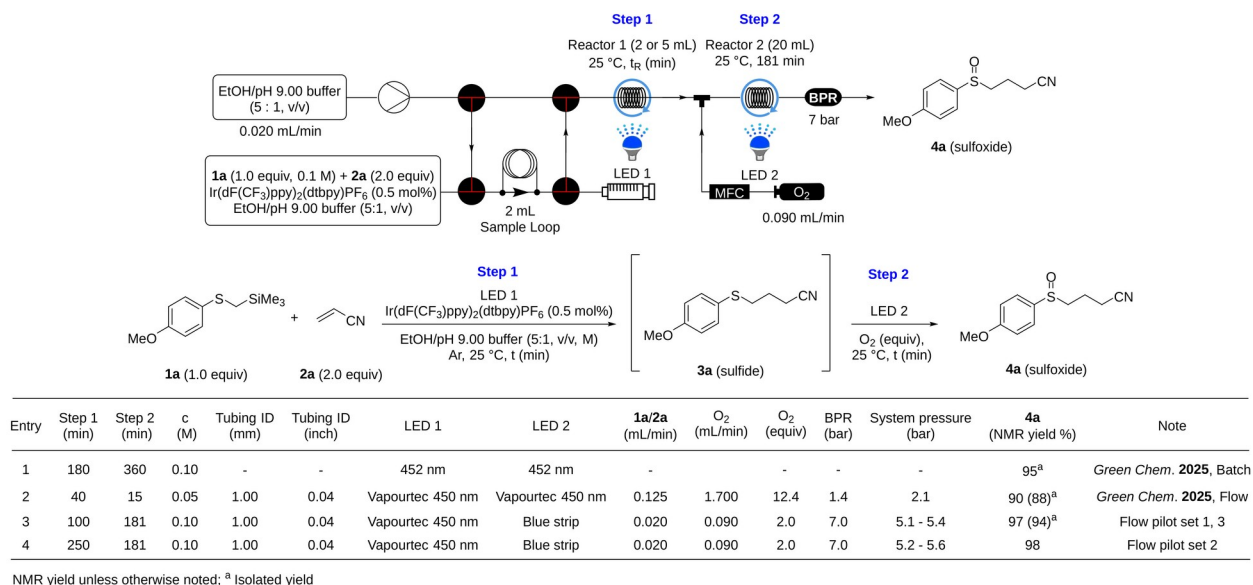
A Vapourtec E-Series flow system (part no. 50-1319) equipped with a Blue V3 pump tube (part no. 50-1301) was used. Reactor 1 was a UV-150 photochemical reactor fitted with a 2 or 5 mL coil of PFA tubing (0.04 in. i.d.) and irradiated with a 450 nm LED (part no. 50-1448; rated electrical input, 60 W). The reactor temperature was maintained at  $25 \pm 1$  °C. Reactor 2 comprised a 20 mL coil of FEP tubing (0.04 in. i.d.) maintained at  $25 \pm 1$  °C using a water bath and fan cooling and irradiated with a blue LED strip.

A solution of 1a (0.2 mmol, 1.0 equiv), 2a (0.4 mmol, 2.0 equiv), and Ir(dF(CF<sub>3</sub>)ppy)<sub>2</sub>(dtbpy)PF<sub>6</sub> (0.001 mmol, 0.5 mol%) in EtOH/pH 9.00 buffer (bicarbonate-based buffer, 5:1, v/v; 1a, 0.10 M) was loaded into a 2.0 mL sample loop and injected into a carrier stream of the same solvent mixture. The liquid flow rate was 0.020 mL min<sup>-1</sup>. Oxygen was introduced through a T-mixer between the two reactors using a mass-flow controller set to 0.090 mL min<sup>-1</sup>. A 7 bar cartridge back-pressure regulator was installed downstream of reactor 2, and the recorded system pressure was 5.1–5.6 bar.

The Stage 1 liquid feed therefore contains nominal concentrations of 0.100 M 1a, 0.200 M acrylonitrile, and 0.000500 M iridium photocatalyst. The 2.0 mL sample loop contains 0.200 mmol 1a and corresponds to a nominal injection duration of 100 min at 0.020 mL min<sup>-1</sup>. This is a finite sample-loop experiment in a continuously flowing carrier, not evidence of indefinitely sustained reagent-fed operation. Exact leading/trailing collection windows and the degassing/argon-handling procedure used during the implemented runs were not specified in the new methods. The frozen design input retains an oxygen-free, argon-prepared first stage; oxygen is introduced only at the interstage T-mixer.

At the stated settings, nominal Stage 1 times are  $2/0.020 = 100$  min and  $5/0.020 = 250$  min. The supplied artwork labels Stage 2 as “181 min”; recalculation from the stated settings gives the nominal inlet-reference index  $20/(0.020 + 0.090) = 181.82$  min. On the proposal's STP convention (273.15 K, 1 atm), 0.090 mL min<sup>-1</sup> of pure O<sub>2</sub> corresponds to 0.004015 mmol min<sup>-1</sup>, or 2.008 equivalents relative to 0.00200 mmol min<sup>-1</sup> of 1a. These are calculated proposal-basis values; the actual MFC reference temperature/pressure must be confirmed. Gas compression, dissolution, consumption, and phase holdup prevent this index from being identified with physical gas or liquid residence time. No tracer or residence-time-distribution measurement was supplied.

An aliquot of the collected reactor effluent was analyzed by <sup>1</sup>H NMR spectroscopy using 1,3-benzodioxole as an internal standard to determine the NMR yield of 4a. The remaining effluent was concentrated under reduced pressure and purified by flash column chromatography on silica gel. KHU reports 97% NMR yield and 94% isolated yield for the grouped Sets 1/3 entry (2 mL R1), and 98% NMR yield for Set 2 (5 mL R1). The recorded system pressures are 5.1–5.4 and 5.2–5.6 bar, respectively. Replicate counts and quantitative-NMR integration/calibration records were not supplied.



**Figure S24.** Original KHU experimental schematic, chemical sequence, and batch/flow comparison for sulfoxide 4a. Source references are Park et al. [4] for the first two rows and the supplied KHU measurements for the FlowPilot rows. The source labels R2 as “181 min”; the stated settings give 181.82 min on the inlet-reference basis, not operating-pressure residence time. Its 0.04 in tubing ID equals 1.016 mm; the displayed 1.00 mm is a nominal label. Literature batch timing (180/360 min) differs from the frozen design input (240/360 min). Source artwork is preserved to keep these distinctions auditable; it is not silently normalized or represented as new batch data.



**Figure S25.** Actual KHU photochemical flow apparatus, as supplied. The labeled photograph identifies the E-series system, UV-150 reactor, downstream coil, MFC, T-mixer, check valve, BPR, cooling module, carrier solvent, and sample loop. The image is not a pressure calibration or a record of startup dynamics; exact check-valve location/orientation and a pressure trace remain to be confirmed.

## 12.4 DPDTC-mediated amidation: preparation and operation

The sequence converts 3-methyl-4-nitrobenzoic acid (1b) to its 2-pyridyl thioester and then to N-benzyl-3-methyl-4-nitrobenzamide (3b), following the chemistry of Saunders et al. [5]. The intermediate is not isolated. **Figure S26** preserves the supplied scheme and reference comparison; **Figure S27** shows the actual apparatus.

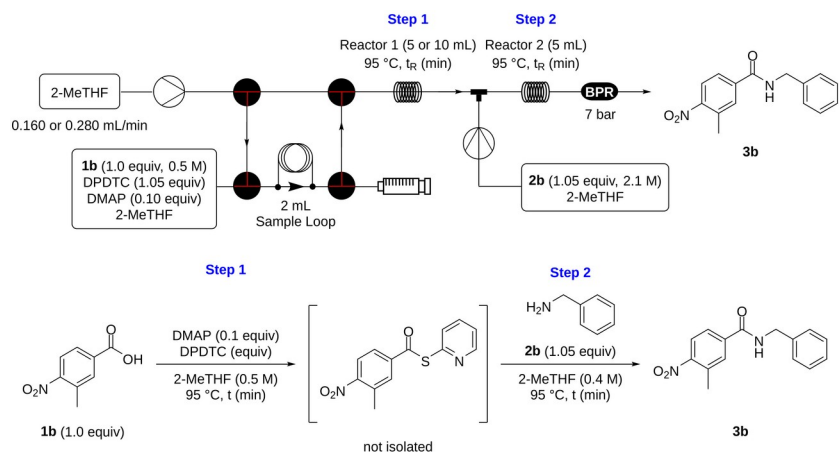
A Vapourtec E-Series flow system (part no. 50-1319) equipped with a Blue V3 pump tube (part no. 50-1301) was used with two pump channels. Reactor 1 comprised a 5 or 10 mL coil of ETFE tubing (0.093 in. i.d.), and reactor 2 comprised a 5 mL coil of ETFE tubing (0.093 in. i.d.). Both reactors were maintained at 95 °C in a water bath.

A solution of 1b (1.0 mmol, 1.0 equiv), DPDTC (1.05 mmol, 1.05 equiv), and DMAP (0.1 mmol, 0.10 equiv) in 2-MeTHF (1b, 0.50 M) was loaded into a 2.0 mL sample loop and injected into a carrier stream of 2-MeTHF. The liquid flow rate through reactor 1 was 0.160 or 0.280 mL min<sup>-1</sup>. At a T-mixer downstream of reactor 1, this stream was combined with a solution of 2b (1.05 mmol, 1.05 equiv) in 2-MeTHF (2b, 2.10 M), delivered through the second pump channel at 0.040 or 0.070 mL min<sup>-1</sup>, respectively. The combined stream then passed through reactor 2. A 7 bar cartridge back-pressure regulator was installed downstream of reactor 2, and the recorded system pressure was 5.2–5.8 bar.

Nominal stock concentrations in Feed A are 0.500 M acid, 0.525 M DPDTC, and 0.0500 M DMAP. A 2.0 mL loop contains 1.00 mmol acid and is displaced in 12.50 or 7.14 min at the two stated flow rates. The benzylamine stock has a nominal concentration of 2.10 M; the specified 1.05 mmol quantity corresponds to 0.500 mL of that stock. The required total feed volume for priming, overlap with the dispersed sample plug, and collection cannot be reconstructed from this nominal reagent amount alone. The new methods do not specify the second-feed start/stop timing or collection window, which are important for interpreting finite-loop stoichiometry.

At Set 1 flows, the acid and benzylamine nominal molar feeds are 0.0800 and 0.0840 mmol min<sup>-1</sup>. At Sets 2/3 flows, they are 0.140 and 0.147 mmol min<sup>-1</sup>. Both give 1.05 equivalents of benzylamine. The combined stream has acid-equivalent concentration 0.400 M and amine-feed-equivalent concentration 0.420 M before allowing for reaction; these are mixing calculations, not measured thioester or product concentrations. R1 times are 31.25 and 35.7143 min; R2 times are 25.00 and 14.2857 min. Actual intermediate conversion was not measured in the supplied data.

An aliquot of the collected effluent was analyzed by <sup>1</sup>H NMR spectroscopy using 1,3-benzodioxole as an internal standard. The remaining effluent was concentrated by rotary evaporation and purified by flash column chromatography on silica gel. Set 1 gave 86% NMR yield and 84% isolated yield (227.0 mg); Sets 2/3 share a reported NMR yield of 68%. The recorded system pressures were 5.2–5.3 and 5.5–5.8 bar, respectively. Both coils were described as operating at 95 °C without an intervening cooling operation. Individual-run identifiers and repeat statistics were not supplied.



| Entry | DPDTC (equiv) | Step 1 (min) | Step 2 (min) | Tubing ID (mm) | Tubing ID (inch) | 1b (mL/min) | 2b (mL/min) | BPR (bar) | System pressure (bar) | 3b (NMR yield %)        | Note                                   |
|-------|---------------|--------------|--------------|----------------|------------------|-------------|-------------|-----------|-----------------------|-------------------------|----------------------------------------|
| 1     | 1.10          | 30           | 30           | -              | -                | -           | -           | -         | -                     | 98 (quant) <sup>a</sup> | ACS sustainable Chem. Eng. 2025, Batch |
| 2     | 1.05          | 30           | 30           | 0.75           | 0.03             | 0.043       | 0.020       | 4.0       | -                     | 97 <sup>a</sup>         | ACS sustainable Chem. Eng. 2025, Flow  |
| 3     | 1.05          | 31.3         | 25           | 2.40           | 0.093            | 0.160       | 0.040       | 7.0       | 5.2 - 5.3             | 86 (84) <sup>a</sup>    | Flow pilot set 1                       |
| 4     | 1.05          | 35.7         | 14.3         | 2.40           | 0.093            | 0.280       | 0.070       | 7.0       | 5.5 - 5.8             | 68                      | Flow pilot set 2, 3                    |

NMR yield unless otherwise noted; <sup>a</sup> Isolated yield

**Figure S26.** Original KHU reaction/setup schematic and batch/flow comparison for amide 3b. The first two rows reproduce the cited Saunders et al. [5] comparison; the FlowPilot rows report new KHU measurements. The source's 0.093 in ID equals 2.3622 mm, whereas 2.40 mm is a nominal inventory label. The literature batch comparison lists 1.10 equivalents of DPDTC; the supplied design input and implemented FlowPilot conditions use 1.05 equivalents. These distinct conditions are retained rather than treated as a matched yield or speed-up comparison. Parenthesized yields in the source are isolated yields; “quant” is the source's qualitative quantitative-yield label.



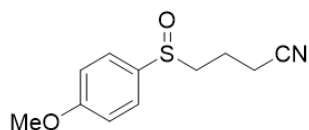


**Figure S27.** Actual KHU two-stage amidation apparatus, as supplied. Two E-series pump channels deliver the sample-loop/carrier stream and the benzylamine stream. The photograph identifies the sample loop, T-mixer, two reactor coils, BPR, and solution containers. It documents the assembled apparatus, not a complete time-resolved collection or calibration record.

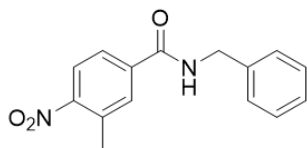


## 12.5 Product characterization and supplied spectra

The characterization below is transcribed from KHU's supplied records with punctuation and section-reference corrections only. Reported masses, yields, shifts, couplings, and observed LRMS values are retained. **Figures S28–S31** are product-characterization spectra; they are not presented as the missing raw quantitative-NMR yield calculations or as independent repeat measurements.

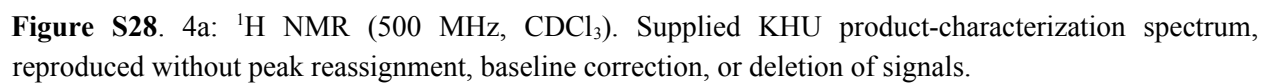


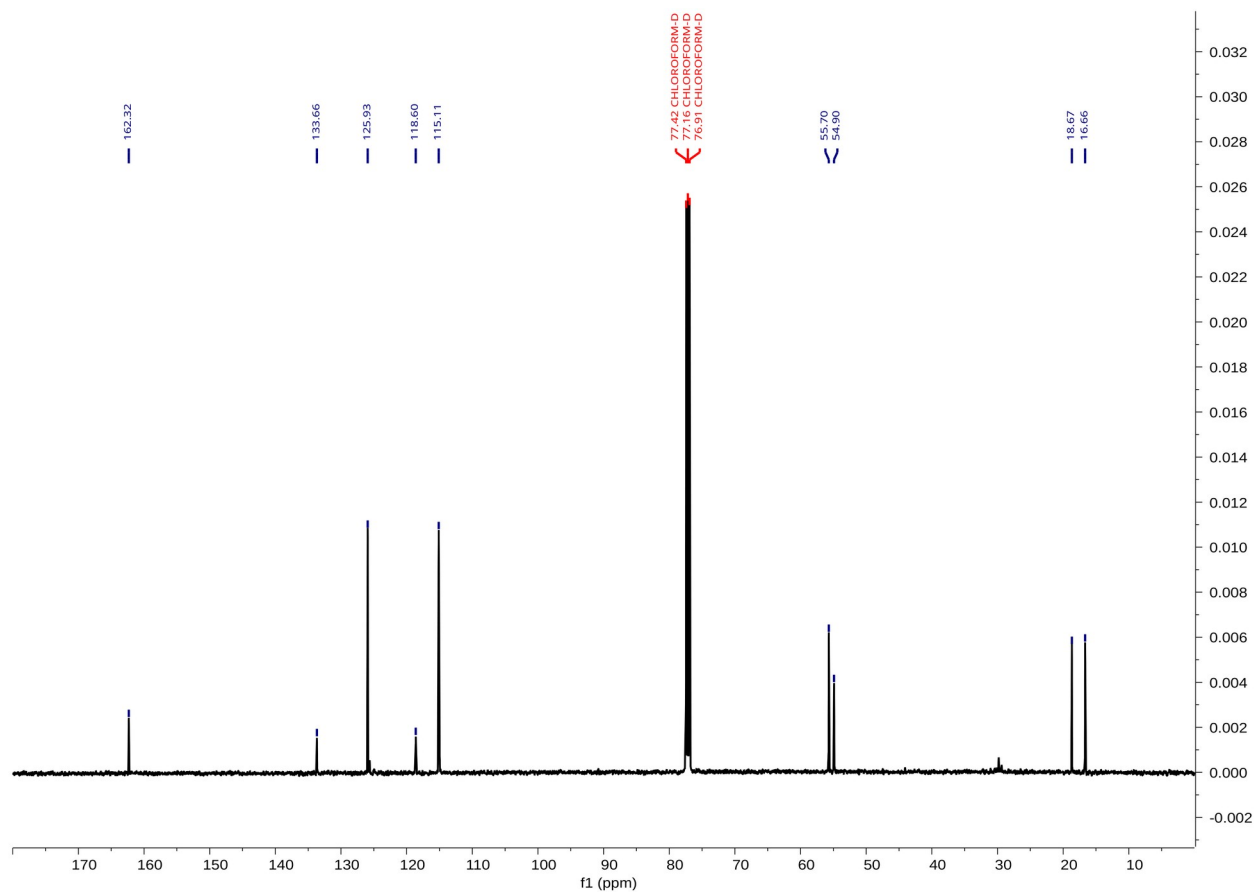
4-((4-methoxyphenyl)sulfinyl)butanenitrile (4a) was synthesized according to the standard flow procedure for **Figure 5** (Section 12.3) and purified by flash column chromatography on silica gel (n-hexane:EtOAc = 1:4) to afford 4a as a colourless oil (42.0 mg, 94% isolated yield).  $R_f = 0.20$  (n-hexane:EtOAc = 1:4).  $^1\text{H}$  NMR (500 MHz,  $\text{CDCl}_3$ )  $\delta$  7.56–7.53 (m, 2H), 7.05–7.02 (m, 2H), 3.85 (s, 3H), 2.95–2.90 (m, 2H), 2.58–2.46 (m, 2H), 2.19–1.94 (m, 2H);  $^{13}\text{C}\{^1\text{H}\}$  NMR (126 MHz,  $\text{CDCl}_3$ )  $\delta$  162.32, 133.66, 125.93, 118.60, 115.11, 55.70, 54.90, 18.67, 16.66; LRMS (ESI)  $m/z$ : calcd for  $\text{C}_{11}\text{H}_{14}\text{NO}_2\text{S}$   $[\text{M}+\text{H}]^+$ : 224.1; found: 224.4; calcd for  $\text{C}_{11}\text{H}_{13}\text{NO}_2\text{SK}$   $[\text{M}+\text{K}]^+$ : 262.0, found: 261.8. [4]



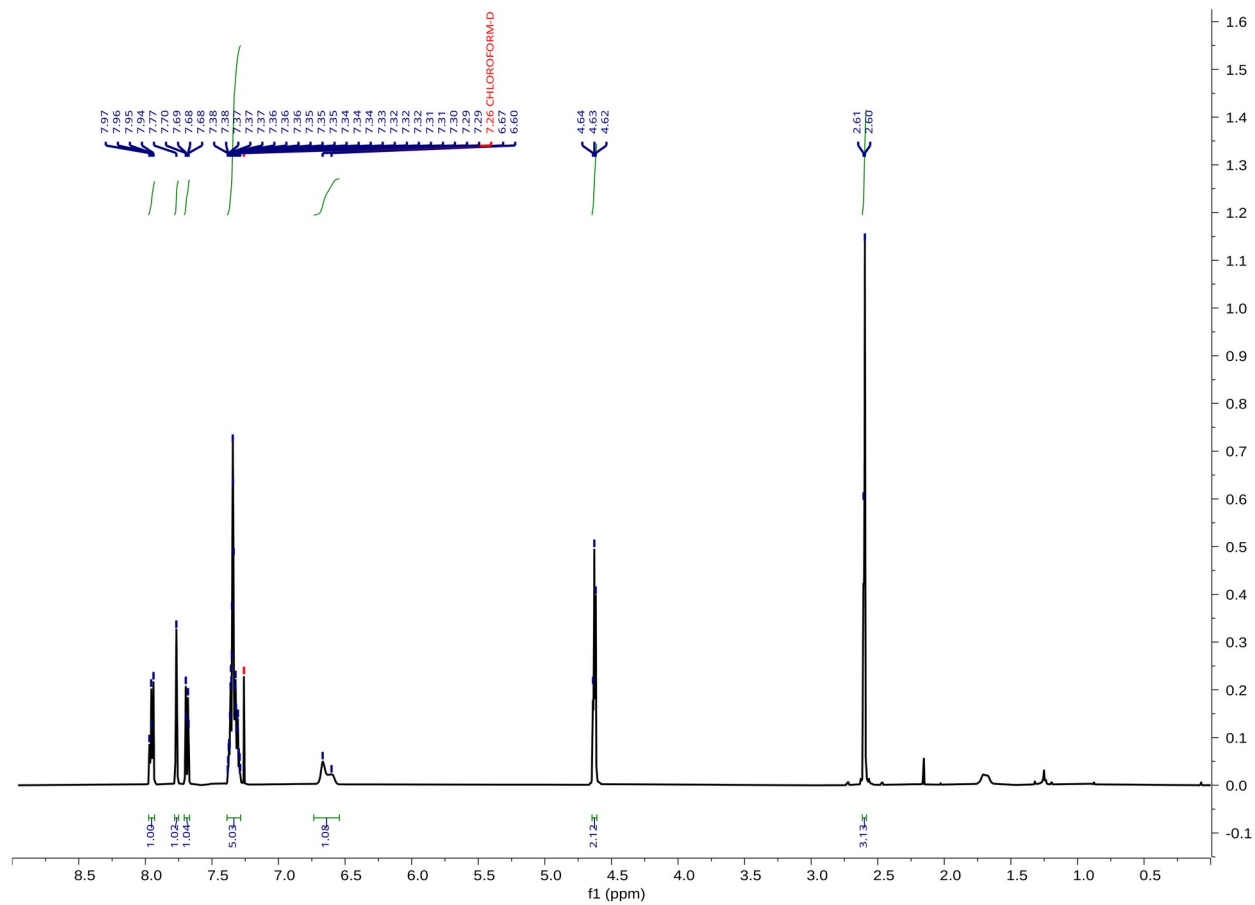
N-Benzyl-3-methyl-4-nitrobenzamide (3b) was synthesized according to the standard flow procedure for **Figure 6** (Section 12.4) and purified by flash column chromatography on silica gel (n-hexane:EtOAc = 1:3) to afford 3b as a pale-yellow solid (227.0 mg, 84% isolated yield).  $R_f = 0.36$  (n-hexane:EtOAc = 1:3).  $^1\text{H}$  NMR (500 MHz,  $\text{CDCl}_3$ )  $\delta$  7.95 (d,  $J = 8.4$  Hz,  $^1\text{H}$ ), 7.77 (s,  $^1\text{H}$ ), 7.68 (dd,  $J = 8.4, 2.0$  Hz,  $^1\text{H}$ ), 7.38–7.29 (m, 5H), 6.67–6.60 (br s,  $^1\text{H}$ ), 4.62 (d,  $J = 5.4$  Hz, 2H), 2.60 (br s, 3H);  $^{13}\text{C}\{^1\text{H}\}$  NMR (126 MHz,  $\text{CDCl}_3$ )  $\delta$  165.64, 150.89, 138.33, 137.66, 134.16, 131.85, 129.02, 128.08, 128.01, 125.46, 125.03, 44.49, 20.44; LRMS (ESI): calcd for  $\text{C}_{15}\text{H}_{15}\text{N}_2\text{O}_3$   $[\text{M}+\text{H}]^+$ : 271.1, found: 271.4; calcd for  $\text{C}_{15}\text{H}_{14}\text{N}_2\text{O}_3\text{Na}$   $[\text{M}+\text{Na}]^+$ : 293.1, found: 293.1. [5,13]

Mass/yield arithmetic agrees with the supplied starting scales: 42.0 mg of 4a (approximately  $223.29 \text{ g mol}^{-1}$ ) from 0.200 mmol corresponds to 94.1%, and 227.0 mg of 3b (approximately  $270.29 \text{ g mol}^{-1}$ ) from 1.00 mmol corresponds to 84.0%. LRMS observed values are reproduced as reported, not recalibrated; original instrument exports and author approval of these assignments remain part of the analytical record to obtain.

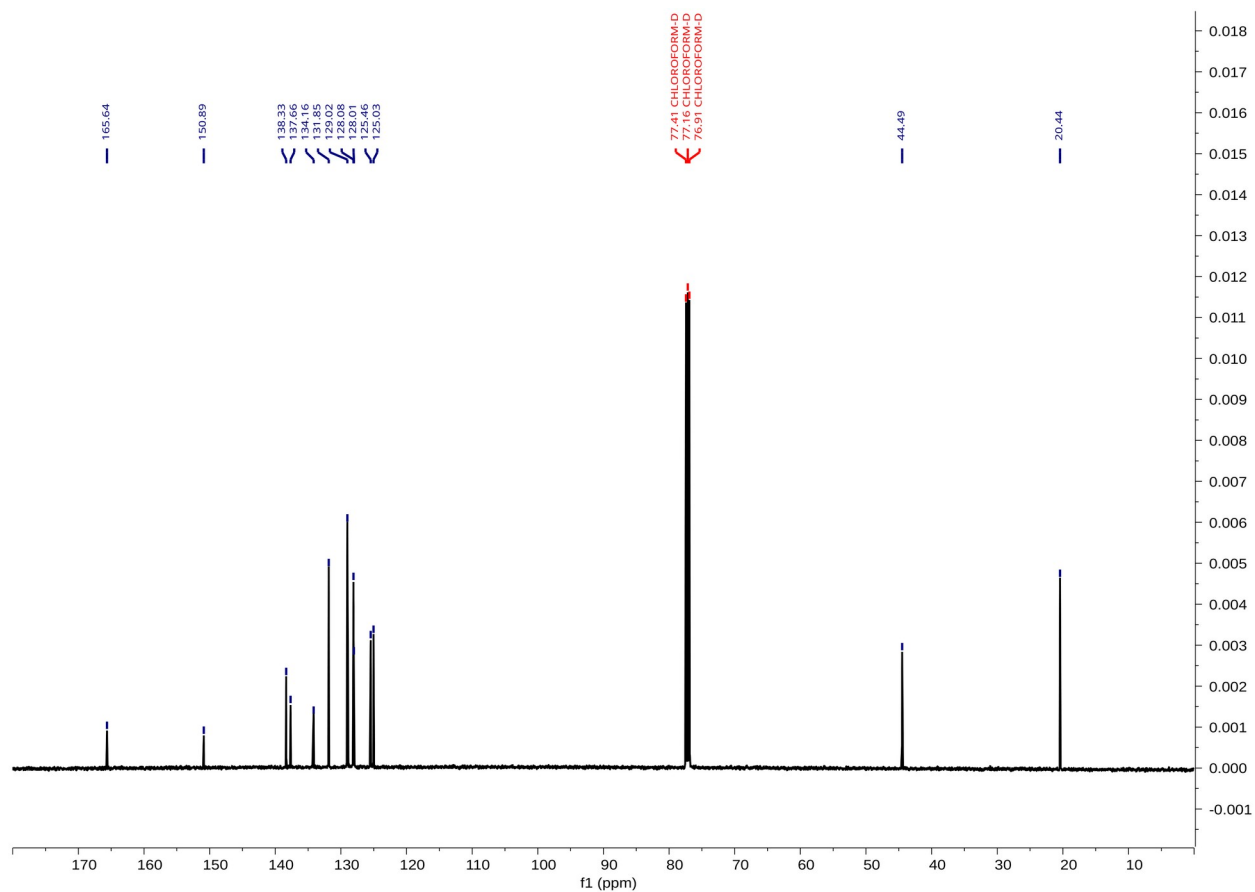




**Figure S29.** 4a:  $^{13}\text{C}\{^1\text{H}\}$  NMR (126 MHz,  $\text{CDCl}_3$ ). Supplied KHU product-characterization spectrum, reproduced without peak reassignment, baseline correction, or deletion of signals.



**Figure S30.** 3b:  $^1\text{H}$  NMR (500 MHz,  $\text{CDCl}_3$ ). Supplied KHU product-characterization spectrum, reproduced without peak reassignment, baseline correction, or deletion of signals.



**Figure S31.** 3b:  $^{13}\text{C}\{^1\text{H}\}$  NMR (126 MHz,  $\text{CDCl}_3$ ). Supplied KHU product-characterization spectrum, reproduced without peak reassignment, baseline correction, or deletion of signals.

## 12.6 Implementation reconciliation and calculation basis

**Table S38.** Proposal-to-implementation reconciliation. Agreement of selected settings does not imply that unreported operational details were verified.

| Feature                             | FlowPilot proposal / input                                                       | KHU implementation / remaining distinction                                                                                                        |
|-------------------------------------|----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Pump configuration                  | Use compatible channels, avoid unnecessary mixing of automated systems.          | Both cases used E-series pumps; <b>Figure 6</b> used two channels of the same system.                                                             |
| Photochemical reactor/light pairing | UV-150-compatible PFA first coil; separately illuminated FEP second coil.        | PFA 2/5 mL with Vapourtec 450 nm LED; FEP 20 mL with blue strip.                                                                                  |
| Oxygen placement and quantity       | Pure O <sub>2</sub> only after Stage 1; 0.090 mL/min at proposal STP convention. | O <sub>2</sub> introduced at the interstage T-mixer; same controller setpoint. MFC reference conditions/calibration not supplied.                 |
| Backflow protection                 | Gas-branch check valve included in revised proposal.                             | Photo labels a check valve; exact branch, orientation, and startup pressure trace need confirmation. No empirical failure probability is claimed. |
| Pressure                            | 7 bar(g) design/BPR target.                                                      | 7 bar cartridge installed; reported pressure 5.1–5.6 bar ( <b>Figure 5</b> ), 5.2–5.8 bar ( <b>Figure 6</b> ). Sensor basis/location unreported.  |
| Feed introduction                   | Nominal component concentrations and pump settings.                              | Finite 2.0 mL sample-loop injection. Pump rates do not establish a constant concentration throughout the dispersed plug.                          |
| Amidation interstage operation      | Direct amine addition, no intermediate isolation.                                | Two heated coils and interstage T-mixer, both at 95 °C. Water bath reported; no cooling operation described.                                      |
| Tubing dimensions                   | Nominal inventory IDs: 1.00 mm and 2.40 mm.                                      | Specified 0.04 in = 1.016 mm and 0.093 in = 2.3622 mm. Use actual tubing dimensions for future hydraulic calculations.                            |
| Performance assessment              | Three response sets per chemistry, some identical conditions.                    | Two distinct implemented settings per chemistry. Shared-set rows are not documented independent repeats.                                          |

**Table S39.** Nominal component feeds from reported stock concentrations and pump settings. Molar feed is  $C \times Q$  ( $M \times \text{mL/min} = \text{mmol/min}$ ). Values describe a stock-filled portion of the sample plug; no constant concentration or intermediate conversion is assumed outside that interval.

| Case / set            | Feed component                   | C (M)    | Q (mL/min) | Molar feed (mmol/min) |
|-----------------------|----------------------------------|----------|------------|-----------------------|
| <b>Figure 5</b> , all | Substrate 1a                     | 0.100    | 0.020      | 0.00200               |
| <b>Figure 5</b> , all | Acrylonitrile 2a                 | 0.200    | 0.020      | 0.00400               |
| <b>Figure 5</b> , all | Ir photocatalyst                 | 0.000500 | 0.020      | 0.0000100             |
| <b>Figure 5</b> , all | O <sub>2</sub> (STP calculation) | Gas      | 0.090      | 0.004015              |

| Case / set    | Feed component | C (M)  | Q (mL/min) | Molar feed (mmol/min) |
|---------------|----------------|--------|------------|-----------------------|
| Figure 6, 1   | Acid 1b        | 0.500  | 0.160      | 0.0800                |
| Figure 6, 1   | DPDTC          | 0.525  | 0.160      | 0.0840                |
| Figure 6, 1   | DMAP           | 0.0500 | 0.160      | 0.00800               |
| Figure 6, 1   | Benzylamine 2b | 2.10   | 0.040      | 0.0840                |
| Figure 6, 2/3 | Acid 1b        | 0.500  | 0.280      | 0.140                 |
| Figure 6, 2/3 | DPDTC          | 0.525  | 0.280      | 0.147                 |
| Figure 6, 2/3 | DMAP           | 0.0500 | 0.280      | 0.0140                |
| Figure 6, 2/3 | Benzylamine 2b | 2.10   | 0.070      | 0.147                 |

The supplied reference tables also compare the current runs with earlier batch and flow procedures (**Figures S24 and S26**). They are contextual literature comparisons, not controlled matched experiments: substrate concentration, oxygen feed, irradiation hardware, reactor size, and reagent ratios differ. No intensification factor, productivity advantage, or causal yield improvement is inferred from those comparisons. Quantitative interpretation of the gas–liquid stage requires the gas reference conditions and actual phase-holdup/transport information; the nominal inlet index alone cannot provide it.

The earlier KHU backflow report concerned pure oxygen at 0.43 mL min<sup>−1</sup> and liquid at 0.020 mL min<sup>−1</sup>, despite the original proposal’s air-based feed calculation. The revised point retained pure oxygen at a reduced controller setting of 0.090 mL min<sup>−1</sup>. A gas-branch check valve resists reverse liquid entry toward the MFC; it does not necessarily prevent gas entering the upstream liquid branch. The assembled photograph identifies a check valve, but the exact branch, orientation, and startup-pressure history remain unconfirmed. No backflow probability or verified hydraulic protection is inferred from the reduced gas setting.

## 12.7 Outstanding experimental metadata and release status

The available record does not include raw quantitative-NMR files and yield-calculation worksheets (including internal-standard amount and purity), independent repeat identifiers, exact collection windows, or second-feed timing. Gas-controller reference conditions and calibration, pressure-sensor reference and location, the implemented oxygen-free feed preparation, and check-valve branch and orientation remain to be confirmed. Original NMR and LRMS instrument exports, the specified tubing dimensions, and the reported 95 °C bath arrangement also require final laboratory confirmation. These gaps are retained explicitly; no measurement or uncertainty has been inferred to fill them.

Source code is publicly accessible at <https://github.com/arntfgh/flowpilot>. Benchmark scripts and the data-access index are available at [https://github.com/arntfgh/flowpilot/tree/master/ablation\\_test](https://github.com/arntfgh/flowpilot/tree/master/ablation_test) and [https://github.com/arntfgh/flowpilot/tree/master/ablation\\_results](https://github.com/arntfgh/flowpilot/tree/master/ablation_results), respectively. The latter is currently an index, not the complete outcome archive. The ESI contains the reported numerical summaries, experimental procedures, and characterization spectra. As of 29 September 2026, the repository is public but has no tagged manuscript release; the complete benchmark archive, raw experimental analytical files, and a persistent data DOI have not yet been deposited. These release items remain to be completed before submission.



## 13 References

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